C#Corner



Software Requirements Engineering

First edition



Sardar Mudassar Ali Khan

Software Requirements Engineering

Sardar Mudassar Ali Khan

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It would be my pleasure and I am feeling ecstatically rapturous to pay the heartiest thanks. First, To Allah (SWT) who is most beneficent and most merciful that he enabled me and blessed me to take the pebble out of his multitude of bounties. To my dear parents, their prayers become true towards the completion of my book. I am greatly beholden to my esteemed honorable motivator and kind-hearted, teachers, friends, and colleagues who supported my idea all the way and never let me down and corporate with me in this way completely sincerely and heartedly.
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— Sardar Mudassar Ali Khan

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1

Welcome to Software Requirements Engineering

Overview

In this chapter, we explore Understanding Software Requirement Engineering. Defining, gathering, and documenting system needs. Importance and reading guide included.

Introduction

Software Requirement is the process of describing the features and functionalities of the targeted system. Software Requirements give us a clear picture of the targeted system i.e., what the expectation of the customer for his/her product requirements can be obvious or hidden, known and unknown expected and unexpected according to the client's point of view.
Example: (Online Law Firm Android App) is a Real-Time Cloud-Based android application that will provide judicial assistance to society. We aim to help people solve their Judicial problems and guide them by providing the best possible solution. Another goal of the application is to bring all the interested parties on a single platform. Notification-based alerts to notify all the parties. Our system will help the Laws and find new cases. The system will have a communication system through which users can communicate with each other. Our system will have a rating system through which the client will rate the users.

What is Software Requirement

Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.

Example:

- Login Functionality of Online Law Firm Android App.
- Admin of the system can suspend and delete any accounts.
- Users of the system can delete any contact.
- Clients can submit /her Case in our system

What is Requirement Engineering

The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.

Example:

Let us consider you want to gather the requirement for (Law Firm Android App) from your client and after completing the requirement elicitation process you analyze them and finely you write the software requirement specification document of the project is called Requirement Engineering process.

Why Software Requirement Engineering is Important

Software requirement engineering is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

How to Read this Book

This book is designed to be read either topic by topic or cover by cover. If you are a primary level programmer, then you should read this book topic by topic OR if you are an intermediate level programmer then you should read this book from chapter 2 OR if you are a High-level programmer then you should read this book from part 2. If you are a general reader, then read the book from start to end.

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.

2

The metaphor of Software Requirement Engineering

Overview

In this chapter, we will explore Software Requirement Engineering. Discover how metaphors illuminate system development, akin to constructing a house, fostering clear understanding and effective communication.

Introduction

In this chapter, we will discuss the metaphors of software requirement engineering and how we can use the metaphor in requirement engineering for the development of software products.

What is a metaphor

A metaphor is the figure of speech that is used to understand and explain something better in software engineering as well as in software requirement engineering. Metaphors clearly define the hidden similarities of the system metaphors define the clear picture of the software system from the idea to the requirement of system software development is same as developing the house in which we first define requirement then we arrange the finance for that then we go to another step-like construction.

Metaphors in Software Requirement Engineering

In software engineering as well as in software requirement engineering metaphors clearly define the hidden picture and similarities of the system metaphors define the clear picture of a software system from idea to the requirement of system software development is the same as developing the house in which we first define requirement then we arrange the finance for that then we go to other steps like construction same in software system we need at first stage idea then we gather requirement for that idea to become a real software system for this process we use some techniques and follow some procedure for the requirement engineering process.
Metaphors are like you want to buy a car you want to know each and everything about that car from the basic requirement of the car to the functionalities as well as the latest feature of the car. But in a software system we follow some procedures for the development first, we define the idea then we gather software requirements for that system after that we follow software development principles for an efficient system.
In later stages, we will use this concept for the requirement engineering process for the project (Online Law Firm Android App).

How to Use Metaphors in Requirement Engineering

In software requirement engineering we use metaphors for defining each little requirement of the system of the software because at a later stage we analyze that requirement and then we start designing and developing the project.

- Software Development is building like a house in which we first chose the place where we want to build the house and we describe the requirements for that house and then we define each step for the completion of the house. In Software development we first define the idea then we describe that idea in the form of functional and non-functional requirements after that we arrange some resources for that project and decide a timeframe for the completion of the software development process and project.
- Software Should be like a real-world object that behaves according to the defined functionality that any customer gives to the development team. The software should be functional.
- A good customer can give you a rich number of requirements that can help you in the software development process but if you are facing a problem with the customer then you use the techniques for the requirement engineering process for any project you are working on.

Importance of metaphors in software Requirement Engineering

Metaphors have great importance in Software requirement engineering because it is the very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of Metaphors is very important for the development of effective and efficient software products.

- Metaphors help us to define hidden similarities of the system.
- Metaphors clearly define the hidden similarities of the system metaphors define the clear picture of the software system from the idea to the requirement of system software development is same as developing the house in which we first define requirement then we arrange the finance for that then we go to another step-like construction.

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.
- Metaphors help us to define hidden similarities of the system.
- Functional Requirements define the functionalities of the system

Example:

- Login Functionality of Online Law Firm Android App.
- Admin of the system can suspend and delete any accounts.
- Users of the system can delete any contact.
- Clients can submit /her Case in our system

Non-Functional Requirement supports the Functional Requirements of the system.

- Availability
- Security
- Modifiability
- Interoperability

3

Prerequisites for Software Requirement Engineering

Overview

In this chapter, we will explore critical aspects of software requirement engineering (SRE) prerequisites, emphasizing their pivotal role in project success. It underscores the importance of meticulous planning, efficient problem definition, and thorough requirement engineering for delivering high-quality software products.

Introduction

In this chapter, we will discuss the prerequisites in software requirement engineering. In software development, the most important thing is an idea and after that, we do requirement elicitation for that idea and then we define pre- and post-requirement for that software idea. In requirement elicitation, the most important thing is quality requirements because these requirements help us in the software construction process and make the project successful. The worst thing in the software development process is to revise development projects two to three times. Before beginning any project, we must be sure about prerequisites, and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC). The term Measure Twice and cut once are very important in software development because Measures all the prerequisites first and then start the development process.

Importance of Prerequisites

In Software development, the prerequisites are very important because if any organization or individual programmer wants to develop a high-quality software product, they must ensure prerequisites and high-quality practice. Such practice emphasizes the beginning of the project.

- If you emphasize these practices at the end of the project, then you emphasize system testing where many people think and ensure quality assurance.
- If you emphasize these practices in the middle of the project, then you emphasize construction practices where the development team starts designing and developing the project.
- If you emphasize these practices at the start of the project, then you need a proper plan for the development process and need high-quality requirements for high-quality products.

Determine the system you are working On.

In Software development, the prerequisites are very important because if any organization or individual programmer wants to develop a high-quality software product, they must ensure prerequisites and high-quality practice. Such practice emphasizes the beginning of the project. Before the start of development determines the project, you are working on what is the nature of the project like life-critical information critical. Determine the time frame and budget for that project. Before beginning any project, we must be sure about prerequisites, and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC).

Cause of Incomplete Planning.

- A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision is the worst part of development.
- Incomplete Requirements may lead a project to be Unsuccessful.
- Before beginning any project, we must be sure about prerequisites and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC) if you forget something in the first stage of software development then your project may lead to an unsuccessful project.

Problem Definition Prerequisites

In Software Development the prerequisites are very important for long-term projects as well as short-term projects. But before the start of any project identify the problem that may cause the project to lead to the failed software product.

Example:

- Incomplete Planning
- Poorly define the project scope
- Ineffective leadership
- Fail to identify the assumptions
- Fail to track the progress.
- Lack of detailed documentation.
- Fail to track requirements
- Fail in the selection of the right technology.
- Inaccurate measurement of time and budget.
- Cultural terms in global terms.
- Lack of datil in project planning.

Requirement Engineering Prerequisites

Complete Planning

A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision this is the worst part of development.

Efficiently define the project scope

In the Requirement Engineering process defining the project, the scope is very much important because if you define the scope of the project as poor or incomplete scope your project leads to unsuccessful.

Effective leadership

In software development, the if project leader is ineffective and poorly manages the project improperly defines the iteration of completion of a project is unsuccessful and useless at the end of the project.

Successfully identify the assumptions

In software development, if you failed to define the assumption about the project then your project in the end will be useless.

Example:

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Track the progress.

In software development, if you failed to track the full progress of the project then your project, in the end, will be useless.

Detailed documentation.

In software development, documentation is a very important phase in which we write software documents like SRS document Design Document in later stages of development we use reverse engineering techniques then we use these documents for updating the software product.

Full track requirements

Software requirement engineering is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

Selection of the right technology.

In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Accurate measurement of time and budget.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

Detailed project planning.

- A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision this is the worst part of development.
- Incomplete Requirements may lead a project to be Unsuccessful.
- Before the beginning of any project, we must be sure about prerequisites and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC) if you forget something in the first stage of software development then your project may lead to an unsuccessful project.

Amount of time you spend on Requirement Engineering.

In Software Development the prerequisites are very important for long-term projects as well as short-term projects. But before the start of any project determine the amount of time you spend on the requirement engineering process as well because software development time and budget are very important for the completion of a successful project.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If you emphasize these practices at the end of the project, then you emphasize system testing where many people think and ensure quality assurance.
- If you emphasize these practices in the middle of the project, then you emphasize construction practices where the development team starts designing and developing the project.
- If you emphasize these practices at the start of the project, then you need a proper plan for the development process and need high-quality requirements for a high-quality product

Note:

- For the successful delivery of the project emphasize these practices at the start of the project, then you need a proper plan for the development process and need high-quality requirements for a high-quality product.

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.
- Metaphors help us to define hidden similarities of the system.
- Functional Requirements define the functionalities of the system.
- Reverse engineering is the process in which when we want to update the software product, we use reverse engineering journey form document to code the product.

4

Key Requirement Decisions

Overview

In this chapter, we delve software requirement engineering (SRE) prerequisites for project success, emphasizing planning, defining scope, leadership, assumptions, progress tracking, documentation, technology selection, budgeting, and programming language choice.

Introduction

Software requirement engineering is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

- Key Decision at the start of the project is very important for the successful delivery of the project.

Some of the key decisions that are very important for successful projects are given below.

Complete Planning

A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision is the worst part of development.

Efficiently define the project scope

In the Requirement Engineering process defining the project, the scope is very much important because if you define the scope of the project as poor or incomplete scope your project leads to unsuccessful.

Effective leadership

In software development, the if project leader is ineffective and poorly manages the project improperly defines the iteration of completion of the project is unsuccessful and useless at the end of the project.

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In software development, if you failed to define the assumption about the project then your project in the end will be useless.

Example:

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Track the progress.

In software development, if you failed to track the full progress of the project then your project, in the end, will be useless.

Detailed documentation.

In software development, documentation is a very important phase in which we write software documents like SRS document Design Document in later stages of development we use reverse engineering techniques then we use these documents for the updating the software product.

Full track requirements

Software requirement engineering is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

Selection of the right technology.

In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Accurate measurement of time and budget.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

Detailed project planning.

- A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision is the worst part of development.
- Incomplete Requirements may lead a project to be Unsuccessful.
- Before the beginning of any project, we must be sure about prerequisites and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC) if you forget something in the first stage of software development then your project may lead to an unsuccessful project.

Choose the programming language.

In software development, if you failed in the selection of programming language which programming language is best for the development of the project then it may also lead to the project being unsuccessful. If you want your project to be successful and delivered on time, then select the right programming language that supports your idea that can easily implement your project functionality.

Cause of Not Choosing the Right Technology.

If at the start of the project you select the wrong programming language, then at the middle of the software development your project will need a revision of the programming language then at that stage you go from middle to start of the project and then start development with a new programming language that supports your idea, and this is the worst type of development.

Choose the Technology you want to use.

In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful. In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Cause of Not Choosing the Right Technology.

If at the start of the project, you select the wrong technology then at the middle of the software development your project will need a revision of technology then at that stage you go from middle to start of the project and then start development with new technology that supports your idea, and this is the worst type of development.

Selection of Major Software Construction Practices.

Coding

- Selection of the right programming language.
- Follow the coding techniques.
- Follow the Software Engineering principals
- Follow the coding convention.
- Coding standards

Teamwork

- Will the Programmer start programming individually or start working with team members to define some certain steps that either team or individual programmer will follow for completion of the project?

Quality Assurance

- Define some quality insurance techniques that will help you to develop the best software product.
- Write test cases for quality assurance.
- Define a testing strategy for your product.
    - Unit Testing
    - System Testing
    - Integration testing
    - Black box testing
    - Whitebox testing

Tools

- Select Right Technology.
- Select Right Programming Language.
- Selection of the Right tools for development.

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.
- Metaphors help us to define hidden similarities of the system.
- Functional Requirements define the functionalities of the system.
- Reverse engineering is the process in which when we want to update the software product, we use reverse engineering journey form document to code the product.
- Programming Language used for the development of software products.

5

Feasibility Study

Overview

In this chapter, we will explore the feasibility explores the feasibility of the Online Law Firm Android App project. It covers its scope, functional requirements, and a detailed feasibility analysis including technical, financial, resource, schedule, and operational aspects. This chapter is crucial for understanding the project's viability and ensuring successful development.

Introduction

In this chapter, we will study the feasibility report of the project when an organization is an approach to the software development company then after the complete analysis of the project a feasibility report is generated feasibility study is the complete report of the project.
Let us make the feasibility report of the Online Law Firm Android App. First, we will discuss the introduction of the project.
Online Law Firm Android App is a Real-Time Cloud-Based android application that will provide judicial assistance to society. We aim to help people solve their judicial problems and guide them by providing the best possible solution. Another goal of the application is to bring all the interested parties on a single platform. Notification-based alerts to notify all the parties. Our system will help the judicial and Judicial Consultant to find new cases. The system will have a communication system through which users can communicate with each other. Our system will have a rating system through which clients will rate the users.

Scope of Project

We aim to help people facing different problems by providing expert advice through our multi-client cloud-based system. The Online Law Firm Android App will provide ease to access and use online platforms for people seeking legal help. The system will facilitate the Law to their tasks and keep track of the complete judicial process (cases). The client's Laws and the judicial consultant will be registered in our system.

- Our system will not register OnlineLawFirm.
- Our system will not register any court.

Functional Requirements

Functional requirements are those requirements in which we specify the project functionalities. These requirements are also called essential requirements.
Let us discuss the functional requirements of the project (Online Law Firm Android App)

Registration of Law, Patient, Admin:

- Users should sign up to use the system.

Manage Account:

- Admin of the system can suspend and delete any accounts.

Manage Contacts:

- Users of the system can delete any contact.

Submission of Cases from Clients:

- Clients will submit /her Case in our system.

Assign the case to the lawyer:

- Judicial Consultants can assign the cases to lawyers according to their area of the field.

Notification system:

- Law and Judicial Consultant send Notification to patient.
- The law experts will send a notification to a lawyer.

GPS Location

- The system will store the GPS location of the Law and development.

Communication System:

- Lawyers and law experts can communicate with clients.

Lawyers' availability confirmation:

- After receiving the notification lawyer can confirm their availability for the Case.

Rating System:

- Users can give a rating to Law

Lawyer promotion:

- Performance-based promotion of Law

Sharing of Legal Documents:

- A Law can share a judicial report with the patient

Save Case History:

- Law and patient can save case history in the system

Feasibility Analysis

Feasibility analysis is the complete and comprehensive analysis of the system. It determines the

- Technical feasibility
- Financial feasibility
- Resource's feasibility
- Schedule feasibility
- Operational feasibility

Let us discuss the feasibility analysis of the project (Online Law Firm Android App)
An overall study of the project has been carried out and the findings have been listed in this report. The feasibility of the "Online Law Firm Android App" can be ascertained from the following assessments:

Technical Feasibility

In technical feasibility, we discuss the technologies that will be used in the targeted system in the first part of the book you have studied that the selection of technology and programming language is the most important part of software development.
Let us discuss the feasibility analysis of the project (Online Law Firm Android App)
The system will be using different technologies for users. Following technologies will be used in the development of the system.

- Java
- XML
- Firebase
- Android Studio

Financial Feasibility

In finical feasibility, we discuss how much finance will be used in the targeted system in the first part of the book you have studied that selection of technology and programming language, and allocation of time and budget is the most important part of software development.
Let's discuss the feasibility analysis of the project (Online Law Firm Android App)
Feasibility of financial aspects e.g., pricing and source of availability and financing of tools and other resources required for the project. As shown in Table 5.1: Financial Feasibility

Table 5.1: Financial Feasibility

Resource Name	Price	Source	Sponsor
Android phone	Rs 18,000/-	Samsung	Self
Laptop	Rs 45,000/- Rs 63,000/-	Acer HP	Self Self
Android Studio	Open Source	https://developer.android.com/studio	Self
Firebase	2500/10GB-	https://console.firebase.google.com/	Self

Resource Feasibility

In resource feasibility, we discuss what type of resources will be used in the targeted system at the first part of the book you have studied that selection of technology and programming language and allocation of time and budget, and selection of resources is the most important part of software development. As shown in Table 5.2: Resource Feasibility.
Let's discuss the feasibility analysis of the project (Online Law Firm Android App)

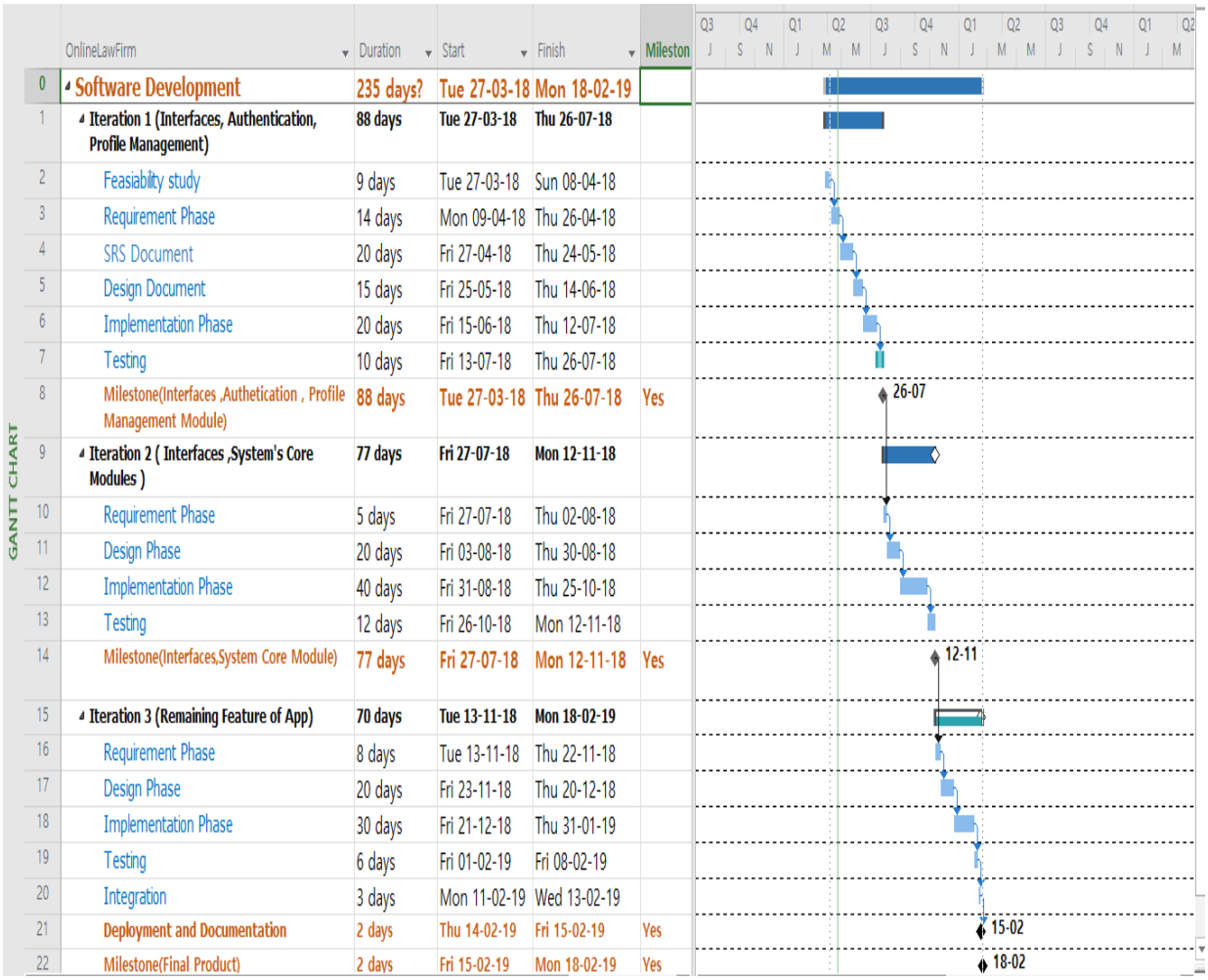
Table 5.2: Resource Feasibility.

Resource Name	Availability
Laptop	Available
Android Phone	Available
Software's	Available
Development Team	3 Members
Budget	Available
Duration	10 Months

Schedule feasibility

In schedule feasibility, we discussed the time and iteration of the project in this part we make the schedule of the project in which we study the start and end of the project as shown in table 5.3: Schedule Feasibility.

As shown in table 5.3: Schedule Feasibility.



Operational Feasibility

In operational feasibility, we discussed how this project will work and what is the outcomes of this project.

- This project is based on real-world implementation and will be deployed to the online community
- The Law who handles Judicial Cases can use this project's Online Law Firm Android app.
- An Online Law can also keep a record of the Cases.

Non-Functional Requirements:

Non-Functional Requirement is those requirements that support the functionalities of the targeted system. Non-Functional requirement targets the quality assurance of the project.

Performance

- The system will give a response within one second.

Availability

- The system will be available 24/7.

Scalability

- The system is scalable to the next level.

Usability

- Desired results, easy to use.

Maintainability

- Easily maintainable.

Assumptions

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Feasibility Study Document

Download the document for practicing and writing the Feasibility study of the Project.
https://github.com/SardarMudassarAliKhan/Fesibility-Study-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.
- Metaphors help us to define hidden similarities of the system.
- Functional Requirements define the functionalities of the system.
- Reverse engineering is the process in which when we want to update the software product, we use reverse engineering journey form document to code the product.
- Programming Language used for the development of software products.

6

Introduction of Requirement Elicitation

Overview

In this chapter, we will explore critical process of requirement elicitation, where techniques to gather, analyze, and document software requirements are explored. From interviews to questionnaires, various methods are discussed, along with challenges, prioritization techniques, and the pivotal role of a business analyst. Key points and practical examples enrich the comprehensive coverage

Introduction

In this part of the book, we will discuss the complete process of requirement elicitation. How we can get requirements about software products by using different techniques in this chapter, we will study about introduction about the requirement elicitation process.

What is Requirement Elicitation

The process to gather the software requirement from customers and analyze those requirements to document them is called requirement engineering or requirement elicitation. If the feasibility report is positive from all aspects, then we start the requirement elicitation process in this process we start requirement engineering we use different techniques to get requirements about the project. We discuss the features and functionalities of the system and identify the user and stakeholders.

Example:

- Let us consider you want to gather the requirement for (Law Firm Android App) from your client and after completing the requirement elicitation process you analyze them and finely you write the software requirement specification document of the project is called Requirement Engineering process.

Process of Requirement Elicitation

Requirement Gathering

- The developer discusses with the client and end-user to know about features and functionalities of the system and wants to know the customer expectations for the software product.

Organizing Requirement

- After the requirement engineering process, the developer prioritizes and arranges the requirement in order of High middle, and low.

Negotiations & Discussion

- If the requirements are ambiguous or there are some conflicts in the requirement of various stakeholders, the developer negotiated and discussed with stakeholders. These requirements come from different stakeholders so to remove the ambiguity and conflicts the developers and stakeholders discussed with each other’s and compromised reasonably.

Documentation

- All the informal, formal functional non-functional requirements are written in the formal document called as Software Requirement Specification (SRS) Document.

Importance of Requirement Elicitation

Software requirement elicitation is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

- Software requirement elicitation is the basis of every software product.

- Software requirement elicitation identifies the Functional and Non-Functional Requirements of the system.
- Software requirement elicitation identifies the business requirement of the system.

Level of Requirement

- Business Requirements
- Functional Requirements
- Non-Functional Requirements

Business Requirements

Business Requirements are those types of requirements that describe why an organization implemented the system. These requirements identify the business logic for the system and describe the high-level view of the system. Business requirements describe the need and expectations of the system.

Example:

- To reduce the number of staff members by some limit/percentage (let's say 30%.
- To reduce the paper handwork of the employees of the organization.

User Requirement

User requirement is those type of requirement that describes the task of the user that the user must be performed with the system

Example:

- Information about courses of the university.
- Information about tickets of the airline.

Functional Requirement

Functional Requirements are those requirements that describe what the developer must be implemented to enable the user to achieve the desired task we identify the features and functionalities of the targeted system functional requirements are very important for the system.

Example:

- The user selects the Airline through the kiosk.
- User Select the No of seats.
- Users select the source and destination for travel.

Techniques and Process of Requirement Elicitation

Business Analyst

A business analyst is an individual who has the primary responsibility to gather requirements, elicit, analyze the document and validate the needs of the stakeholders. BA acts as the bridge between customers and the developers' team. BA does bi-directional communication to cover up the requirements risk. The business analyst communicates with the project manager and communicates with the development team and with the testers and the project sponsor.

Function and Responsibilities of Business Analyst

- A business analyst plans the required approach for the requirement engineering process.
- The business Analyst identifies the stakeholders who give us the requirement about the targeted system.

- The business Analyst identifies the elicitation techniques and elicits the requirements. Business Analyst analyzes the requirement and handles conflict.
- The business Analyst documents the requirement and makes the formal document for requirements which is an SRS document.
- The business Analyst communicates the requirement between all the stakeholders.
- The business Analyst leads the requirement validation process.
- Business Analyst facilitates requirements prioritization.
- Business Analyst manages requirements and approved requirements.

Who can be a business User?

Former User of the Organization

We can make the former user of the organization or ex-employee of the organization the business analyst

Pros

Pros of making the former employee of the organization a business analyst.

- Former user of the organization under the working environment of the organization.
- Former users speak the language of the organization.
- Former users know the existing system.

Cons

Cons for making the former employee of the organization a business analyst.

- Former users maybe not be well in software engineering concepts.
- Former users may have less technical knowledge.
- Former users might not consider input from joiner's staff.

Former developers /testers of the Organization

We can make the former developers and testers of the organization business analysts.

Pros

Pros for making the former developers and testers of the organization business analysts.

- Understand how the system is in development.
- Software engineering concepts may be betters.
- Developers or testers are good at finding errors and bugs in underdeveloped systems.

Cons:

- Developers and testers usually lack communication skills.
- Focus more on the functionality of the system and ignore the customer choice that leads to the overlooked stakeholder.

Former project manager of the Organization.

We can make the former project manager of the organization a business analyst.

Pros

Pros of becoming a project manager as a business analyst.

- Former project managers have a lot of experience in working with teams.
- Strong organizational skills and strong communication skills.

Cons:

Cons in making project manager as the business analyst.

- Formers project manager lack in software engineering concepts.
- Former project manager lacks modeling skill.

Subject matter expert

We can make the subject matter expert of the organization a business analyst subject matter expert also known as a domain expert.

Pros:

Pros in making SME as the business analyst.

- SME has a lot of experience.
- SMEs have a lot of understanding of the domain.

Cons:

Cons in making SME as the business analyst.

- SMEs may not give an innovative idea.
- SMEs in old the requirement according to their preference

Rookie

We can make a rookie a business analyst rookie is a fresh graduate.

Pros:

Pros in making Rookie as the business analyst.

- No preconceived notions (No idea) clean about the statement.
- Rookie gives Innovative ideas.

Cons:

- Rookie has no experience in the domain.

Elicitation Techniques

Elicitation techniques are those techniques in which we study how to elicit and gather the requirements for the targeted system. Some of the elicitation techniques are discussed below.

Interview

In an interview face to face, interaction occurs with the clients and stakeholders. In the interview we have two types of interviews closed interviews and open interviews. In closed interviews, we have a pre-defined set of questions and in Open interviews no pre-defined set of questions. Interviews give us qualitative data, but a questionnaire gives us quantitative data.

Pros:

- From interviews we collect a rich amount of data.

Cons:

- Qualitative data are difficult to analyze.

Questionnaires

Questionnaires give us quantitative data because we give a single paper of multiple statements to a large audience, and everyone gives us different types of views on those statements. Before developing any type of questionnaire, we must focus on these precautions which are given below.

- The purpose of the survey should be defined.
- The sampling group means the targeted group.
- Provide clear instructions on how to complete the questionnaires.
- Keep the questionnaires brief and user-friendly.
- Prepare and develop the questionnaires in such a way that can gather a rich amount of data.
- conduct the survey and analyze the results.

Types of questionnaires

Closed-ended

Closed-ended means limits the answer of the respondent to the option given.

Dichotomous

- (Yes/No), (True/False), (Agree/Disagree)

Multiple choice questions.

- In multiple choices, we give different types of questions.

like red Scale

- On the red scale, we give the rating scale to the audience.

Example:

- Size Strongly Satisfied Agrees on Disagree.
- Color
- Shape

Open-ended

- No predefined set of options like taking feedback from the users of the system.

Matrix Questions

- Closed-ended questions but arranged in the form of a table. See example no 1.

Contingency

- Questionnaires were answered based on the previous response.

Example:

If you are using iPhone 7 and why you are using it, I phone 7.

Challenges in Requirement Elicitation

The yes-but syndrome

- The yes-but syndrome means requirements are clear to all stakeholders but not completely clear.

The undiscovered ruins syndrome

- The undiscovered ruins syndrome means the identification of the right stakeholders that give us the richest data.

The user and developer syndrome

- The user and developer's syndrome mean a communication gap between users and developers.

Challenges in Requirement Elicitation

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Classification of the data.

Qualitative data

- The data gives us information about the quality of the software and hardware product.

Example:

Color, text, Appearance.

Quantitative data

- The data gives us information about the quantity of the product, and part of the product .this data deals with numbers.

Example:

Number, weight, Area, volume.

Structured data

- The data is in the organized form while unstructured data is in the form of unorganized form.

Five-Step of Problem Analysis

Gain agreement on problems

- Describe the problem.
- Affects Identify stakeholders.
- End Results Describe the input of the problem on stakeholders.
- Benefits of solution Indicate the proposed solutions and key benefits.

Understand the Root cause.

- Too Much scrape customer returns.
- Manufacturing Defects.
- Damage in shipment.
- Obsolete Goals.
- Inaccurate sales Orders.

- Understand the root cause of why loss in benefits occurs.
- Create the Roadmap to understand the root cause and then find the solution to that problem.

Identify stakeholders

- Identify stakeholders that are directly related to our system and that support our system.

Define system boundaries

- Identify the geographical boundaries of the system and actors that are external to the other system.
- Define the system boundaries of your system.
- And actors that are external to the other system.

Constraints

- Define the economic constraints.
- Define technical constraints.
- Define schedule.
- Environments.

Background Reading

The elicitation technique gathers requirements by reading the various documents associated with the organization.

Source of the background reading.

- Company reports, Organization, Charts, Policy manuals, reports, and existing system documents.

The problem of Background reading.

- Very long and tedious technique.
- Documents might not match with the reality of the organizations.

Benefits of Background reading.

- A good starting point when organizations are unknown

Introspection

In introspection, Business analysts think about the requirements and features that must be possessed by the system, but these are added by the business analyst itself. This might lead to gold plating.

Social Analysis

Active Analysis.

- Inactive analysis business analysts physically and practically observe the working of the customer organization.

Passive Analysis:

- In passive analysis business, analysts are not physically present in the customer organization but observe the working through the technology.
- Like: CCTV Cameras.

Explanatory:

- In explanatory business, the analyst must present in the customer organization but in a different form. In explanatory business, analysts interact with the employees of the organization.

Workshop

Workshop in which key stakeholders gather for a short intensive period, not more than 1 or 2 days for gathering techniques of requirement engineering.

Advantages of Workshop

- Stakeholders are physically present in the organization.
- Face-facing communication between all stakeholders.
- Brainstorming session together techniques for Requirement engineering Process.

Preparing of Workshop

Selling the concept

- All stakeholders share ideas with each other and find a general objective.

Ensuring the participation of the right stakeholders

- All Right stakeholders collect the rich data for Requirement Engineering.

Attending the Logistics

- Logistics means Resources requirement to support the Workshop Process.

Project-Specific Information

- A document that is related to the system you are going to develop is called project-specific information

Out of Box thinking

- Will help all participants to find the process of creativity rule of Brainstorming.

Facilitators

Facilitators or mostly the management people can be outside of the organization.

Roles of Facilitators

- The person who is responsible for the workshop.
- Establish the professional tone for the workshop.
- Start and stop the meeting on time.
- Establish and enforce the rules for meetings.
- Introduce the agenda.
- Facilitate the decision-making process without participating in context.
- Control descriptive behavior.

Brainstorming and Techniques of Brainstorming

Brainstorming is the process of generating ideas to solve a requirement engineering problem. It is a group activity in which efforts are made to reach a conclusion.

Rules of Brainstorming

Idea Generation

Do not allow criticism.

- During the idea, the generation does not criticize any participants.

Let your imagination soar.

- Listen to all stakeholder statements in a flow during the idea generation process.

Generate ideas as many as possible.

- During the idea generation, all stakeholders come up with new ideas.

Mutate and combine.

- Change the ideas and combine similar ideas during the idea generation process.

Idea Reduction

Pruning

- To remove the invalid ideas from all ideas that are generated during the idea generation.

Grouping

- Merging similar ideas that are generated during the idea generation.

Feature Definition

- Feature definition to summarize idea into 2 and 3 lines.

Prioritization

- Give the priority to the rest of the ideas.

Hundred Dollar Test

In this test, every stakeholder has 100 dollars, and when the test starts then every stakeholder gives money against each idea this process repeats every idea in the last step, we make a prioritization list and decide which idea should be taken first based on the priority list.

Critical Important and Use Test

Critical:

- Means Indispensable Feature.

Important:

- Means important to some features but not all.

Useful:

- Useful means are nice if we take the idea then it’s nice to our system

Story Boarding and his Types and techniques

Introduction

Story Boarding describes the system's functionality by telling the story. In story Boarding, we make a story about all system features and functionalities in the form of a story.

Types of Storyboarding

Passive Story Boarding

- In a passive story, the Boarding Business Analyst tells a story to the user with photo sketches and Screen Shorts.

Active Story Boarding

- Inactive story Boarding Bossiness Analyst tells a story with the help of animation presentation and with videos.

Interact Story Boarding

- In the interact story Boarding, a user interacts with mockup prototypes and asks the user to get involved in the prototyping process.

Advantages of Storyboarding

Provides an early review of user interfaces.

- Inexpensive.
- Easy and modify.
- Overcome yes but syndrome and Blank Syndrome.

Blank Syndrome

- Users have no idea what they want.

Prioritization techniques

In or Out

- Refer Requirement to Bossiness Objectives (High-Level Objective) based on the size of iteration (time resource). In this technique, we can go back to the requirements and repeat the process.

Drawback

- If we have more requirements, then we repeat or out process that's the drawback.

Three Level Scale

- In three-level scales, we have three categories High, Medium, Low, and three Dimensions important Urgent and Not so important.
- If the size of chunks is greater than the size of the iteration, we split again the iteration into categories of high medium low.

Moscow Techniques

Moscow techniques are divided into four parts.

- Must: The Requirements must be satisfied by the system to consider it successful.
- Should: The requirements are important but not mandatory.
- Could: The requirement should be implemented only if the time and resources are present.
- Want: The requirement will not be implemented.

Drawback.

- Want is unclear and ambiguous.
- No dimensions.

Quality Function Development

Customer: → Relative Benefits

Scale: 1→9

Relative Penalty

Scale: 1→9

Development: → Relative Cost Scale

Scale: 1→9

Relative Risk Scale.

Scale: 1→9

Description of Quality Function Analysis

Relative Benefits

Every new feature would be provided to customers and Bossiness.

Scale: 1→9

1: This means No one finds it to use the full feature of the system.

9: This means No one Extremely values able and useful features for the system.

Relative Penalty

How customers and Bossiness will suffer if the feature is not added.

Scale: 1→9

1: No one will be upset if the feature is not added.

9: Indicate serious downsides.

Relative Cost Scale

Cost of implementation on each feature.

Scale: 1→9

1: Quick and Easy.

9: Time-consuming.

Relative Risk Scale

The probability of not getting the right feature.

Scale: 1→9

1: Program it easily.

9: Serious concerns about feasibility.

Software Requirement Engineering Document

Download the document for practicing and writing the SRE document of the Project.
https://github.com/SardarMudassarAliKhan/Software-Requirement-Engineering-Document.git
Submit your document for feedback from our Requirement engineer at the email address
mudassarali.official@gmail.com

Key Points

- SRE stands for Software requirement Engineering.
- Software Requirement is a Condition and Capability that must be possessed by a system according to some standards.
- The process to gather the software requirement from customers and analyze those requirements, and documenting those requirements is called requirement engineering.
- Metaphors help us to define hidden similarities of the system.

- Functional Requirements define the functionalities of the system.
- Reverse engineering is the process in which when we want to update the software product, we use reverse engineering journey form document to code the product.
- Programming Language used for the development of software products.
- Moscow Techniques tell us which requirement IOS must be in the system and which requirement it should be in the system.
- Storyboarding is the requirement for elicitation techniques
- Interviews and Questioners are the two requirement elicitation techniques
- Brainstorming is the process of generating ideas to resolve conflicts in the requirement elicitation.
- Background Reading
  The elicitation technique gathers requirements by reading the various document associated with the organization.
- Business Analyst
- A business analyst is an individual who has the primary responsibility to gather requirements elicit analyze the document and validate the needs of the stakeholders. BA acts as the bridge between customers and the developers' team. BA does the bi-directional communication to cover up the requirements risk. The business analyst communicates with the project manager and communicates with the development team and with the testers and with project sponsor.
- Pre-Condition are those requirements that must be true.
- Post- Conditions are those requirements that must be true after the pre-condition.

7

Requirement Validation and Verification

Overview

In this chapter Validation and Verification, we will explore requirement validation and verification in software engineering. It elucidates the importance of clear, complete, correct, coherent, and feasible requirements, outlining methods for verification, including model creation, logical argumentation, inspection, analysis, and expert review, ensuring adherence to quality criteria.

Introduction

In this chapter, we will study how we can validate and verify the requirements that are gathered for the software product. In this chapter, we will study the characteristics of requirements and discuss a different type of requirement test that helps us to verify and validate the requirements.

What is Requirement Validation

Requirement validation is the process of confirming the completeness and correctness of the requirement. Validation also ensures the requirement.

- Clearly define the business objective.
- Clearly define the idea of the object.
- Clearly define the scope of the project.
- Verify that requirements meet the needs of customers.
- Verify that requirements are clearly understood by the developer

Objective:

The objective of the validation is essential to the identification of the missing requirements and to ensure that requirements meet the quality criteria. Validation targets the individual requirement.

Characteristics of Requirement

Clear

- The requirement should be clear and crystal and should not be ambiguous.

Complete

- The requirement should be complete means not more and not less.

Correct

- Accurately define the customer external needs

Coherent

- The requirement should be coherent according to all aspects.

Valid

- Validation is essential to the identification of the missing requirements and ensures that requirements meet the quality criteria

Feasible

- The requirement should be feasible and implementable with known constraints

Modifiable

- Can easily be changed when necessary.

Necessary

- The important requirement should be documented according to customer need

Prioritized

- Prioritized requirements according to rank.

Traceable

- Can be linked with the system requirements and design code and testing.

Verifiable

- Correct implementation of requirements can be determined by testing and analysis.

What is Requirement Verification

Software requirement verification is the process of checking the designed and fully developed software product according to our requirements that are collected in the first phase of development. Verification involves various inspection tests and analysis of thoughts and product life cycle. The requirement must be verified before designing and developing the software product.

How to verify Software Requirements

We can verify requirements by following ways

Create a Model

- Create the authentic Model and best describe the software system and verify the required form that model requirement must be verified before designing and developing the software product.

Verify Requirements form Customers

- Verify requirements from customers and all stakeholders. The requirement must be verified before designing and developing the software product.

Verify requirement from Document

- Verify requirement from documents that truly represents the software system like software requirement specification (SRS) and design document.

Logical Argument

- Logical Argument is the process of generating a statement from the existing requirement

Inspection

- Inspection is the most efferent and organized way of verifying requirements. In this process, two to three team members work to gather for the verification of software requirements for the software product.

Analysis

- The analysis is the process in which we analyze the requirement after the elicitation process. In this process we use conduct different tests for checking our requirements is fully describe the software product.

Expert Review.

- Expert review means verifying requirements from business Analysts, Domain Experts, and Senior Software developers.

Key Points

- SRE stands for Software requirement Engineering.
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- Pre-Condition are those requirements that must be true.
- Post- Conditions are those requirements that must be true after the pre-condition.
- Verification means building the product right.
- Validation means building the right product.

8

Requirement Management

Overview

In this chapter, we delve into managing software requirements, emphasizing the significance of continuous requirement management throughout the project. It discusses requirement characteristics, the importance of setting requirement baselines, reasons for project failures, and strategies for successful requirement management, including meticulous planning, effective leadership, and accurate tracking.

Introduction

In this chapter, we will study how we can manage and the requirements that are gathered for the software product. In this chapter, we will study the characteristics of requirements management which is continuous throughout the project. We will study why requirement engineering is important.

What is Requirement Management

Requirement management is the process of documenting analyzing tracing and reviewing and then agreeing on the valid requirements and then applying the requirement change process and communicating to all relevant stakeholders requirement change process in which users can change the requirement technology can change the requirement environment can change the requirement (Here we have requirement baseline which has two parts correct baseline and review baseline) Requirement baseline is the process in which when we do work on iteration and that step we work on set of approved requirement that makes requirement baseline. The requirement baseline provides ease in changing the requirement.

Why set a requirement baseline?

Answer: Requirement baseline specifies the set of requirements approved by all the stakeholders for the specification iteration.
The requirement baseline evaluates the impact of change on the system.
The requirement baseline reflects change throughout the requirement process.

Importance of Requirement Management

The importance of requirement management is to ensure that an organization's document verifies and validates the requirement and expectations of its customers and all stakeholders. Requirement management begins with analysis and elicitation of the objectives and constraints of the organization.

- The requirement management process supports the traceability of the requirements which is very important to meet the customer expectations.
- The traceability also supports the requirement change process.
- The requirement management process includes planning for requirements, integrating requirements, and organizing for working with them.
- Requirement management involves communication between the project team and stakeholders of a software project.

Why Project Fails

If the organization's concerns are not clear about the requirement management process, then software product/project leads to failing reason of project failures are.

In Complete Planning

A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision this is the worst part of development.

Inefficiently define the project scope

In the Requirement Engineering process defining the project, scope is very important because if you define the scope of the project poorly or incompletely scope your project led to unsuccessful.

Ineffective leadership

In software development, the if project leader is ineffective and poorly manages the project improperly defines the iteration of completion of the project is unsuccessful and useless at the end of the project.

Not Successfully identifying the assumptions

In software development, if you failed to define the assumption about the project then your project in the end will be useless.

Example:

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Not Tracking the project progress.

In software development, if you failed to track the full progress of the project then your project, in the end, will be useless.

No Detailed documentation.

In software development, documentation is a very important phase in which we write software documents like SRS document Design Document in later stages of development we use reverse engineering techniques then we use these documents for the updating the software product.

No track requirements

Software requirement engineering is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

Not Selection of the right technology.

In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Not Accurate measurement of time and budget.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

Not Detailed project planning.

- A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision this is the worst part of development.
- Incomplete Requirements may lead a project to be Unsuccessful.
- Before the Beginning of any project, we must be sure about prerequisites and after checking that all the permits have been obtained and make sure that all the requirements are obtained then move to the next step of the software development life cycle (SDLC) if you forget something in the first stage of software development then your project may lead to an unsuccessful project.

How we can Manage Software Requirements

We can manage requirements in the following ways.

- By Documenting all the project progress
- By Detailed Documentation.
- Software Requirement Specification (SRS) Documentation.
- By Writing Design Documentation
- By Writing All project reports from the project proposal to the end of the project.

Software Requirements Management Document

Download the document for practicing and writing the Requirement Management of the Project.
https://github.com/SardarMudassarAliKhan/Software-Requirement-Management-Document.git
Submit your document for feedback from our Requirement engineer at the email address
mudassarali.official@gmail.com

Key Points

- Programming Language used for the development of software products.
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- Verification means building the product right.
- Validation means building the right product.

9

Introduction of Software Process Models

Overview

In this chapter, we will see software process models, defining their significance in software development. It covers activities like software specifications, design, implementation, verification, validation, and evolution. These models ensure efficient development, quality assurance, adaptability to modern trends, and overall project success, emphasized through key points and examples.

Introduction

In this chapter, we will discuss software process models, why software models are important for the development of software processes, and how they can help the development team as well as all stakeholders of the software product.

What are Software Process Models?

The software model is the set of related activities that leads to the software products of the software product. These activities involve the software development process from scratch or modeling existing systems. Every software model provides the speedy development of any software product and supports every activity of software development from scratch to the end of development. Every software model is a simplified representation of the software process. Every software model represents the process from a different perspective.Every software development model must include the following activities.

Software specifications

This activity defines the product functionalities of the targeted system and the constraints involved in software development.

- Functionalities can be functional or non-functional of the targeted system.
- In Functional Requirement we discuss the essential properties of the system without these properties system can’t be functional
- In non-functional requirements, we discuss the properties of the system which can support the essential requirements of the system.

Software Design and implementation

This activity involves the implementation of the software product, this is the second step of software development.

- This step involves the implementation of the essential requirements of the system.
- This step involves the assurance of quality attributes.

Software Verification and Validation

This activity involves the verification of the customer's needs and expectations. This step involves the validation and verification process of the first two-phase software specification and software design and implementation likewise we check the system meets the customer requirement that we gather in the first phase of the requirement elicitation process.

Software Evolution

This activity involves the modification of software to meet the customer and market new trends. Modification means improving the system to such standards that can support today's technologies and meets the customer requirement efficiently.

Importance of Software Process Models

Software Process models have great importance in the development of software products. Every software model controls, manages, guides, and monitors the software development process. Software process models give a clear way for the development process. We discussed some important reasons why software process models are important.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.
- Software process models create the new logic for problem-solving based on open-ended programming environments for the high computer system.
- Guides the software development teams to make new methodology for the development and construction of software products.
- Software process models provide a flexible and supportive development environment.
- Using Software process models every organization can run more than one task at a time.

Key Points

- The software model is the set of related activities that leads to the software products of the software product.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Programming Language used for the development of software products.
- Moscow Techniques tell us which requirement IOS must be in the system and which requirement it should be in the system.
- Storyboarding is the requirement for elicitation techniques
- Interviews and Questioners are the two requirement elicitation techniques
- Brainstorming is the process of generating ideas to resolve conflicts in the requirement elicitation.
- Background Reading
  The elicitation technique gathers requirements by reading the various document associated with the organization.
- Business Analyst business analyst is an individual who has the primary responsibility to gather requirements elicit analyze the document and validate the needs of the stakeholders. BA acts as the bridge between customers and the developers' team. BA does the bi-directional communication to cover up the requirements risk. The business analyst communicates with the project manager and communicates with the development team and with the testers and with project sponsor.
- Pre-Condition are those requirements that must be true.
- Post- Conditions are those requirements that must be true after the pre-condition.
- Verification means building the product right.
- Validation means building the right product.

10

Software Process Models

Overview

In this chapter, we will see software process models' significance in software development, exploring the Software Development Life Cycle (SDLC) and its phases like Project Planning, Requirements Definition, Design, Development, Testing, and Deployment. Various models such as Waterfall, Iterative, Spiral, V-Model, Agile, and RAD are discussed, emphasizing their applications and advantages. The chapter underscores the importance of efficient planning, requirement gathering, design, and testing methodologies to ensure the successful development and deployment of software products, catering to stakeholders' needs and enhancing product quality.

Introduction

In this chapter, we will discuss software process models, why software models are important for the development of software processes, and how they can help the development team as well as all stakeholders of the software product. In the software development life cycle, SDLC consists of a series of planned activities to develop the software product.

What is SDLC?

The software development life cycle is the process followed within a Software Organization. It is a detailed process that describes how to develop, maintain and replace the software product. The following figure describes the graphical representation of a process of SDLC.

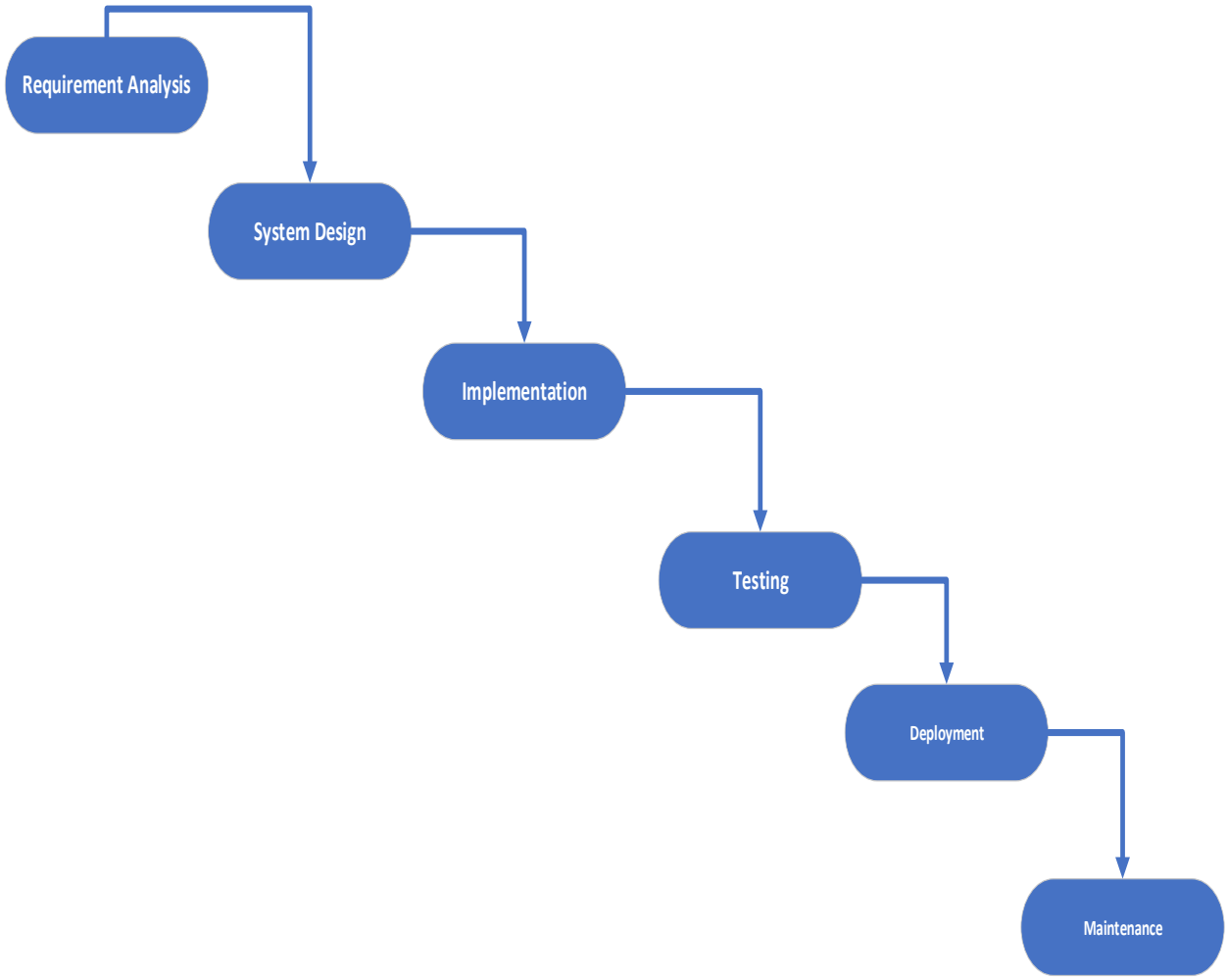


Figure 10.1: Software Development Life Cycle

Step 1: Project Planning

This step is the most important step of the software development life cycle (SDLC). This step is performed by senior members of the team because a big project requires more efficient planning for making the project successful likewise small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times of revision this is the worst part of development. In this step, we identify quality assurance procedures and risks associated with the project. The outcome of this step is used in

the feasibility study of the project. Project feasibility defines the various procedures that can be followed for the successful delivery of a project with minimum risks.

Step 2: Defining Requirements

Defining requirements is the most important part of the software development life cycle. The process to gather the software requirement from customers and then analyze those requirements and document them is called requirement engineering or requirement elicitation. If the feasibility report is positive from all aspects, then we start the requirement elicitation process in this process we start requirement engineering we use different techniques to get requirements about the project. We discuss the features and functionalities of the system and identify the user and stakeholders.

Step 3: Designing and Architecture

The designing and product architecture means clearly defining the structure of modules along with communication among modules and data flow representation with the external third-party module. The internal structure of the module should be clearly defined. This step of the software development life cycle is reviewed by all the stakeholders based on various parameters such as risk assessment, product robustness, time and budget, etc.

Step 4: Building and Developing the product

After defining the clear and rich requirements the fourth step is building and developing the software product according to the defined standards. Software organizations start coding and designing products this step involves the implementation of all the pre-and post-condition of a project like the development of complete functionalities of the software product according to the requirement verified by senior members of teams and all the stakeholders of the project.

Step 5: Testing the product.

In this step of the software development life cycle, we define the testing strategies in which we do the testing of different modules, and we check that either every module is behaving and functioning according to the standard definition or according to the requirements define in the first phase of the SDLC process.

Step 6: Deployment

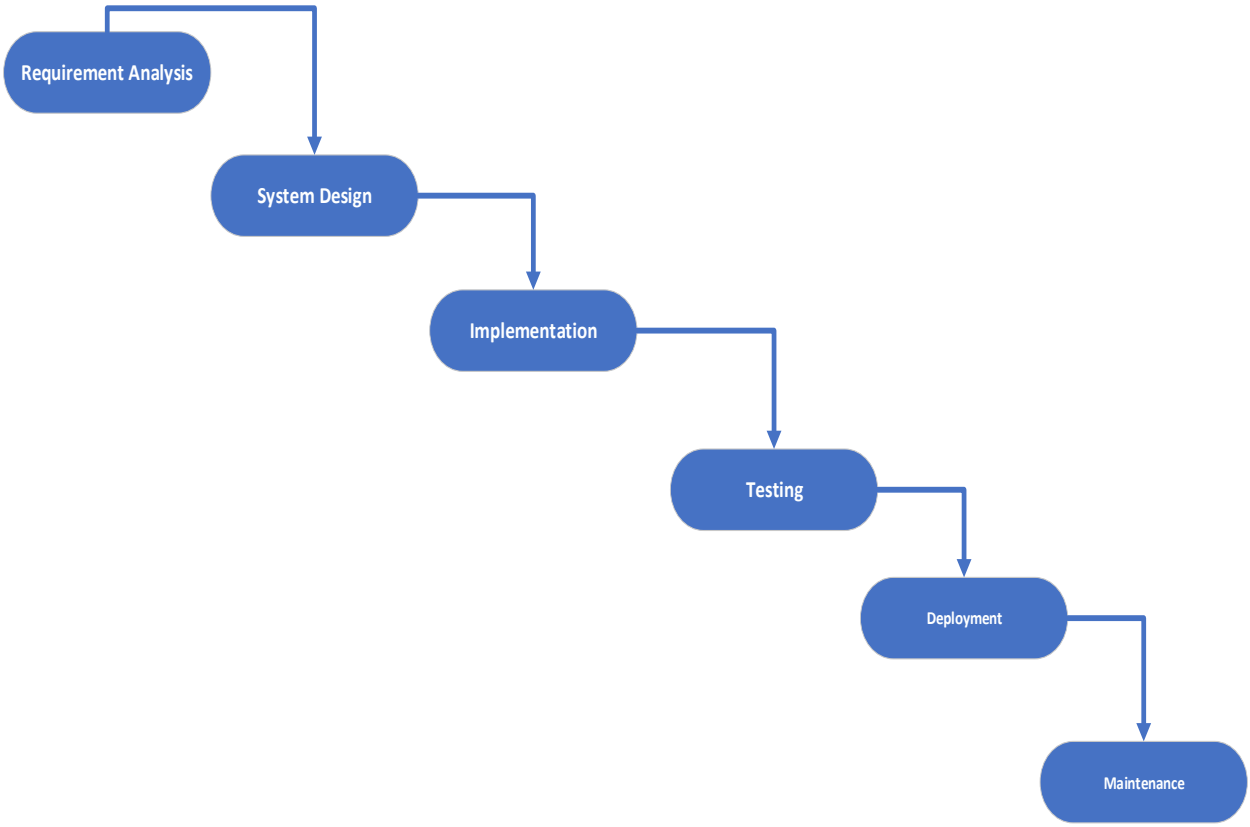
After the complete testing of the product and the product is formally ready for deployment then some of the time the product is released in the stages according to the deployment plans of the organization. And some time product deployment is dependent on the feedback of the customers.

Waterfall Model

The waterfall model is the first model in software development history. It is also referred to as the (Linear sequential life cycle model) this model is simple to understand and use. In the waterfall model, each phase is completed before moving to the next phase. The waterfall model is the first and earliest approach used in SDLC for the development of software products in this model do not overlap.

Waterfall Model Design

The waterfall approach was the first approach in the software development life cycle and was widely used in the development of software products and for the success of the project.



The sequential phases of waterfall models are.

Requirement Analysis.

- All the key requirements of the system development and captured in this phase and we document them in the software requirement specification document.

System Design.

- The requirement specification from the first phase is studied in this phase and these requirements help us to define the overall system design and architecture of the system.

Implementation.

- After defining the clear and rich requirements the fourth step is building and developing the software product according to the defined standards. Software organizations start coding and designing products this step involves the implementation of all the pre-and post-condition of a project like the development of complete functionalities of the software product according to the requirement verified by senior members of teams and all the stakeholders of the project.

Integration and testing

- In this step of the software development life cycle, we define the testing strategies in which we do the testing of different modules, and we check that either every module is

behaving and functioning according to the standard definition or according to the requirements define in the first phase of the SDLC process.

Deployment

- After the complete testing of the product and the product is formally ready for deployment then the of time product is released in the stages according to the deployment plains of the organization. And some time product deployment is dependent on the feedback of the customers.

Maintenance

- This activity is involving the modification of software to meet the customer and market new trends. Modification means that improve the system to such standards that can support today's technologies and meets the customer requirement efficiently.

Applications

The waterfall model is widely and extensively used in the software industry some given below point explains the usage of the Waterfall model.

- The requirement is very well clear and fixed
- Product Definition is stable and well defined
- No ambiguity in the requirements
- Due to the rigidity of the model projects are short

Advantages of Waterfall models

- Simple and Easy to understand
- Clearly and defined stages
- Easy to arrange tasks
- Process and results are well documented
- Works well when the project is small
- Easy to manage due to the rigidity of the model

Dis-Advantages of Waterfall models

- The high number of risks and uncertainty
- Poor model for long and ongoing projects
- Not suitable for life-critical systems because of the rigidity
- Difficult to manage projects progress within stages
- Not suitable for the project that they are following an Object-oriented approach.
- Cannot accommodate the requirement change process.

Iterative Model

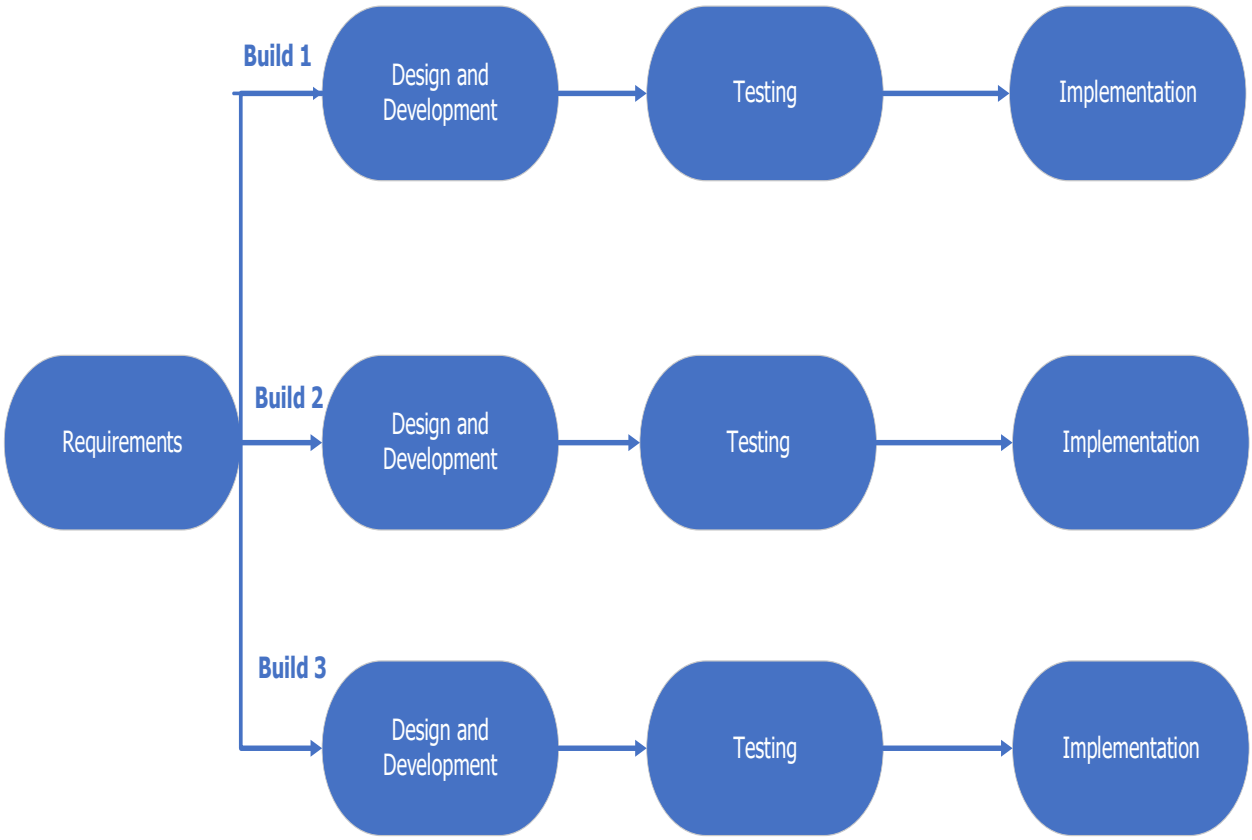
The iterative model starts the process with a simple implementation of a small set of requirements and iteratively enhances the evolving versions of the system that is complete and ready to be deployed. The iterative model does not attempt to start the development after the complete specification of the system instead development starts with just part of the software product.

Iterative Model Design

The iterative process starts with a small representation of the software product. In this model, we start working on a project with a small chunk of requirements. In this model, we repeat

certain steps like design and development testing and implementation in each iteration. Iterative enhance the evolving versions of the system are complete and ready to be deployed.

The following picture illustrates the Iterative Model



Applications

The iterative model is widely and extensively used in the software industry some given below point explains the usage of the iterative model.

- The requirement of the complete system is clearly defined.
- Major requirements must be defined however some requirements and functional may need modification and upgrading with time
- There are some high-level risks, features, and goals involved that may be changed in the future.

Advantages of Iterative Model

- Some working functionality can be developed quickly at the early stages of the Iterative project
- Clearly and defined stages
- Easy to arrange tasks
- Progress can be measured
- Results are obtained at the early stages
- Easy to manage due to the rigidity of the model
- Less costly to change the requirements at any stage
- Easier to manage high risks
- Parallel development can be planed.

Dis-Advantages of Iterative Model

- More resources may be required.
- More Management attention is required.
- The cost of change is less but it is not suitable for changing requirements again and again.
- Not suitable for smaller projects
- Management Complexity is more.
- A highly skilled resource is required.
- Project progress is highly dependent on the risk analysis phase.

Spiral Model

The spiral model combines the idea of iterative development with the controlled and systematic aspects of the first model waterfall.

Spiral Model Design

The Spiral Model has four phases. The software development process has repeatedly passed from these phases called iterations. It allows us to release the product in iteration.

First Step: Identification

This phase includes the identification of system requirements by continuously communicating with customers and developers and Business analysts (BA). This phase begins with the gathering of business requirements of the system in the baseline of the spiral. In the subsequent spiral when the product grows up the subsystem requirement and unit requirements gather in this phase.

Second Step: Design

The design phase begins with the conceptual design in the baseline of the spiral model and involves the architectural and logical design of the module physical design and final product in the subsequent spiral.

Third Step: Build

The constructed phase refers to the production of the actual product in the baseline we just construct the conceptual design of the product from customer feedback. Then the subsequent spiral of the clarity of requirement and design details a working model of software is called build with versions number.

Fourth Step: Evaluation of Risks

Risk Analysis involves identifying and estimating and monitoring the technical feasibility and management risks. At the end of the software development, process testers test the build and give it to the customer for evaluation and the customer provides feedback.

Applications

Spiral Model is widely used in the software development industry given below points explains the usage of the spiral model in the industry.

- For medium to high-risk software projects.
- Major changes are expected during the development phase

- Requirements are complex and need evaluation to get more clarity

Advantages of Iterative Model

- Changing requirements at any stage can be accommodated
- The spiral model allows us to extensive use of the prototype.
- Users see the actual outcomes early due to the stage of the spiral
- Development is divided into smaller tasks and risky parts this makes the development easy and better for risk management
- The requirement can be gathered more accurately

Dis-Advantages of Iterative Model

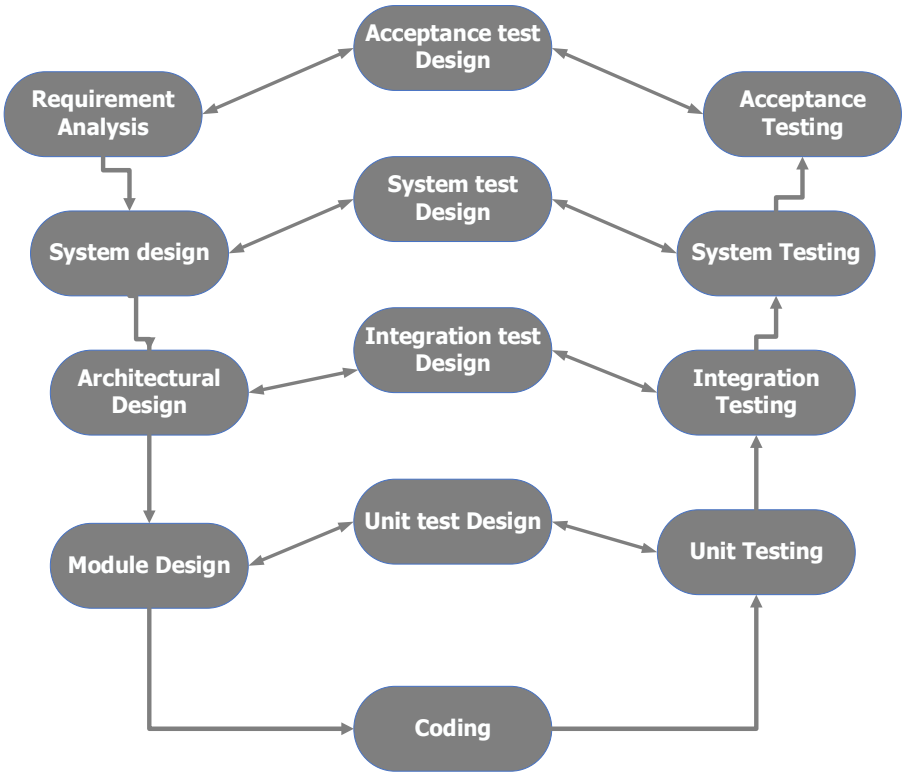
- Management of all activities of the software product is more complex
- Software development is more complex
- Many stages of the spiral model need more and excessive documentation
- Not Suitable for small and less risky projects.

V-Model

The V model is the SDLC Model where execution of the process happens sequentially. V model is also known as validation and verification. The V model is the extension of the Waterfall model based on the associated testing phase and development stages. This means that every phase in the V model is directly associated with the testing phase this is a highly disciplined phase.

V Model Design

The V-model is the testing phase of the development phase is designed in parallel, so the validation is one side of the V-Model and Verification is Another side of the V-Model, and the coding phase joins these two phases.



Business Requirement Analysis

This phase includes the identification of system requirements by continuously communicating with customers and developers and Business Analysts (BA). This phase begins with the gathering of business requirements of the system this is the most important activity and need to be managed well most of the customer is not sure about what exactly they need. The system test plan is based on the system design.

System Design

Once you have clear and crystal requirements then it's time to design the complete system. The system design is the understanding of the complete product under development.

Architectural Design

In architectural design, one technical approach is proposed which based on this technical and financial decisions will be taken. In architectural design, the system design is divided into more modules taking up different functionality. This is also called high-level design. In this phase, we study how the communication occurs between the modules and with the other software system is understood clearly in this stage of the V-Model.

Module design

In this phase, the detailed internal design for all the systems modules is specified referred to as low-level design. It is very important that the design is compatible with other module designs in system architecture as well as other external system designs. The unit test is an important part of the development phase. This helps us to find the maximum error and faults in the early stage of the development process. These unit tests can be designed based on internal module design.

Coding Phase

The actual coding of a module is designed in the design phase is taken up to the coding phase the most suitable programming language is decided based on architectural design. The coding is based on coding standards. The code is going through different code reviews and code optimization tests so that performance of the code is checked for the final build and external delivery.

Validation Phase

The different phases of the validation of the V model are explained below.

Module testing:

Unit testing is associated with the testing of modules. In module testing, we will test each module and ensure that every module and check whether the functionalities work properly in the way that we want. We created a separate and small testing procedure for every single module and tried to test each module of the application separately. Testing is done on the module level by dividing the module into chunks and checking the quality of the module separately. The following module is being tested under the umbrella of Module testing.

Example:

- Accounts Module

- Manage Account Module
- Manage Contact Module

Integration testing:

Integration testing is associated with Architectural design. After we tested each module separately then we tried the fundamental principle of integration testing in which we combine individual units are combined and tested as a group to identify the faults and bugs from the result of integration testing. We used test drivers and test stubs to identify the faults and bugs from the group of units and clarify how individual units interact with each other and work as expected.

System Testing:

System testing is directly associated with the system design phase. After Integration testing then we tried the fundamental principle of system testing in which we test complete software systems. The purpose of this testing is to evaluate the compliance of the software product with the given software requirements.

Acceptance testing:

Acceptance testing is directly associated with the requirement analysis phase. After Integration testing then we tried the fundamental principle of acceptance testing in which we test the system for acceptability and test the complete software system the purpose of this testing is to evaluate the compliance of the software product with the given business requirements and assess that whether this system or product is acceptable for delivery.

Applications

V model is the same as the waterfall model, both are sequential models and requirements are very clear before the start of the project. V-Model is widely used in the software development industry given below points explains the usage of the spiral model in the industry.

- Requirements are well defined and clearly defined in the document.
- Product definition is stable.
- There is no ambiguity in the requirements.
- V model is suitable for short projects.

Advantages of V-Model

- This is a high declined model so stages are completed one at a time.
- V-Model works well in small projects where requirements are well defined.
- Simple to understand and easy to use
- Easy to manage due to the rigidity of the model each phase has a specific deliverable and review process.

Dis-Advantages of V-Model

- High risks and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for ongoing and long projects.
- No working software is produced until late during the life cycle.

Big-Bang Model

The Big-Bang model is the Software development life cycle model where the development team does not follow any specific process. The development of the software product just starts when finance and efforts are available the input and output of the software may not be as per customer requirements. The Big-Bang model does not follow any specific process/procedure and very little planning is required for the development process so in this model, very little planning is required. The big bang model is best for small types of projects.

Big-Bang Model Design and Application

The Big-Bang model is focused on all possible resources in software development and coding with very little planning or with no planning. The requirement understood and implemented as they come. Any change may or may not affect the whole product. The big bang model is suitable for a small project or two developers working together on a small project. This model is useful for academic projects and practice projects. The big bang model is very best for those projects where requirements are not understood, and the project's final date is not given.

Advantages of the Big-Bang Model

- The big advantage of the Big-Bang model is it focuses on all possible resources in software development and coding with very little planning or with no planning.
- A very simple model to understand and use.
- Very few resources are required.
- The big-Bang model provides flexibility to developers.
- This model is good for academic and practice projects.
- Big Bang Model provides flexible management of the software development process.

Dis-Advantages of Big-Bang Model

- Management of all activities of the software product is more complex.
- Software development is more complex.
- Many stages of the spiral model need more and excessive documentation.
- Not Suitable for small and less risky projects

Agile Model

The agile model is the combination of the incremental and iterative model that focuses on product quality and adaptability and customer satisfaction by rapid delivery of working and fully functional software products. The agile process breaks the software product into small builds this build provides in iteration. Each iteration life cycle is two to three weeks, and every iteration involves the core functionalities of the system and a complete team working on these iterations in various areas.

- Planning
- Requirement Analysis
- Design
- Coding
- Unit Testing
- Acceptance Testing

Agile Model Design

The agile model believes that every project needs to be handled differently and existing methods need to be checked to see that best suit software requirements. In agile development, each task is divided into chunks with a time frame for the successful delivery of a specific feature. An iterative approach is taken, and a working software build is delivered after each iteration each build is incremental and the final build holds the features of the whole product according to the customer requirements.

Agile Manifesto

Individual and Interactions

- In agile development, self-organization and motivations are very important areas of interactions like co-location and pair programming.

Working Software

- The demos of the working software are the best way of communication in the software development with stakeholders of the software product instead of just describing the documentation with all stakeholders.

Customer Collaborations

- In software development, the requirements are very important for the development of the software product to gather complete, clear, and rich requirements for software to interact with customers.

Responding to change

- In software development responding to change is a challenge and the agile model is focused on quick changes in continuous development.

Agile vs traditional development of SDLC

Agile is based on adaptive software development methods and whereas the other model follows the traditional approach based on the predictive approach.

Predictive approach

In the Predictive approach development of the software product only depends on the requirements and analysis that’s done at the beginning of the software.

Adaptive Approach

But in the Adaptive Approach to software development, there is no need for detailed planning is required. But we focus on clarity of the features that we want to develop. There are feature development planning and product requirement changes dynamically. The product is tested repeatedly in the iteration so in the end quality product is produced.

Customer Interaction

Interaction in software development is the backbone of the agile methodology open communication and less documentation is the best and most typical feature of agile development.

Applications

The agile model is widely used in the software development industry given below points explains the usage of the spiral model in the industry.

Advantages of Agile Model

- A very realistic approach for software development.
- Promotes teamwork and cross-training.
- Resource requirements are limited.
- The requirement can be gathered more accurately.
- Good software process model for the environment that changes steadily.
- Little or no planning is required.
- Easy to manage.
- Gives flexibility to developers.

Dis-Advantages of Agile Model

- Not suitable for handling complex dependencies.
- More risk of sustainability, maintainability, and extensibility
- Depends heavily on customer interaction.
- Transfer of technology to a new team member or another team may be quite challenging due to a lack of documentation.
- Very minimum documentation is generated during the development of software.

RAD Model

The RAD (Rapid Application Development) is based on prototyping and iterative development with no specific planning required. The process of writing the software itself involves planning for the development of software products. RAD (Rapid Application Development) is focusing gathering requirements through workshops and focus groups during the early stage of the testing. Rap uses the iterative approach for software development and reuses the prototype in later stages of development.

What is RAD (Rapid Application Development)

Rapid Application development is the software development methodology that uses minimal planning in favor of rapid prototyping a prototyping is the working model of the software in the rad functionalities of the system is developed in parallel as a prototype and integrated at the end for the complete product for faster product delivery. RAD uses the incremental and iterative approach for the development of software products for this purpose we have small teams and domain experts, and IT experts working to gather on software prototyping for faster and quality products the most important part of the software is to develop the best and reusable prototype.

RAD Model Design

The Rad model distributes the analysis, Design, Build, and test phase into the short and iterative development cycle. Like

First Step: Business Modeling

The business modeling for the software product under development is designed in terms of the flow of information and distribution of information between the different business channels. In

this stage, complete business analysis for vital information for business but this information can be gotten from the successful flow of the information between different distribution channels.

Second Step: Data Modeling

The information gathered from different distribution channels in the business modeling phase is analyzed in the data modeling phase in the form of sets. The attributes of all data sets. In this phase, we defined and identified all the data sets and the relation between these data objects established and defined in detail in the business model.

Third Step: Process Modeling

The data object defined and identified in the data modeling is converted to establish the business information flow needed to establish the business objectives as per the business model designed in the first phase of the RAD Model. The process modeling describes any change and enhancement to the data sets adding deleting updating and modifying the data objects described by process modeling.

Fourth Step: Application Generation

The actual product is built-in in the coding phase using the automation phase to convert the process data into an actual prototype.

Fifth Step: Testing and Turnover

The overall testing time in the RAD model is reduced because RAD us prototypes so the testing process is continuous, and prototypes are tested in every iteration. But the data flow and the interface between the components are thoroughly tested with complete test coverage. So, most of the components of the software product are tested already so the issues and risks are minimal.

Applications

RAD Model is widely used in the software development industry given below points explains the usage of the RAD model in the industry.

- Rad Model is used when the project can be modularized and delivered in an incremental manner
- It should be used with the high availability of the designer for design modeling.
- It should be used when the domain experts are available with maximum business knowledge.
- Should be used when the requirements change during the project and prototypes are presented to the customer in every iteration.

Advantages of Rad Model

- Changing requirements at any stage can be accommodated.
- Progress can be measured.
- Reduce development time.
- Encourage customer feedback
- A quick review of the initial software product

Dis-Advantages of Rad Model

- Management of all activities of the software product is more complex

- Highly depend on the effective and efficient team members for describing system requirements
- Suitable for the system that is components based and scalable
- Suitable for a project that takes short development time
- Management complexity is more

Software Prototype

Software Prototypes is the process of building the working model of software that is under development. But the prototype may not be holding the exact functionality of the real software product. Software prototyping is most of the most popular techniques in software development. It enables the development team to understand the customer requirements. It helps us to get valuable feedback from customers and helps the development team's designers of the product about what is the actual scope of the product and what we are going to develop.

What is Software Prototyping?

Prototyping is the working model of software with limited functionality. The prototype of the software product always does not hold the exact logic of the software. Software prototyping is most of the most popular techniques in software development. It enables the development team to understand the customer requirements. It helps us to get valuable feedback from customers and helps the development team's designers of the product about what is the actual scope of the product and what we are going to develop. It will help us to develop the initial product that is described in the proposal.

Some of the important steps are given below.

Basic Requirement Identification

This phase includes the identification of system requirements by continuously communicating with customers and developers and Business Analysts (BA). This phase begins with the gathering of business requirements of the system this is the most important activity and need to be managed well most of the customer is not sure about what exactly they need. The system test plan is based on the system design.

Developing the initial prototype

After defining the basic requirements of the system, design the showcase the requirement into the initial user interface of the system.

Review the prototype

After defining the initial interface present it to the customer and other all stakeholders and collect the feedback from the customer then review the prototype and then redesign the prototype and then use feedback for the product under development.

Revise and enhance the prototype

The feedback of customers and comments are discussed during the Review of the prototype stage some negotiations happen with the customer on some factors. Time and budget constraints, feasibility constraints, and technical feasibility constraints of the actual product implementation. So, the prototype cycle is changed again and again until the customer's expectation is met.

Software Prototype Types

We discuss the following types of software prototypes.

Through Away/ Rapid Prototyping

Through away prototyping is rapid prototyping or close-ended prototyping. This type of prototype uses little effort and minimum requirement analysis. Once the actual requirements are understood then a prototype is discarded, and the actual product is developed based on clear and understood requirements.

Evolutionary Prototyping

Evolutionary development is called breadboard prototyping which means building the functional prototype with minimal functionality at the beginning of the software development process. By using evolutionary prototypes, we use the prototype in later stages, and in this type of prototype the requirements are well-understood are included in the prototype and other requirements are added when they are well understood.

Incremental Prototyping

Incremental prototyping refers to incrementally building multiple fully functional prototypes integrating all the prototypes at the end of the system development.

Extreme Prototyping

The extreme prototype is used when in the website development domain. It consists of three sequential phases. First, the prototype is made with all existing resources and presented in HTML pages then data is simulated using prototypes. In the final phase, the services are implemented with final prototypes.

Applications of prototyping

Prototyping has great importance in software development. It increases the user involvement in the system, so the quality and user expectations are met very efficiently. Software prototype provides an initial preview of the system so that is why it is widely used in the modern software development industry.

Advantages of Software Prototyping

- It increases the user interaction with software development teams before and in the middle of the system development.
- It helps the user to understand the system well before it is developed the system.
- Reduce development time and cost because the defects can be detected much earlier.
- Encourage customer feedback.
- Missing functionality can be identified early.
- A quick review of the initial software product
- Confuse and hidden functions can be identified.

Dis-Advantages of Software Prototyping

- Risk is insufficient requirement analysis on the dependency of the prototype.
- Users may get confused about the prototype and actual system.
- Developers may use the existing prototype but technically it is not feasible.

Conclusion.

We conclude in this chapter that software process models play a key role in software development. Software process modes increase the efficiency of a software system and decrease the development time. Software Process models have great importance in the development of software products. Every software model controls, manages, guides, and monitors the software development process. Software process models give a clear way for the development process. We discussed some important reasons why software process models are important.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.
- Software process models create the new logic for problem-solving based on open-ended programming environments for high computer systems.
- Guides the software development teams to make new methodology for the development and construction of software products.
- Software process models provide a flexible and supportive development environment.
- Using Software process models every organization can run more than one task at a time.

Key Points

- The software model is the set of related activities that leads to the software products of the software products.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Programming Language used for the development of software products.
- Prototyping is the working model of the software
- RAD is a Rapid Application Development Software Model
- The software model is the set of related activities that leads to the software products of the software products.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.

- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
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11

Importance of Software Models

Overview

In this chapter, we will see the pivotal role of software process models in modern software development, emphasizing efficient planning, requirement management, iterative development, testing environments, effective communication, and predictive and adaptive approaches for successful project delivery.

Introduction

In this chapter, we will discuss the importance of software products and why the software process model is important for the development of software products. How software process model plays a key role in the software product. How modern software is more efficient using the process model. We will study all these points in this chapter.

Why the software process Model is important.

Software Process models have great importance in the development of software products. Every software model controls, manages, guides, and monitors the software development process. Software process models give a clear way for the development process.

We discuss some important reasons why software process models are important given below.

- Every Software model provides the software development cycle that helps us to efficient and effective development.
- By using the Software development process model, we can get the requirements from the customer very effectively.
- The software process model provides an effective way of requirement management.
- The software Process model defines the product business modeling.
- It provides the testing environment throughout the development cycle.
- It provides the complete details of the project by documenting all the processes during development.
  Like: SRS, Design Document, Feasibility Reports, Testing Reports
- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.
- The software process Model Provides quick and early delivery of software products by reducing development.
- The software process model provides the Architectural Design and System design of the software systems.
- In the Software process model, the development of self-organization and motivations are very important areas of interactions like co-location and pair programming.
- In the Software process model, the development of the demos of the working software is the best way of communicating in the software development with stakeholders of the software product instead of just describing the documentation to all stakeholders.
- In software development, the requirements are very important for the development of the software product to gather complete, clear, and rich requirements for software to interact with customers.

- In software development responding to change is a challenge and the agile model is focused on quick changes in continuous development.

Why Process Model important for the Modern Software Development

Here we are discussing some important points in which we identify why the software process model is important for software development.

Efficient Software Development

Software process models provide an efficient software development environment in which different team members work on different tasks that are mentioned in software documentation. Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques and by using software process models.

- The software process model provides incremental and iterative development for a software project.
- Software Process

Provides Effective Way of Requirements Elicitation

The software process model provides an effective way of software requirement elicitation and elicitation techniques. Software requirement elicitation is a very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

- Software requirement elicitation is the basis of every software product.
- Software requirement elicitation identifies the Functional and Non-Functional Requirements of the system.
- Software requirement elicitation identifies the business requirement of the system.

Provides Effective Way of Requirements Management

The software process model provides an effective way of requirement management. Requirement management is the process of documenting analyzing tracing and reviewing and then agreeing on the valid requirements and then applying the requirement change process and communicating to all relevant stakeholders requirement change process in which users can change the requirement technology can change the requirement environment can change the requirement (Here we have requirement baseline which has two parts correct baseline and review baseline) Requirement baseline is the process in which when we do work on iteration and that step we work on set of approved requirement that makes requirement baseline. The requirement baseline provides ease in changing the requirement.

Why set a requirement baseline?

Answer: Requirement baseline specifies the set of requirements approved by all the stakeholders for the specification iteration. The requirement baseline evaluates the impact of change on the system. The requirement baseline reflects change throughout the requirement process. The importance of requirement management is to ensure that an organization's document verifies and validates the requirement and expectations of its customers and all stakeholders. Requirement management begins with analysis and elicitation of the objectives and constraints of the organization.

- The requirement management process supports the traceability of the requirements which is very important to meet the customer expectations.
- The traceability also supports the requirement change process.
- The requirement management process includes planning for requirements, integrating requirements, and organizing for working with them.
- Requirement management involves communication between the project team and stakeholders of a software project.

Provides Incremental and Iterative Development Environment

Software process models provide the iterative and Incremental development environment In which we divide the whole project into modules and design and development the whole project concerning customer requirements and this process increases the efficiency of the project and reduces the workload from development teams.

Provides Testing Environment

The software process model provides the testing environment throughout the project life cycle. Some of the testing techniques we do in the development period are given below.

Module testing:

Unit testing is associated with the testing of modules. In module testing, we will test each module and ensure that every module and check whether the functionalities work properly in the way that we want. We created a separate and small testing procedure for every single module and tried to test each module of the application separately. Testing is done on the module level by dividing the module into chunks and checking the quality of the module separately. The following module is being tested under the umbrella of Module testing.

Example:

- Accounts Module
- Manage Account Module
- Manage Contact Module
- Add to Cart Module
- Online Billing Module

Integration testing:

Integration testing is associated with Architectural design. After we tested each module separately then we tried the fundamental principle of integration testing in which we combine individual units are combined and tested as a group to identify the faults and bugs because of integration testing. We used test drivers and test stubs to identify the faults and bugs from the group of units and clarify how individual units interact with each other and work as expected.

System Testing:

System testing is directly associated with the system design phase. After Integration testing then we tried the fundamental principle of system testing in which we test complete software systems. The purpose of this testing is to evaluate the compliance of the software product with the given software requirements.

Acceptance testing:

Acceptance testing is directly associated with the requirement analysis phase. After Integration testing then we tried the fundamental principle of acceptance testing in which we test the system for acceptability and test the complete software system the purpose of this testing is to evaluate the compliance of the software product with the given business requirements and assess that whether this system or product is acceptable for delivery.

Provides Effective Way of Communication

In Software development, communication is very important for efficient software development. The software process model provides an effective and efficient way of the communication channel between the customer and the development team Interaction in software development is the backbone of the agile methodology open communication and less documentation is the best and typical feature of agile development.

Customer Collaborations

In software development, the requirements are very important for the development of the software product to gather complete, clear, and rich requirements for software to interact with customers.

Quality Product

Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.

Effective Planning

Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.

Reduce Workload

The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.

Provide Effective Logic

Software process models create new logic for problem-solving based on open-ended programming environments for the high computer system. Guides the software development teams to make new methodology for the development and construction of software products.

Flexible Supportive Development Environment

Software process models provide a flexible and supportive development environment. Using Software process models every organization can run more than one task at a time.

Predictive approach

In the Predictive approach development of the software product only depends on the requirements and analysis that's done at the beginning of the software.

Adaptive Approach

But in the Adaptive Approach to software development, there is no need for detailed planning is required. But we focus on clarity of the features that we want to develop. There are feature development planning and product requirement changes dynamically. The product is tested repeatedly in the iteration so in the end quality product is produced.

Customer Interaction

Interaction in software development is the backbone of the software model's methodology. Open communication is the best and typical feature of the software process model in software development.

How Process Models play a key role in Software Development

The software process model plays a key role in software development some of the key roles are given below.

Complete Planning

Software process models provide the complete planning for the software project. A big project requires more efficient planning for making the project successful likewise small projects also require planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times of revisions the worst part of development.

Define complete the project scope

The software process model defines the complete project scope. In the Requirement Engineering process defining the project, scope is very important because if you define the scope of the project poorly or incompletely scope your project led to unsuccessful.

Effective Team leadership

The software process model defines the different steps of the software development process in each step different teams work to gather for the successful delivery of the project. In software development, the if project leader is ineffective and poorly manages the project improperly defines the iteration of completion of the project is unsuccessful and useless at the end of the project.

Successfully identify the assumptions

In software development, if you failed to define the assumption about the project then your project in the end will be useless.

Example:

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Tracking the project progress.

Software process models provide the complete project process trail. In software development, if you failed to track the full progress of the project then your project, in the end, will be useless.

Detailed documentation.

The software process model provides the complete documentation of the project from start to end. In software development, documentation is a very important phase in which we write software documents like SRS document Design Document in later stages of development we use reverse engineering techniques then we use these documents for the updating the software product.

Track requirements

In every software project model Software requirement engineering is the very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

Selection of the right technology.

The software process model at the start of the project guides us that which technology is best for our project. In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Accurate measurement of time and budget.

The software project model provides the exact time frame and information about finance required for the development.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

Conclusion

We conclude in this chapter that software process models play a key role in software development. Software process modes increase the efficiency of a software system and decrease the development time. Software Process models have great importance in the development of software products. Every software model controls, manages, guides, and monitors the software development process. Software process models give a clear way for the development process. We discussed some important reasons why software process models are important.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers

and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.

- Software process models create the new logic for problem-solving based on open-ended programming environments for the high computer system.
- Guides the software development teams to make new methodology for the development and construction of software products.
- Software process models provide a flexible and supportive development environment.
- Using Software process models every organization can run more than one task at a time.

Key Points

- The software model is the set of related activities that leads to the software products of the software product.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Programming Language used for the development of software products.
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12

Selection of Process Models

Overview

In this chapter, we will explore the vital role of software process models in development, detailing their importance in planning, scope definition, leadership, assumption identification, progress tracking, documentation, technology selection, budgeting, and programming language choice. The chapter elucidates why and when software models are essential, how to select the right one, and the optimization possibilities. Key points highlight the impact of software models on product efficiency, quality, and development methodology.

Introduction

In this chapter, we will discuss a selection of process models and the importance of software process models why software models are important for the development of software processes, and how they can help the development team as well all stakeholders of the software product. In the software development life cycle, SDLC consists of a series of planned activities to develop the software product.

Why we use software Model.

We use the software process model because Models give us a way of development to define each activity of software development.

Complete Planning

Software process models provide the complete planning for the software project. A big project requires more efficient planning for making the project successful likewise a small project also requires planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two to three times revision this is the worst part of development.

Define complete the project scope

The software process model defines the complete project scope. In the Requirement Engineering process defining the project, scope is very important because if you define the scope of the project poorly or incompletely scope your project led to unsuccessful.

Effective Team leadership

The software process model defines the different steps of the software development process in each step different teams work to gather for the successful delivery of the project. In software development, the if project leader is ineffective and poorly manages the project improperly defines the iteration of completion of the project is unsuccessful and useless at the end of the project.

Successfully identify the assumptions

In software development, if you failed to define the assumption about the project then your project in the end will be useless.

Example:

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Tracking the project progress.

Software process models provide the complete project process trail. In software development, if you failed to track the full progress of the project then your project, in the end, will be useless.

Detailed documentation.

The software process model provides the complete documentation of the project from start to end. In software development, documentation is a very important phase in which we write software documents like SRS document Design Document in later stages of development we use reverse engineering techniques then we use these documents for the updating the software product.

Track requirements

In every software project model Software requirement engineering is the very important phase of the software development process or simply Software development life cycle (SDLC). This phase is used to translate the unclear and incomplete idea of the customer into complete precise and formal specification. This document is very must important and acts as a bridge between the Customer, User, and Developers. Therefore, the importance of software requirement engineering is very important for the development of effective and efficient software products.

Selection of the right technology.

The software process model at the start of the project guides us that which technology is best for our project. In software development, if you fail in the selection of technology which technology is best for the development of the project then it may also lead to the project being unsuccessful.

Accurate measurement of time and budget.

The software project model provides the exact time frame and information about finance required for the development.

- If the time exceeds the completion of the project, then your project and development process lead to an unsuccessful project.
- If the budget exceeds the completion of the project, then your project and development process lead to an unsuccessful project.

When to use the Software Model

In software engineering as well as in software requirement engineering software process model clearly define similarities of the system and defines the clear picture of the software system from idea to the requirement of the system software development is the same as developing the house in which we first define requirement then we arrange the finance for that then we go to another step like construction same in software system we need at first stage idea then we gather requirement for that idea to become a real software system for this process we use some techniques and follow some procedure for the requirement engineering process.
But the question is when to use the software process model.

- When we cannot estimate the correct budget for the targeted system.
- When we cannot estimate the exact time and budget of the system.
- When the system is big and can’t be manageable without proper management.
- When we cannot select the right technology for the system at the early stages of development.
- When we cannot define the exact scope of the project.
- When we cannot the requirement engineering process of the system.
- When we cannot manage the complete requirement track of the system requirement.

How to use Software Model

Step 1: Project Planning

This step is the most important step of the software development life cycle (SDLC). This step is performed by senior members of the team because a big project requires more efficient planning for making the project successful likewise small projects also require planning but if at the start of the project you made a mistake and miss any prerequisites then your project will require two

to three times of revision this is the worst part of development. In this step, we identify quality assurance procedures and risks associated with the project. The outcome of this step is used in the feasibility study of the project. Project feasibility defines the various procedures that can be followed for the successful delivery of a project with minimum risks.

Step 2: Defining Requirements

Defining requirements is the most important part of the software development life cycle. The process to gather the software requirement from customers and then analyze those requirements and document them is called requirement engineering or requirement elicitation. If the feasibility report is positive from all aspects, then we start the requirement elicitation process in this process we start requirement engineering we use different techniques to get requirements about the project. We discuss the features and functionalities of the system and identify the user and stakeholders.

Step 3: Designing and Architecture

The designing and product architecture means clearly defining the structure of modules along with communication among modules and data flow representation with the external third-party module. The internal structure of the module should be clearly defined. This step of the software development life cycle is reviewed by all the stakeholders based on various parameters such as risk assessment, product robustness, time and budget, etc.

Step 4: Building and Developing the product

After defining the clear and rich requirements the fourth step is building and developing the software product according to the defined standards. Software organizations start coding and designing products this step involves the implementation of all the pre- and post-condition of a project like the development of complete functionalities of the software product according to the requirement verified by senior members of teams and all the stakeholders of the project.

Step 5: Testing the product.

In this step of the software development life cycle, we define the testing strategies in which we do the testing of different modules, and we check that either every module is behaving and functioning according to the standard definition or according to the requirements define in the first phase of the SDLC process.

Step 6: Deployment

After the complete testing of the product and the product is formally ready for deployment then some of the time the product is released in the stages according to the deployment plans of the organization. And some time product deployment is dependent on the feedback of the customers.

How to select the Software Model for the Software project.

Every Software model provides the software development cycle that helps us to efficient and effective development. By using the Software development process model, we can get the requirements to form the customer very effectively. The software process model provides an effective way of requirement management. The software Process model defines the product business modeling. It provides the testing environment throughout the development cycle. It provides the complete details of the project by documenting all the processes during

development. Selecting the software product is a challenging task in the software development process. Select the process model according to the nature of the software product given below. We will discuss how we can select the right software process model in the right way.

But the question is when to use the software process model. The answer to this question is given below.

STEP 1: Learn about SDLC Models

The software development life cycle is the process followed within a Software Organization. It is a detailed process that describes how to develop, maintain and replace the software product. Before Using the SDLC Process you should have a clear understanding of the SDLC Model and experience in how to use the SDLC Modes in software development.

STEP 2: Assess the needs of Stakeholders

Defining requirements is the most important part of the software development life cycle. The process to gather the software requirement from customers and then analyze those requirements and document them is called requirement engineering or requirement elicitation. If the feasibility report is positive from all aspects, then we start the requirement elicitation process in this process we start requirement engineering we use different techniques to get requirements about the project. We discuss the features and functionalities of the system and identify the user and stakeholders. We must study the business domain, stakeholders' concerns, and requirements business priorities, our technical capability and ability, and technology constraints to be able to choose the right SDLC against their selection criteria.

STEP 3: Define the criteria

Some of the selection criteria or arguments that you may use to select an SDLC.

- Is the SDLC suitable for the size of our team and their skills?
- Is the SDLC suitable for the selected technology we are using and implementing a solution?
- Is the SDLC suitable for clients' and stakeholders' concerns and priorities?
- Is the SDLC suitable for the geographical situation (Distributed Team)?
- Is the SDLC suitable for the size and complexity of our software?
- Is the SDLC suitable for the type of projects we do?
- Is the SDLC suitable for the software engineering capability?
- Is the SDLC suitable for the project risk and quality insurance?

STEP 4: Decide

When you define the criteria and the arguments you need to discuss with the team you will need to have a decision matrix and give each criterion a defined weight and score to each option. After analyzing the results, you should document this decision in the project artifacts and be aware of it with related stakeholders.

STEP 5: Optimize

You can always optimize the SDLC during the project execution. You may notice upcoming changes do not fit with the selected SDLC, it is okay to align and cope with the changes. You can even make your own SDLC Models which optimize your organization to the type of project you are involved in.

Conclusion

We conclude in this chapter that software process models play a key role in software development. Software process modes increase the efficiency of a software system and decrease the development time. Software Process models have great importance in the development of software products. Every software model controls, manages, guides, and monitors the software development process. Software process models give a clear way for the development process. We discussed some important reasons why software process models are important.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.
- Software process models create the new logic for problem-solving based on open-ended programming environments for the high computer system.
- Guides the software development teams to make new methodology for the development and construction of software products.
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- Using Software process models every organization can run more than one task at a time.

Key Points

- The software model is the set of related activities that leads to the software products of the software product.
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13

Essential and Accidental Requirements

Overview

In this chapter, we will explore software process models, highlighting their significance in guiding software development. It categorizes requirements into essential and accidental, emphasizing their pivotal roles. The chapter concludes with key insights on modern software development practices and offers resources for practicing requirements engineering.

Introduction

In this chapter, we will discuss a selection of process models and the importance of software process models why software models are important for the development of software processes, and how they can help the development team as well all stakeholders of the software product. In the software development life cycle, SDLC is consists of a series of planned activities to develop the software product.

What are Essential Requirements

Essentials Requirements are those requirements that are a critical part of the system. Without these requirements we even can't combine the overall functionality of the system.

Example:

- Auth functionality of the system.
- Account management in a multi-client system.

What are Accidental Requirements

Accentual Requirements are those requirements that can be changed during the development process OR the Design Phase OR Requirement analysis phase simply at any stage of software development.

Example:

- The government laws changed for the payments process.

Conclusion

We conclude that essential and accentual play an important role in software development. Essential Requirements are the lifeline of a software product without these requirements we can't build a functional system at any cost. Accentual Requirements are those requirements that can be changed during the development process OR the Design Phase OR Requirement analysis phase simply at any stage of software development.

Some of the important points about software development are given below.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
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- Guides the software development teams to make new methodology for the development and construction of software products.

- Software process models provide a flexible and supportive development environment.
- Using Software process models every organization can run more than one task at a time.

Essential and Accidental Requirements

Download the document for practicing and writing the Essential and Accidental Requirements of the Project.
https://github.com/SardarMudassarAliKhan/Essential-and-Accidental-Requirements-.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Key Points

- The software model is the set of related activities that leads to the software products of the software product.
- Software Specification defines the functionalities of the targeted system.
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- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Essential Requirements are the lifeline of a software product without these requirements we cannot build a functional system at anycast.
- Accentual Requirements are those requirements that can be changed during the development process OR the Design Phase OR Requirement analysis phase simply at any stage of software development.

14

System Functional Requirements

Overview

In this chapter, we will explore system functional requirements, distinguishing between functional and non-functional requirements. It explores gathering methods such as studying existing systems, meetings, and analysing organizational documents, highlighting their importance in software development. Key points and resources for practicing functional requirements are also provided.

Introduction

In this chapter, we will study functional and non-functional requirements of the system. Functional Requirements define the behavior of the system means what the system should do Like, and Code Compiler should complete the Machine Code.

What are Functional Requirements?

Functional Requirements of the system are a very critical part of the system in which we defined the overall functionality of the system. Functional Requirements define the behavior of the system means what the system should do Like, and Code Compiler should complete the Machine Code.

Some of the Functional Requirement's examples are listed below.

- Business Rules.
- Transaction corrections, adjustments, and cancellations.
- Admin Functionality of the system.
- Authentication rule.
- Payment Functionality of the system.
- Log in and Log out Functionality.

How are Functional Requirements Gathered?

Functional Requirements of any system can be gathered in the following ways.

Study the existing system.

- Functional Requirements of the system can be extracted by studying the existing system. Study the existing module and test every functionality of the system for extracting the functional Requirements of the system.

Meetings

We can gather the functional requirements of the system by arranging back-to-back meetings with clients and discussing every functionality of the system.

- Study the Organizational Documents
- Study the User Manual of the Existing system.

Conclusion.

After studying this chapter, we concluded that Functional Requirements and gathering functional requirements is a very important part of software development. Functional Requirements define the behavior of the system means what the system should do Like, and Code Compiler should complete the Machine Code.

System Functional Requirements

Download the document for practicing and writing the Essential and Accidental Requirements of the Project.
https://github.com/SardarMudassarAliKhan/Essential-and-Accidental-Requirements-.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Key Points

Study the existing system.

- Functional Requirements of the system can be extracted by studying the existing system. Study the existing module and test every functionality of the system for extracting the functional Requirements of the system.

Meetings

We can gather the functional requirements of the system by arranging back-to-back meetings with clients and discussing every functionality of the system.

- Study the Organizational Documents
- Study the User Manual of the Existing system.

15

System Non-Functional Requirements

Overview

In this chapter, we will explore significance of Non-Functional Requirements (NFRs) in software systems, covering aspects like availability, interoperability, performance, reliability, usability, maintainability, portability, scalability, and reusability, crucial for project success and user satisfaction. Download and submit your NFR document for expert feedback.

Introduction

In this chapter, we will study how and why the Non-functional requirements of the system are important for the development of any software system. We will study how Non-Functional requirements support the overall functional requirements of the system.

What is a Non-Functional Requirement?

Nonfunctional Requirements define how the system works. Non-Functional requirements describe the overall behavior of the system like robustness scalability and maintainability of the system.

A simple definition of a Non-functional requirement is any requirement that specifies the overall behavior of the system is called a non-functional requirement.

Non-functional requirements are significant because they have the power to make or break the success of a software system or a product.

Even if a system satisfies all expected functional requirements, users will not hesitate to outright reject it if it fails to produce the desired quality results.

Many functional needs that are extremely important and need to be expressed in the proper amount of detail in the requirements papers are derived from non-functional requirements. Additionally, they play a crucial role in several important architectural decisions and ensure that important design considerations are made early on rather than when costs and the risk of modification are higher.

Some of the Non-Functional Requirements are

- Performance
- Scalability
- Capacity
- Availability
- Reliability
- Recoverability
- Maintainability
- Serviceability
- Security
- Regulatory
- Data Integrity
- Usability
- Interoperability.

Availability

The degree of a solution's operability and accessibility at a given time is referred to as availability. It is a time during which the system is completely operational, i.e., accessible for usage, and it is occasionally included as a service level agreement due to the importance of the system to the company.

Aspects of availability that need to be considered during elicitation include—but are not limited to periods during which availability is essential to achieve business objectives, the effects of a system's downtime, scheduled maintenance windows, and notifications during a system outage.

Interoperability/Compatibility

The level of interoperability indicates how well the solution works with other parts. It is a gauge of how well the system works with other software programmers and how simple it is to interact with other hardware.

Performance

Performance is the degree to which a solution or a part of it carries out its functions while using the fewest resources possible. Due to its significant design consequences and influence on hardware device selection, it is a very significant quality attribute.

Reliability

A solution or one of its components must be reliable if it is to carry out its intended functions under predetermined circumstances for a predetermined amount of time.
Reliability may be defined in terms of the average number of cycles a system runs before failing, the percentage of successfully performed operations over time, the maximum allowable failure probability, or the total number of failures over time.
During elicitation, reliability-related topics will be covered, including but not limited to evaluation of the system to be regarded as reliable, reliability categorization that distinguishes between occasional failures and regular failures, and the effect of failure on business activities.

Usability

Usability refers to how easily a user can pick up utilizing the solution and how easily he or she can use the system. Usability encompasses a variety of factors in addition to ease of learning and usage, such as memory ease, error avoidance & handling, and accessibility, among others. The number of human-system touchpoints necessary to complete a task, the average time required to complete a task, the average number of errors made during task completion, the average number of attempts before a task is completed, and other factors related to usability will all be discussed during the elicitation process.

Maintainability/Modifiability

The term "maintenance-friendliness" describes how easily a solution or a part of it may be mended, improved to suit business needs or boost productivity, or adjusted to a changing environment.
It represents how simple it is to comprehend, modify, and extend the software architecture and code. Maintainability is crucial for systems that must be routinely modified or improved. The percentage of patches that are made right the first time, the percentage of changes that are deployed correctly the first time, and the typical amount of time needed to repair a problem correctly are among the maintainability aspects to be reviewed during elicitation.

Portability

The capacity to move a system or one of its components from one operating environment to another is known as portability. Its significance has grown over time as a result of the need for apps to function across several environments, including Windows, Mac, and Linux, as well as mobile operating systems like Android and iOS.
During elicitation, portability issues will be raised concerning, but not limited to, various platforms on which the system must run, system components, and/or data pieces that must be explicitly developed for greater portability.

Scalability

Scalability is a term used to describe how well a solution can change over time to accommodate growing workloads. The user base, transactions, data, network traffic, or other reasons may have increased the workload.
The system should be able to scale up, which may entail adding more servers, faster processors, more disc space, or more network capacity to the hardware or utilizing software-specific techniques like data compression, code and performance optimization, and other similar methods.

Reusability

The ability of software or a software component to be reused in various applications is referred to as reusability. Which system elements/components should be developed in a way that enables their reuse across different applications must be defined?
Aspects of reuse that will be discussed during elicitation include but are not limited to functionality or artifacts from other applications that can be used across applications that are being built, as well as functionality or artifacts from the application that is being built that can be used across other applications, either current or future.

Importance Non-Functional Requirements

Non-functional requirements are significant because they have the power to make or break the success of a software system or a product. Even when a system satisfies all expected functional criteria but fails to produce the desired quality results, users will not hesitate to outright reject it.

Conclusion

Users frequently have expectations for how well a software system should function in terms of availability, usability, speed of functional execution, the likelihood of failure, security against unauthorized access, and effectiveness in handling unforeseen circumstances when they do arise.

Software Non-Functional Requirements

Download the document for practicing and writing the Software Non-Functional Requirement of the Project.
https://github.com/SardarMudassarAliKhan/Software-System-Non-Functional-Requirements-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

16

Introduction of System

Overview

In this chapter, we will explore the creation of Online LawFirm, a cloud-based Android app offering real-time legal assistance. It covers project objectives, motivations, feasibility, technical aspects, financial considerations, and operational aspects essential for success. The chapter provides a comprehensive overview of the development process, emphasizing key decisions and feasibility assessments.

Introduction

Online LawFirm is a Real-Time Cloud-Based android application that will assist society. We aim to help people solve their problems and guide them by providing the best possible solution. Another goal of the application is to bring all the interested parties on a single platform. Notification-based alerts to notify all the parties. Our system will help lawyers and Law Experts to find new cases. The system will have a communication system through which users can communicate with each other. Our system will have a rating system through which the client will rate the users.

Objective

The objective of this project is to develop a Multi-Client Cloud-based law Firm Android application to provide law assistance services. Different Problems occur in daily life, which require expert advice. Our system will have experts that will advise the people that are facing the problems. If a problem exceeds to such an extent that assistance is required, then the case will be forwarded to a law expert or lawyer. The different lawyers will be a part of the system.

Motivation

The Problem has become an integral part of our society. Society has been struck with many Problems like Domestic Problems, Relationship issues, Property Issues, Robbery Cases, Murder Cases, Terrorism, Finical Issues, Human rights violation cases, Fraud Cases, Cyber Crime Problems, and Other daily life Issues. We aim to provide a platform that can efficiently manage work against such Problems. Our system will help people to find a lawyer on a single platform.

Feasibility Report

An overall study of the project has been carried out and the findings are listed in this report. The feasibility of “ONLINELAWFIRM” can be ascertained from the following assessments.

- Technical feasibility
- Financial feasibility
- Resource’s feasibility
- Schedule feasibility
- Operational feasibility

Technical Feasibility:

The system is technically feasible, and it successfully satisfies the user's basic needs. The system will be using different technologies for users. Following technologies will be used in the development of the system.

- Java
- XML
- Firebase
- Android Studio
- Google Maps
- Google Services
- Marshmallow 6.0.1
- Android APIs

Financials Feasibility:

Feasibility of financial aspects e.g., pricing and availability of resources financing of tools, and other resources that are required for the development of the project. Provide tabulated information in the given format as shown in table 1.1 (Financial Feasibility).

Table 1.1: Financial Feasibility

Resource Name [Hardware/software]	**Price [Monitory price]**	**Source [Website/Dealer Address etc.]**	**Sponsor**
Laptop	Acer 45000 Hp 50000	Hp, Acer	Self
Android Studio	Open source	https://developer.android.com/studio	Self
Mobile	18000	Samsung	Self
GPS	Free	https://google.gps.com/Location	Self
Firebase	2500/10GB	https://console.firebase.google.com/	Self

Resource Feasibility:

Software

- Android Studio
- Google Map (API)
- Firebase
- GPS

Hardware

- Laptop
- Mobile

Development Team

- 3 Member

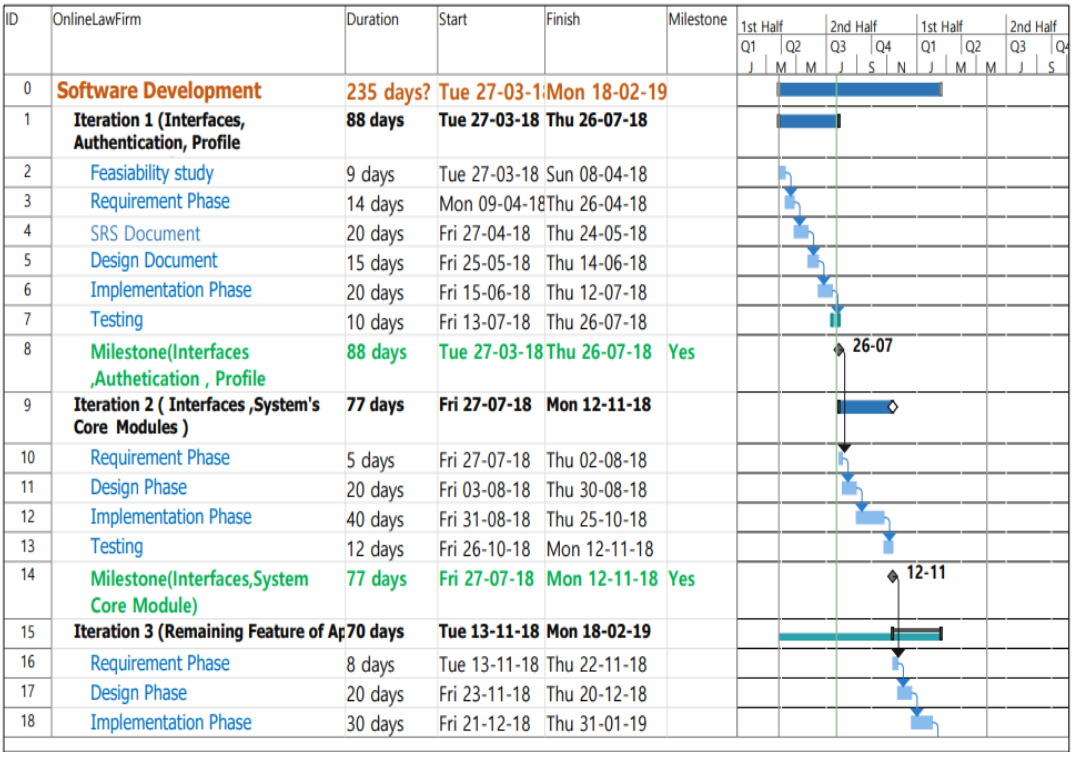
Duration

- 10 Months

Schedule Feasibility

A giant chart that shows all the sequence of activities concerning time and duration. Schedule Feasibility includes the sequence of iteration and details of alliteration and their sub-tasks and duration of subtasks. Details of Milestone and detail of Final iteration in which complete system will have given to stakeholder as a shown in Figure 1.1 (Schedule Feasibility).

Figure 1.1: Schedule Feasibility



ID	OnlineLawFirm	Duration	Start	Finish	Milestone
0	**Software Development**	235 days?	Tue 27-03-1	Mon 18-02-19	
1	**Iteration 1 (Interfaces, Authentication, Profile**	**88 days**	**Tue 27-03-18**	**Thu 26-07-18**	
2	Feasiability study	9 days	Tue 27-03-18	Sun 08-04-18	
3	Requirement Phase	14 days	Mon 09-04-18	Thu 26-04-18	
4	SRS Document	20 days	Fri 27-04-18	Thu 24-05-18	
5	Design Document	15 days	Fri 25-05-18	Thu 14-06-18	
6	Implementation Phase	20 days	Fri 15-06-18	Thu 12-07-18	
7	Testing	10 days	Fri 13-07-18	Thu 26-07-18	
8	Milestone(Interfaces ,Authetication , Profile	88 days	Tue 27-03-18	Thu 26-07-18	Yes
9	**Iteration 2 (Interfaces ,System's Core Modules)**	**77 days**	**Fri 27-07-18**	**Mon 12-11-18**	
10	Requirement Phase	5 days	Fri 27-07-18	Thu 02-08-18	
11	Design Phase	20 days	Fri 03-08-18	Thu 30-08-18	
12	Implementation Phase	40 days	Fri 31-08-18	Thu 25-10-18	
13	Testing	12 days	Fri 26-10-18	Mon 12-11-18	
14	Milestone(Interfaces,System Core Module)	77 days	Fri 27-07-18	Mon 12-11-18	Yes
15	**Iteration 3 (Remaining Feature of Ap**	**70 days**	**Tue 13-11-18**	**Mon 18-02-19**	
16	Requirement Phase	8 days	Tue 13-11-18	Thu 22-11-18	
17	Design Phase	20 days	Fri 23-11-18	Thu 20-12-18	
18	Implementation Phase	30 days	Fri 21-12-18	Thu 31-01-19	

The tool Used for making the project plan is Microsoft Project
Link: https://www.microsoft.com/en-us/microsoft-365/project/project-management-software

Operational Feasibility

The operational Feasibility of the project is based on real-world implementation and will be deployed to the online community.

- The lawyer who handles Law Cases can use this project 'OnlineLawFirm'
- Lawyers can also keep a record of the Cases, which are heard by the Court
- The product will be deployed on the Google Play store

Assumptions

- The project will work on Android Based Mobile Technology
- The project will work in presence of Wi-Fi, 3G, or 4G technology.

Conclusion

Some of the important points about software development are given below.

- Modern software products are faster and more efficient because they were built using certain engineered techniques these techniques are software process models.
- Modern software products are quality products because they were built using certain formal techniques during the development phase production house follows the software process models and test each step again and again to improve the quality of the product.
- Life Critical systems need more effective and efficient planning to complete the software product. In this situation, software process models give us the way of planning to complete such types of software systems.
- The modern software product is used to reduce the workload of a company and some origination needs software systems because they want the exact statistics of customers

and if the software product failed to satisfy the needs of customers, then this type of software development is worse because this process had not used the software process models and leads to business failure.

- Software process models create the new logic for problem-solving based on open-ended programming environments for the high computer system.
- Guides the software development teams to make new methodology for the development and construction of software products.
- Software process models provide a flexible and supportive development environment.
- Using Software process models every organization can run more than one task at a time.

Project Proposal Document

Download the document for practicing and writing the Software Project Proposal document of the Project.
https://github.com/SardarMudassarAliKhan/Project-Proposal-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Key Points

- The software model is the set of related activities that leads to the software products of the software product.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Programming Language used for the development of software products.
- Prototyping is the working model of the software
- RAD is a Rapid Application Development Software Model
- The software model is the set of related activities that leads to the software products of the software product.
- Software Specification defines the functionalities of the targeted system.
- Software Design and Implementation this activity involves the implementation of the software product this is the second step of software development.
- Essential Requirements are the lifeline of a software product without these requirements we can’t build a functional system at anycast.
- Accentual Requirements are those requirements that can be changed during the development process OR the Design Phase OR Requirement analysis phase simply at any stage of software development.

17

Requirement Analysis

Overview

In this chapter, we will explore the requirements analysis phase for OnlineLawFirm, emphasizing product scope, functionalities, and specific needs. It defines user classes, interfaces, operating environment, constraints, assumptions, and dependencies. Detailed descriptions of functional and non-functional requirements ensure clarity and precision in system development for optimal performance and user satisfaction.

Introduction:

This chapter includes and focuses on a brief introduction of the requirements analysis phase and focuses on the product scope and functionalities of the system. This chapter covers the functional and non-functional requirements of the system and specific requirements of OnlineLawFirm.

Product Scope:

We aim to help people facing different problems by providing expert advice through our multi-client cloud-based system. The online law firm will provide ease to access and use of online platforms for people seeking legal help. The system will facilitate the law firm to handle these clients and allocate lawyers to their tasks and keep track of the complete legal process (cases). The client, lawyer, and law experts will be registered in our system.

This system will perform the following functions.

- Users should sign up to use the system.
- Admin of the system can suspend and delete any accounts.
- Users of the system can delete any contact.
- Clients can submit /her Case in our system.
- Law Experts will receive the case according to their area of the field and then verifies each case after verification law expert will proceed and reject the case.
- Notification-based alerts to notify all the parties.
- The system will store the GPS location of the Lawyer and expert.
- The system will have a communication system for legal discussion and sending legal documents.
- A lawyer can confirm their availability for the Case
- The client can give a rating to a lawyer and law expert.
- Performance and rating-based promotion of lawyer.
- Lawyers and Law Experts can save case history in the system.
- The system will categorize the lawyer based on results and ratings.

Definitions, Acronyms, Abbreviations:

An acronym is a word or name formed as an abbreviation from the initial's components of a phrase or a word discussed in this Table 2.1 (Definitions, Acronyms, Abbreviations)

Table 2.1: Definitions, Acronyms, Abbreviations

Term	Acronyms	Abbreviation	Definitions
1	Android Studio		It is an app development platform
2	Internet		It is used to get global data
3	Constraints		Limitations or Restrictions
4	XML		Language for Designing UI
5	Firebase		Cloud Database for storing Data
6	Java		High-level Programming language for backend Coding
7	Configuration		It means a product, which is available
8	I/O	Input /Output	Input and out operation with the communication device

9	UI	User Interfaces	Standard way for positioning the usability options
10	SDK	Software Development kit	Used for developing the software product
11	GUI	Graphical User Interface It is an interface provided to the user.	A standard way of using system control in the form of View in the system
12	OLF	OnlineLawFirm	An android-based application used for assistance
13	SRS	Software Requirements standards	SRS is a standard way of mentioning techniques for gathering software requirement
14	PC	Personal Computer	A platform for communication
15	App	Android Application	Mobile-based Application
16	DB	Database	It is an organized collection of data
17	API	Application Programming Interfaces	A set of functions and procedures that allow for the creation of an application
18	GPS	Global positioning system	GPS used to identify the location
19	APK	Android package kit	Extensions of Android Applications

Document Convention:

This document uses the following conventions. Throughout the document, the same convention will be used in this document. Set of rules choose for character sequence to be used for identifier, which donates variables, words, functions, types, and other entities.

Formatting Convention:

This section includes the detail of the convention used in this document overview of the texture format written in a standardized way. The font style must be Time New Roman heading will be bold with standard font size. The details of the standard format are mentioned in given Table 2.2 (Formatting Convention).

Typographical conventions used in this document are as follows:

- Bolding is used to emphasize headings and subheadings.
- The theme font is Times New Roman (Headings CS)
- Font size is 18 and 14 for headings and subheadings respectively.
- Font size for text is 12.
- High level requirements are numbered continuously as 1,2,3,4,5,6
- Inherited detailed requirements are numbered in a float as 1.1,1.2,1.3 and so on for each high-level requirement.
- Double inverted commas are used for pre-known words or words which can conflict depending on the audience.
- Bullet points are used for defining a single topic of a whole description.
- Underlined Words with blue font color are used to identify the web addresses.

Table 2.2: Formatting Convention

Name	Convention
Font	Time New Roman 12 point
Heading 1	Bold 16-point size
Heading 2	Bold 14-point size
Heading 3	Bold 12-point size
Sections	Number 1-5
Sub Sections	Section No and subsection No
Copied	Italics
Line space	1 space
Justified	Yes

Naming Convention:

The naming convention is the set of rules for choosing the character sequence to be used for identifiers, which denote variables, types, functions, and other entities.

- Terms requiring explanations are explained in “()” in front of the term.
- Abbreviations are worried in the upper-case latter
- References Definitions, statements, or numbered as “[1]” subscripts

Overall Description:

This section will give an overview of the whole system. The system will be explained in its context to show how the user will interact with the system and perform its basic functionality. It will also explain how the user will use the features of the system and what functionality is available next. Finally, the constraints and assumptions will be presented. Product perspective includes the figure that will show the working of the system. Product functionality includes the functionalities of “ONLINELAWFIRM” concerning all users. User characteristics include details regarding the interfaces of the system operating environment and explaining the specification and requirements of the system. Design and implementation portions will cover the possible constraints of “ONLINELAWFIRM “assumptions and dependencies of the system will be covered.

Product perspective:

This App is not dependent on any existing system it is the first step of introducing android base technology for the registration of problems and the system will facilitate the law firm to handle these clients and allocate lawyers to their tasks and keep track of the complete legal process (cases). The client, lawyer, and law experts will be registered in our system. This system will be user-friendly and available 27/7so that users can access the data at any time from cloud storage. The product perspective is clearly defined in Figure 2.1 (Product Prospective)

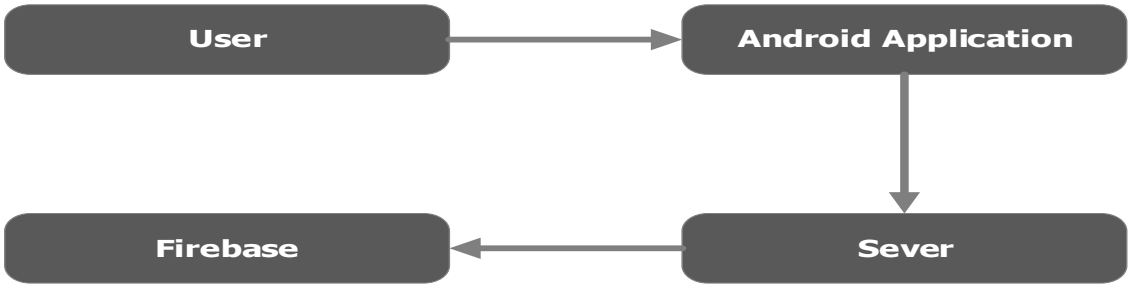


Figure 2.1: Product Prospective

Product Functionality:

This project will include the following functionalities.

- Registration of Clients, Experts, and lawyers on the cloud base system
- Submission of the problem by the client
- Multi-Client System
- Generation of a list of experts that can give assistance
- Request Analyzer to analyze the case to suggest a lawyer accordingly
- Ability to edit the saved profile information
- Contact Management
- Status Management
- Legal Document Management
- Court hearing Calendar
- Location Expert to save, and show the GPS location of users
- A notification system that can send a notification to the users
- Lawyer's availability confirmation.
- Case allocation to experts and Lawyer
- Ratings and Feedback system
- Communication between Expert and client
- History Saver to save Case History
- Successful and unsuccessful cases will be shown on the lawyer's Profile
- Categories of a lawyer based on the previous history

Users and Characteristics

The following are the user classes:

- Admin Class
- Client Class
- Lawyer Class
- Law Expert Class

ADMIN Characteristics: The user of this class will have the rights to all the elements of the product

CLIENT Characteristics: The user of this class will be able to Register his/her case and can communicate with a lawyer and law Expert and use the specified functionalities of the product

LAWYER Characteristics: The user of this class will be able to Receive Requests from the Client and will send the availability notifications on that Request and use the specified functionalities of the product.

LAW EXPERT Characteristics: The user of this class will be able to Receive Requests from Clients Receive Case Registration Requests, send the notifications on that Request and use the specified functionalities of the product.
The interfaces of the “ONLINELAWFIRM” are user-friendly and easy to use a normal user who has basic knowledge of mobile can easily interact with the system

Operating Environment:

The product will operate on the following platform.

- Mobile
- Tablet

The operating environment detail of the system is as shown in the given below Table 2.3 (Operating Environment)

Table 2.3: Product perspective

Requirements	Specifications
Hardware platform	Mobile, Tablets
Operation Environment	Android Technology
Versions	6.0.1

Design and Implementation Constraints:

This section includes the details of constraints for design and implementations are declared under which this application will be developed. An example of what the constraints are where the hardware is concerned. What are the constraints when the database and software are concerned?

The detail is as given below

- The Product requires Google Play Geo APIs Embedded in the App and there is an internet access.
- The Product requires Google Play Auth APIs will be used in the App and there is an internet access.
- It requires the service of a Firebase server.
- It requires the language support of Java, XML, Firebase other android technologies.
- It will follow the Patterns of Software Engineering
- It will follow the principles and standards of Software Engineering

User Documents:

A textual document for the help of the user is provided which gives the starting tutorial for the use of the Product modules.
Manuals of two pages will provide that contain the following details.

- How to install the application from the Google play store?
- How to use the application according to the desired status?
- How do find the desired lawyer and law experts?
- How to be registered according to the status?
- How to get benefits from “ONLINELAWFIRM”?
- How to send a Case to lawyer and Law Experts?
- How to Use the communication system of application?

- How to Use GPS (API) to send location to the user?
- How do give a rating to Lawyers and law experts?

Assumption and Dependencies:

Here are some limitations and dependencies of the system as given below.

- The product is dependent on the services of Google maps (APIs), the internet, and Firebase.
- Response time of the system will be in seconds.
- To get better results users must have better internet connections
- Google Play services must be up to date
- Time delay occurs during the fetching of the data from firebase depending on the internet connection

Specific Requirements:

This Section will include a detailed and point-to-point description of essential requirements of “ONLINELAWFIRM” like inputs and outputs of the system and also the detail of tools used in the designing of the interfaces.

User Interfaces:

- The GUI will include the home screen.
- The home screen will include the Signup module and login Module
- When Users register their account in the system then the system will show options to select their status
- The home screen will include Menu options that will change their behavior according to the status of users
- Menu Option will include My-Profile Connects, Communication, History, Logout
- My-Profile will show the profile Information of the Signed in User and there will be an option to edit his/her profile.
- The Connects Screen will show all related contacts.
- The Communication option will have the ability to send Messages and Show received messages. Legal documents will share through this system. Notifications will also be shown here
- The History Option will contain a record of cases
- Case-Handler Module will have options. This module will change its behavior according to the status of the users
- To Register a new case the client will go to the case handler and state his problem
- The law-Expert module will receive all client Requests.
- The lawyer Module will receive all case requests from the law-expert module.
- Admin Module will have the different Interface from others and have the authority to delete and suspend any account

Hardware Interfaces:

- Android Technology Mobile device

Software Interfaces:

- Java
- Firebase
- XML
- Android Studio

Communications Interfaces:

- We will only use the android firebase library to communicate with the server

The details of the requirements of the software are shown in the given below table 2.4 (Requirements of software).

Table 2.4: Requirements for Software

Requirements	**Specifications**
OS	Android, Tablets
OS Versions	Android 6.0.1
Database	Firebase

Functional Requirements:

This section includes the details of the Functional Requirements of the system, without these requirements the system will not perform the basic functionalities. The essential requirements of the system are as given below.

Registration of Client/Lawyer, Experts, Admin:

- Users should sign up to use the system

Manage Account:

- Admin of the system can suspend and delete any accounts

Manage Contacts:

- Users of the system can delete any contact

Submission of Cases from Clients:

- Clients will submit /her Case in our system

Verification of Cases:

- Law Expert will receive the case according to their area of the field and then verifies each case after verification law expert will proceed and reject the case

Assign the case to the lawyer:

- Law Expert can assign the cases to law according to their area of the field

Notification system:

- Law expert and Lawyer send Notification to Client
- A law expert will send a notification to a lawyer

GPS Location

- The system will store the GPS location of the Lawyer and expert

Communication System:

- Lawyers and law experts can communicate with clients

Lawyers' availability confirmation:

- After receiving the notification lawyer can confirm their availability for the Case

Rating System:

- Users can give a rating to lawyers and law expert

Lawyer promotion:

- Performance-based promotion of lawyer

Sharing of Legal Documents:

- A lawyer can share the legal document with clients

Save Case History:

- Lawyers and Law Expert can save case history in the system

Non-Functional Requirements:

For the betterment of the system, we need to add some non-functional requirements that support the functionalities of the system.

Performance Requirements:

- The product will perform well if the software and hardware requirements are following the assumed hardware and software requirements mentioned in the external interface requirements section.
- The system will send, Receive, or exchange data within 1 second

Start uptime.

The app starts with the reality of the situations that will become obvious eventually somewhat more. Due to interacting and logging in to firebase to authenticate the user and get the data from firebase. The time taking of the application will be three to five sec.

Safety Requirements:

- Application is not creating any safety hazards for the user because it will not be critical in the operating environment, and it will be not harmful to anyone

Security Requirements:

- The Product back-end server should authenticate all users 100% correctly
- The Product will provide security on the data of all users

Software Quality Attributes:

Reliability:

- Provides the best solution to the users on every problem.
- Error-free data availability
- Update data instead of updating the database.
- The system will be reliable and will perform its intended functions.
- It does not meet any failure or system crash.

Availability:

- This product will perform without 1 second of delay.
- The Product will give a response with desired time on every action of the users.
- The Product will be available to the users 24/7

Usability:

- As the product will be easy to handle and works most expectedly with no failure
- This application will be built on the latest GUI and design patterns, e.g., Minimum taps to achieve a function.
- This application will give accurate and desired results.
- This application follows Software Engineering Standards, so users will enjoy using this app

Maintainability:

- As the admin team, we will be providing our services as required like updating and keeping a record of the system.
- This system is easy to maintain all data of users on the server.
- Easy to maintain the entire database.

Reusability:

- Re-use this product in any domain having the required specifications.

Portability:

- This application will work on all android technology.
- This application will be effective the android versions 6.0.1

Adaptability:

- Since it is simple to use for any android mobile user
- This software product shall not require an acute knowledge of android and it will be user-friendly.

Testability:

- The system can easily have tested using standard software techniques including quality testing and usability testing.

Robustness:

The app should be more powerful by placing the coding expense of every capacity at least. Correspondence effectiveness and reaction time will be inside as for as possible as how and how must time the application will take and will be recovered from any failure.

Extendibility

This application can easily be extended because of the modular approach used in the development process. This application built using android allows us to separate the business logic and User interfaces easily.

Software Requirement Analysis Document

Download the document for practicing and writing the Software Requirements engineering document of the Project.
https://github.com/SardarMudassarAliKhan/Software-Requirement-Engineering-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

18

System Design

Overview

In this chapter, we will explore explores detailed design's theoretical framework, emphasizing system architecture and its components. It addresses the Use Case View, Primary Actors, Fully Dressed Use Cases, and Operation Contracts, offering practical examples and diagrams for clarity. Additionally, Composition Viewpoints are discussed, detailing system structure and elements interaction. Supplementary resources, such as package diagrams, class diagrams, and system sequence diagrams, are provided to facilitate understanding. Readers can access downloadable documents for practicing and receive feedback from a requirement engineer via email for enhanced learning.

Introduction:

Here in this section, we will briefly discuss the framework of theoretical account in detailed design. We discuss the working model of the system in detail and the complete architecture of the system and every one of its constituents and capacities is discussed in this chapter.

Use Case View:

A method through which we analyze, Clarify, Identify, and organize the system requirements User Complete Operating Process is defined in Figure 3.1 (Use Case Diagram).
Primary Actors:

- Admin
- Law Expert
- Lawyer
- Client

Use Case Diagram:

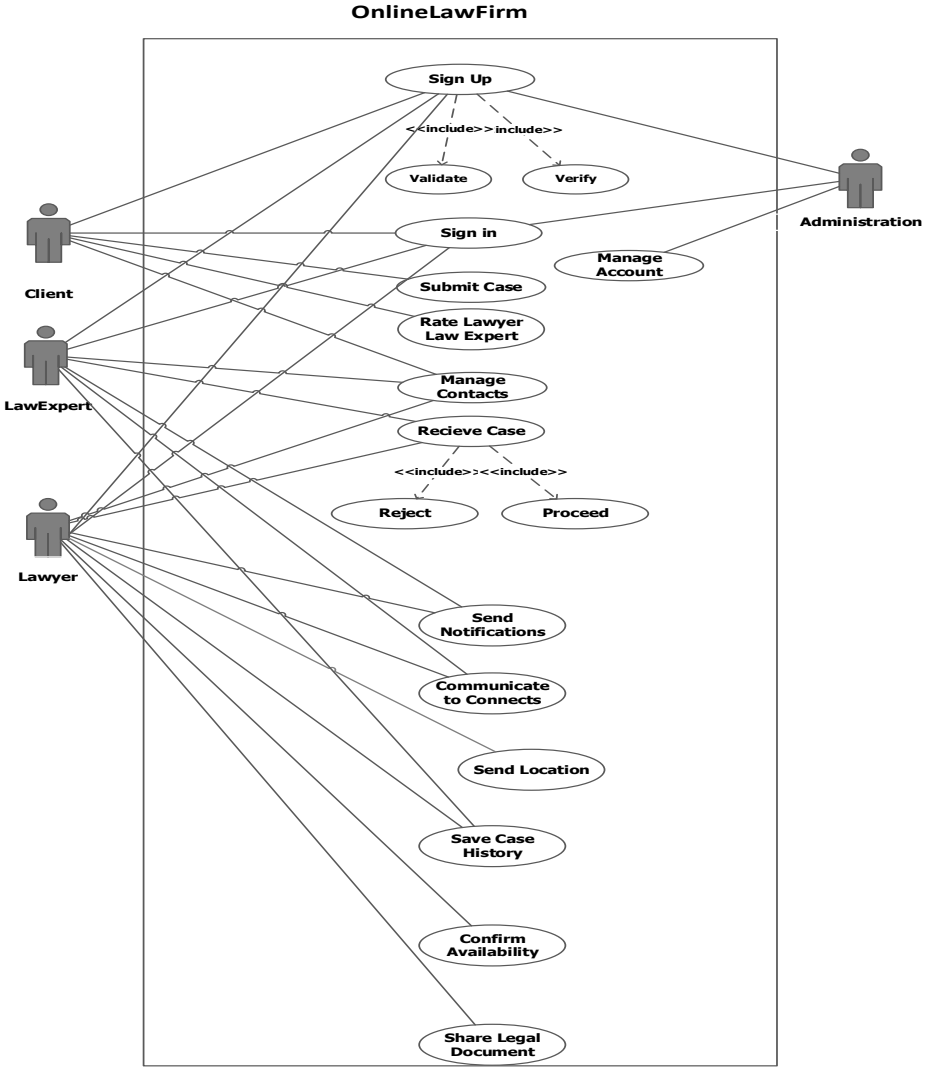


Figure 3.1: Use Case Diagram

The tool used for making the diagrams of the project during this book writing.
Link: https://www.microsoft.com/en-ww/microsoft-365/visio/flowchart-software

Fully Dressed Use Cases:

The use case is the list of actions or event steps typically defining a role and interaction between a system and user to achieve the desired goal.

Use case: Sign up:

Table 3.1: Sign up

<table>
<tr><td>Name:</td><td>Signup</td></tr>
<tr><td>Scope:</td><td>OnlineLawFirm</td></tr>
<tr><td>Level:</td><td>Primary</td></tr>
<tr><td>Primary Actors:</td><td>Client, Lawyer, Law Expert, Admin</td></tr>
<tr><td>Pre-Condition:</td><td>None</td></tr>
<tr><td>Post-Condition:</td><td>a. The user will have registered to the system</td></tr>
<tr><td colspan="2">Main Success Scenario:
<table>
<tr><th>User</th><th>System</th></tr>
<tr><td>1) Use the signup Option</td><td></td></tr>
<tr><td></td><td>2) A form will be shown to the user</td></tr>
<tr><td>3) Fill out the account registration Form</td><td></td></tr>
<tr><td>4) Press the form submission Option</td><td></td></tr>
<tr><td></td><td>5) Validate the form</td></tr>
<tr><td></td><td>6) Create a user in the database</td></tr>
<tr><td></td><td>7) Success Message Will be Shown</td></tr>
</table></td></tr>
<tr><td>Extensions:</td><td>5a) if the Inputs are not valid then the system will show a message to the user to fill in the form again and then resubmit.

5b) if any input field is empty a message will be shown to fill in data properly and submit the form</td></tr>
<tr><td>Special Requirements:</td><td>The list of experts will be shown within 1 second</td></tr>
<tr><td>Technology:</td><td>None</td></tr>
<tr><td>Miscellaneous Issue:</td><td>None</td></tr>
<tr><td>Frequency of use:</td><td>Only the time of sign up</td></tr>
</table>

Software System Use Case Document

Download the document for practicing and writing the Software system use case document of the Project.
https://github.com/SardarMudassarAliKhan/Software-System-Use-Case-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Operation Contract:

Table 3.2: Operation Contract Sign Up

Id	Olf1
Operation	create Account ()
Cross Reference	Fill the Form
Pre-Condition	None

Use Case: Sign Up:

Table 3.3: Sign In

Name:	Sign in
Scope:	OnlineLawFirm
Level:	Primary
Primary Actors:	Client, Lawyer, Law Expert, Admin
Pre-Condition:	a) The User must have an account
Post-Condition:	b) The User gets access to the Application

Main Success Scenario:

User	System
1) Use the Sign in Option	
	2) Show the pop-up for account selection
3) Select the Account	
4) Press sign-in button	
	5) Validate the Account
	6) Sign in Successfully
	7) Success Message Will be Shown

Extensions:	5a) if the account is not present in the system, then Message will be shown you are not registered.

Special Requirements:	The User will be signed in within 1-3 seconds.
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Every Time

Operation Contract:

Table 3.4: Operation Contract Sign In

Id	Olf2
Operation	sign in ()
Cross Reference	Choose the Account for access to the system
Pre-Condition	a. The user Must have an account b. The user must have Google Account
Post-Condition	The User has been granted access to the Application

Use Case Manage Account:

Table 3.5: Manage Account

Name:	Manage Contact
Scope:	Online LawFirm
Level:	Secondary
Primary Actors:	Administration
Pre-Condition:	a) The Administration Member Must have an account of Admin status b) The admin must be signed In
Post-Condition:	a) The Requested account is suspended.

Main Success Scenario:

User	System
1) Use the Delete and Suspend option	
	2) Show a form asking the user to enter the name email of the account
3) Enters the Email of the account	
4) Click the desired option	
	5) Account will be deleted or suspended
	6) A message will be shown

Extensions:	5a) if the Entered email does not exist in the system, then a message will be shown. 5b) if the input field is empty a message will be shown to fill in data properly and resubmit
Special Requirements:	None
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only when the suspicious activity is detected

Operation Contract:

Table 3.6: Operation Contract Manage Account

Id	OLF3
Operation	manage Account ()
Cross Reference	Fill the Email Fields
Pre-Condition	a) The account of the user must exist in our database The entered Email must be existing in the system
Post-Condition	The entered account is suspended

Use Case: Manage Contacts:

Table 3.7: Manage Contacts

Name:	Manage Contacts
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Client, Lawyer, Law Expert
Pre-Condition:	a) The desired contact must be connected to the user b) The user must be signed In
Post-Condition:	a) A selected account was deleted from the Connects list.

Main Success Scenario:	
User	**System**
1) Use the Case Menu Option	
	2) a) show the case menu if the client sends the case, then he will be chosen the case send option b) show the case menu if the lawyer and law expert use the case menu then they both use the case
3) a) Client sends the case c) Lawyers and law Expert use the case Request and ongoing option	
4) a) Client clicks the case send option c) Lawyers and law experts use the request accept/reject option	
	5) The system added the contact according to the decision of the user
	6) Success Message Will be Shown

Extensions:	5a if the selected contact does not exist in the user connects then a message will be shown.
Special Requirements:	a) The account deleted within 1 second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only Time of managing contact

Operation Contract:

Table 3.8: Operation Contract Manage Contact

Id	OLF4
Operation	manage Contacts ()
Cross Reference	Select the contacts
Pre-Condition	a) The Client must be signed In
Post-Condition	The selected account was deleted from the user's connects list.

Use Case: Submit Case:

Table 3.9: Submit Case

Name:	Submit Case
Scope:	OnlineLawFirm
Level:	Primary
Primary Actors:	Client
Pre-Condition:	a) The Client Must have an account b) The Client must be signed In
Post-Condition:	a) A list of relevant experts will be shown to the users

Main Success Scenario:

User	System
1) Select the submission of use case option to submit his/her problem	
	2) Shows the case submission form
3) Fill out the Case Submission Form	
4) Select the lawyer or law expert	
	5) Validate the form
	6) The case sends to a particular user
	7) Success Message Will be Shown

Extensions:	5a) if the Inputs are not valid then the system will show a message to the user to fill in the form again and then resubmit. 5b) if any input field is empty a message will be shown to fill in data properly and submit the form
Special Requirements:	The list of experts and lawyers will be shown within 1 second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only the time of case submission

Operation Contract:

Table 3.10: Operation Contract Submit case

Id	OLF5
Operation	submit Case ()
Cross Reference	Fill the Form
Pre-Condition	a) The Client Must have an account b) The Client must be signed In
Post-Condition	A list of relevant experts has been shown to the users

Use Case: Set Rating:

Table 3.11: Set Rating

Name:	Set Rating
Scope:	OnlineLawFirm
Level:	Primary
Primary Actors:	Client
Pre-Condition:	a) The user must be logged in to the system. b) The case showed be completed
Post-Condition:	a) The given rating added to the target profile

Main Success Scenario:

User	System
	1) The input rating screen is shown to the use after the completion of the case.
2) Fill in the stars input to give a rating to the user.	
	3) Convert rating into numeric form.
	4) Add the rating to the target profile.
	5) Show Success Message

Special Requirements:	The list of experts will be shown within 1 second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	After the completion of the Case

Operation Contract:

Table 3.12: Operation Contract Set Rating

Id	Olf6
Operation	set Rating ()
Cross Reference	Fill in the star rating
Pre-Condition	a) The Client Must have an account b) The Client must be signed In c) The case was completed
Post-Condition	The user was registered to the system

Use Case: Verify Case:

Table 3.13: Verify Case

Name:	Verify Case
Scope:	OnlineLawFirm
Level:	Primary
Primary Actors:	Client, Lawyer, Law Expert
Pre-Condition:	a) The user must have logged in to the system. b) The case was submitted to the system.
Post-Condition:	b. The case status and information changed to verified.

Main Success Scenario:

User	System
	1) Show the complete information on each user profile
2) Check the case description and documents	
3) Fill out the case description form	
4) Check the case information and verify each case and update the status	
	5) Validate the new information
	6) Change the case status
	7) Submit the new case information
	8) Success Message Will be Shown

Extensions:	None
Special Requirements:	None
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Whenever the case is submitted

Operation Contract:

Table 3.14: Operation Contract Verify case

Id	Olf1
Operation	create Account ()
Cross Reference	Fill the Form
Pre-Condition	None
Post-Condition	The user was registered to the system

Use Case: Send Location:

Table 3.15: Send Location

Name:	Send Location
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Lawyer, Law Expert
Pre-Condition:	The user must be logged in to the system.
Post-Condition:	The current expert location sent to the selected client

Main Success Scenario:	
User	**System**
1) Uses the send current location option	
	2) A list of connected clients is shown to the E
3) Selects the client from the list to send the location	
4) Press the send location button	
	5) The current location longitude and latitude fetched
	6) The location values send to the selected clie
	7) Success Message Will be Shown

Extensions:	If the GPS of an android device is turned off first, turn the device GPS and get the location again.
Special Requirements:	Device GPS Should be on
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Every time a face-to-face meeting is held

Operation Contract:

Table 3.16: Operation Contract Send Location

Id	OLF8
Operation	send Location ()
Cross Reference	Press the send Current Location option
Pre-Condition	a) The Client Must have an account b) The Client must be signed in
Post-Condition	The location was sent to the correct targeted profile

Use Case: Send Notification:

Table 3.17: Send Notification

Name:	Send Notification
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Lawyer, Law Expert
Pre-Condition:	a) The Users Must have an account b) The Users must be signed In

Post-Condition:	a) The Notification Box will be shown to the relevant users

Main Success Scenario:

User	**System**
1) Accept the case	
	2) Send the notification to the user
3) Reject the Case	
	4) Send the notification to the user
5) Reject the Request or accept the request	
	6) The system sends the Notification
7) Add the New Information	
	8) Send the notification to the user
	9) Success Message will be shown

Extensions:	5a) if the notification is not sent then the system asks the user to send the notification again
Special Requirements:	a) The Notification will be sent within one second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only the time of sending notification

Operation Contract:

Table 3.18: Operation Contract sends Notification

Id	OLF9
Operation	send Notification ()
Cross Reference	Fill out the Notification Form
Pre-Condition	c) The Client Must have an account d) The Client must be signed In
Post-Condition	The Notification form has been shown to the relevant users

Use Case: Communicate to Connect:

Table 3.19: Communicate to connect

Name:	Communicate to connects
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Client, Lawyer, Law Expert
Pre-Condition:	a) The Users Must have an account b) The Users must be signed In
Post-Condition:	a) The Chat Box will be shown to the relevant users

Main Success Scenario:

User	System
1) Use the Chat option	
	2) Show the chat box
3) Fill the desired fields	
4) Press the send chat option	
	5) Show the message in the chat box
	6) Save the chat history in the database
	7) Success Message Will be Shown

Extensions:	3a) if the message field is empty then a message warning will be shown select the file again
Special Requirements:	The message will be sent within one second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only the time of sending a message

Operation Contract:

Table 3.20: Operation Contract Communicate to Connect

Id	OLF10
Operation	send Message ()
Cross Reference	Fill the Message Field

Pre-Condition	a) The Client Must have an account b) The Client must be signed in
Post-Condition	The Chat Box has shown to the relevant users

Use Case Saves Case History:

Table 3.21: Save case history

Name:	Save case history
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Lawyer, Law Expert
Pre-Condition:	a) The Users Must have an account b) The Users must be signed in
Post-Condition:	a) The user will have registered to the system

Main Success Scenario:

User	**System**
1) Use the case menu option	
2) Click the case requests option	
3) Open the case	
4) Use the case history option	
5) Add/delete the information	6) Validate the form
	7) A success message will be shown

Extensions:	5a) if the form will not fill properly then a warning message will be shown
Special Requirements:	None
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only the time of sign Up

Operation Contract:

Table 3.22: Operation contracts save case history

Id	OLF11
Operation	Save case History ()
Cross Reference	Generate the Notification
Pre-Condition	a) The Client Must have an account b) The Client must be signed In
Post-Condition	a) Save case history option has shown to the user

Use Case: Send Availability:

Table 2.23: Send Availability

Name:	Send availability
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Lawyer, Law Expert
Pre-Condition:	a) The Users Must have an account b) The Users must be signed in
Post-Condition:	a) A notification option will be shown to the user b) Notification will be sent to a law expert

Main Success Scenario:

User	System
1) Use the send availability option	
	2) Show the desired option
3) Check the check box	
	4) Validate the decision
	5) Change the status
	6) Success Message Will be Shown

Extensions:	None
Special Requirements:	The availability sent within 1 second
Technology:	None

Miscellaneous Issue:	None
Frequency of use:	Only the Time of send availability

Operation Contract:

Table 3.24: Operation Contract Send Availability

Id	OLF12
Operation	send Availability ()
Cross Reference	Generate the Notification
Pre-Condition	a) The Client Must have an account b) The Client must be signed in
Post-Condition	a) Case history save and shown on user profile

Use Case: Send Documents:

Table 3.25: Send Document

Name:	Send Document
Scope:	OnlineLawFirm
Level:	Secondary
Primary Actors:	Client, Lawyer, Law Expert
Pre-Condition:	a) The Users Must have an account b) The Users must be signed In
Post-Condition:	a) The Chat Box will be shown to the relevant users

Main Success Scenario:

User	System
1) Use the chat option	
	2) The system shows the chat option
3) Select the File from the system	
4) Press the send button	
	5) Validate the file attachment
	6) Show in the chat box
	7) Success Message Will be Shown

Extensions:	5a) If the input file attached is not compatible then the system will show a message to the user to fill the form again and then re-submit. 5b) If any input file attach is not compatible will be shown to fill data properly and submit the form
Special Requirements:	The file will be sent within one second
Technology:	None
Miscellaneous Issue:	None
Frequency of use:	Only the time of sending an attachment

Operation Contract:

Table 2.26: Operation Contract Send Legal Document

Id	OLF13
Operation	Send legal Document ()
Cross Reference	Select the attachment from the system
Pre-Condition	The user must be signed in
Post-Condition	Attach box shown to the user

Composition Viewpoints:

The composition shows the structure of the system and its elements. In the section design elements are explained briefly how the system structure and its elements interact with each other, and which software engineering principle is used in the system.

Package Diagram:

This diagram is based on the “Unified Modeling Language” this section shows the structure of the system, how system elements depend on each other and how the system elements interact with each other as shown in figure 3.2 (Package Diagram).

Logical Viewpoint:

This diagram is based on the “Unified Modeling Language” this section shows the functionality of the system how system classes depend on each other and how the system classes interact with each other and how the system calls the methods from another class and how the system will use the server for CRUD operation in database.

Class Diagram:

The purpose of this diagram is to show the functionalities of the system as shown in figure 3.2 (Package Diagram) and how the system performs the functionality for the user when the user interacts with the system.

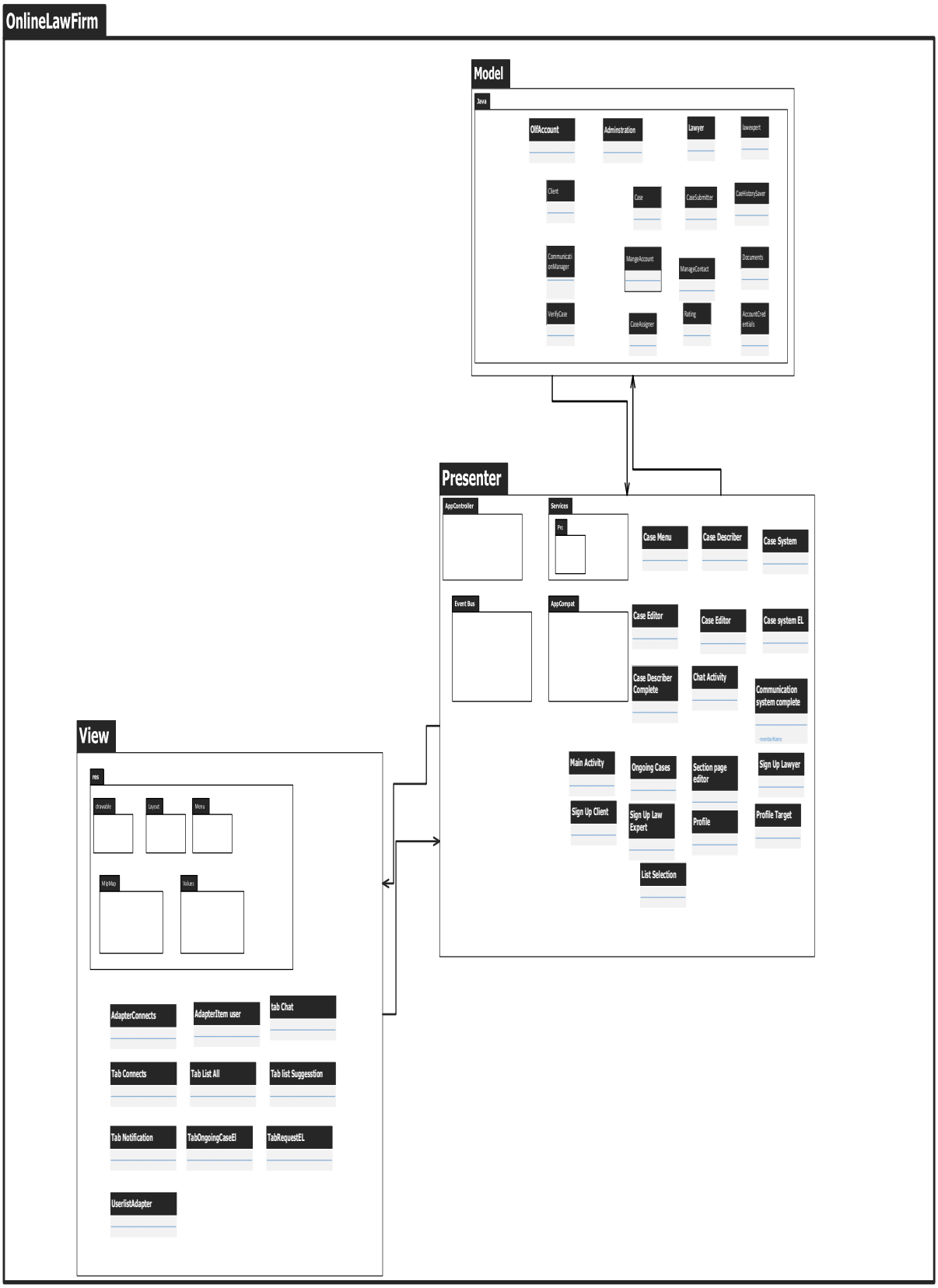


Diagram 3.2: Package Diagram

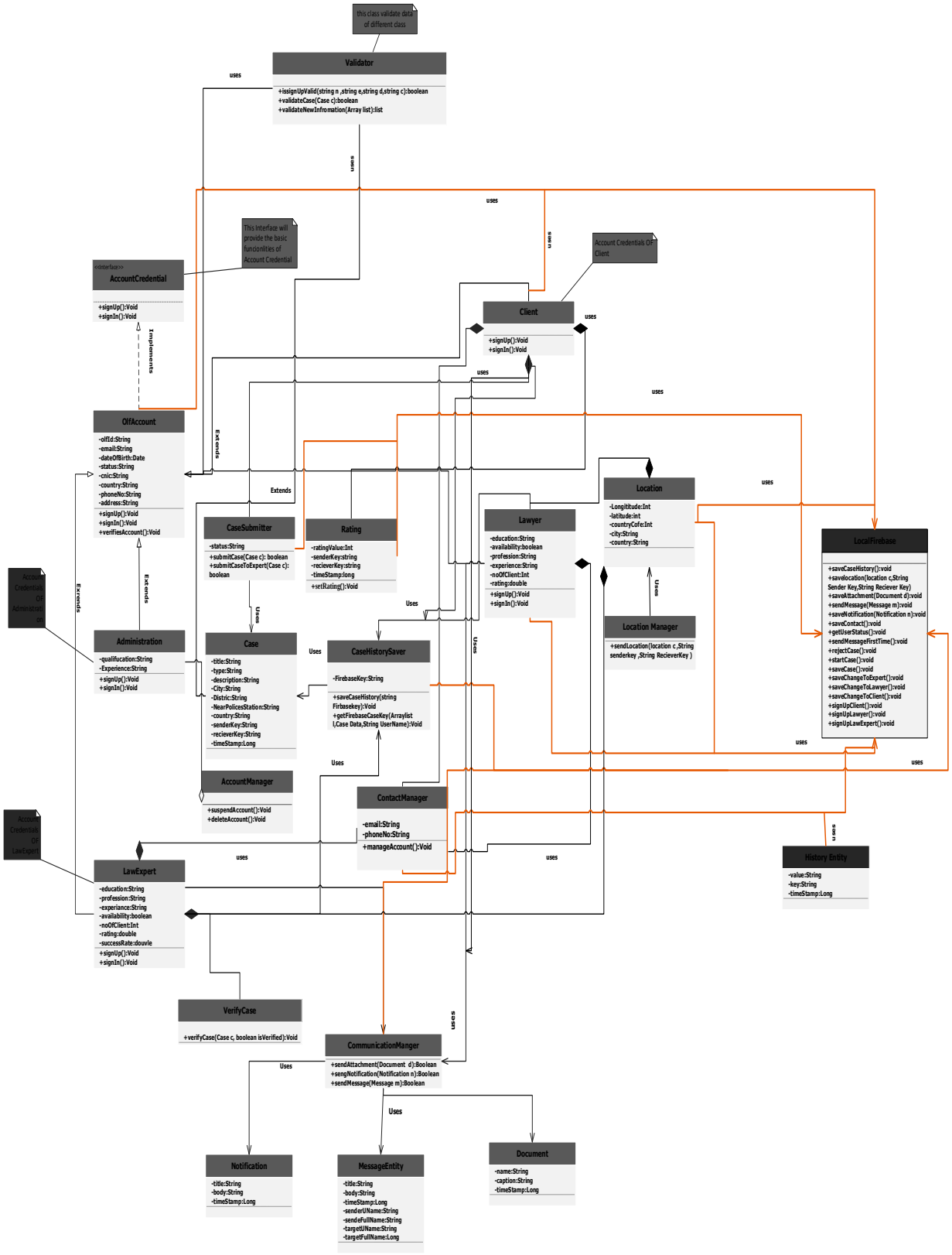


Diagram 3.3: Class Diagram

System Sequence Diagram:

The system sequence diagrams are based on the “Unified Modeling Language” this diagram visualizes the use case against the canaries.

System Sequence Diagram Sign Up:

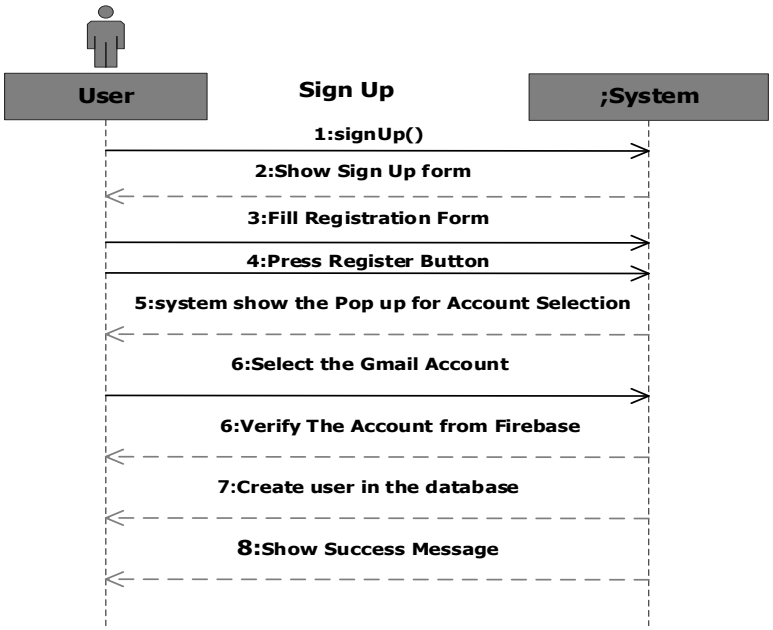


Diagram 3.4: SSD-Sign up

System Sequence Diagram Sign In:

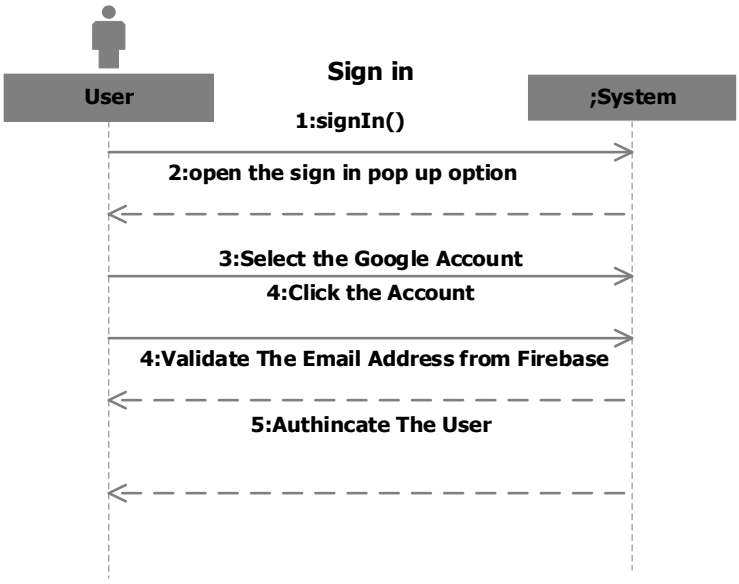


Diagram 3.5: SSD-Sign In

System Sequence Diagram Manage Account:

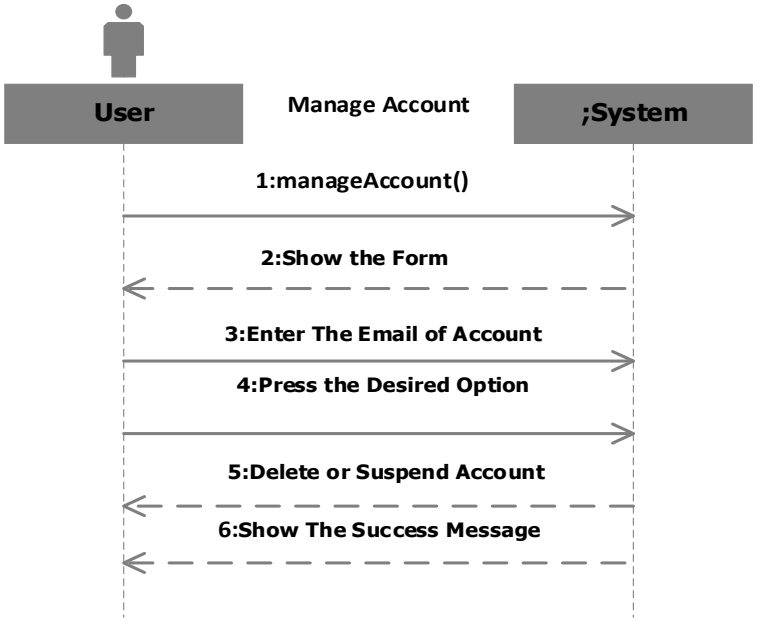


Diagram 3.6: SSD-Manage Account

System Sequence Diagram Manage Contacts:

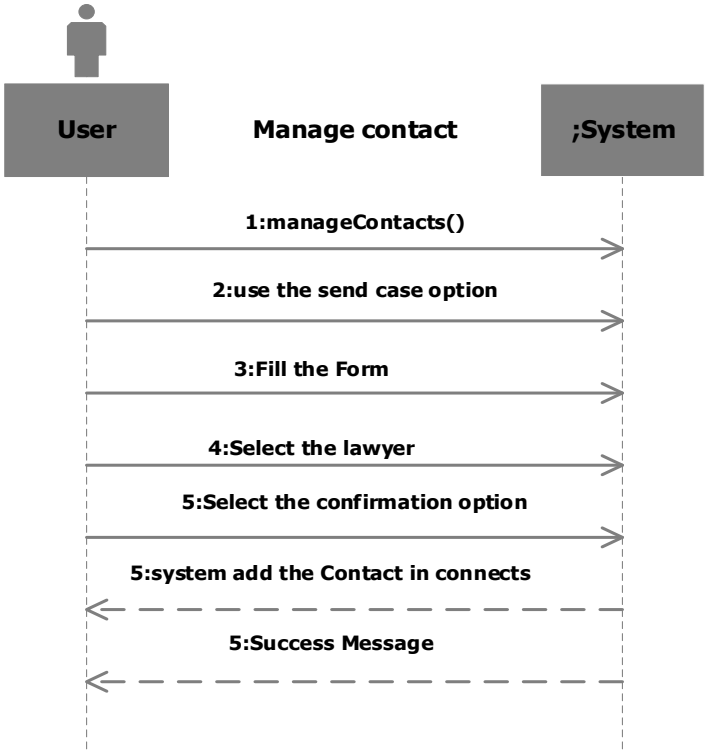


Diagram 3.7: SSD-Manage Contacts

System Sequence Diagram Submit Case:

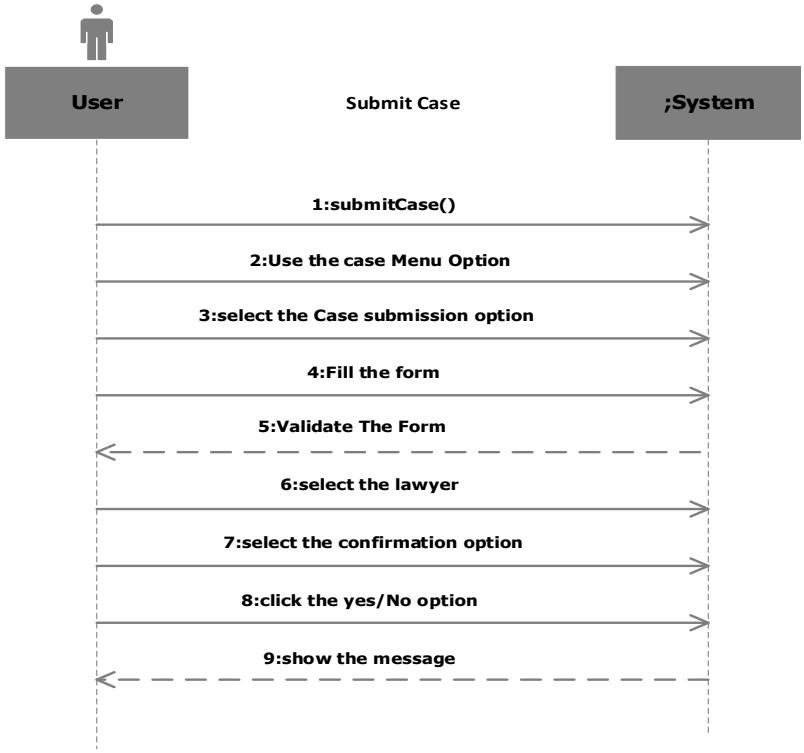


Diagram 3.8: SSD-Submit Case

System Sequence Diagram Set Rating:

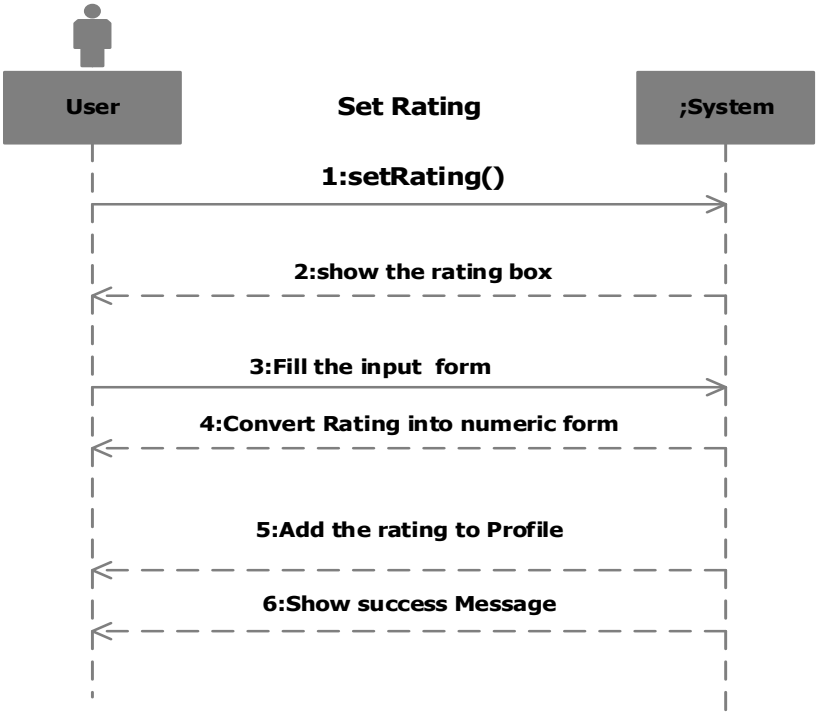


Diagram 3.9: SSD-Set Rating

System Sequence Diagram Share Legal Document:

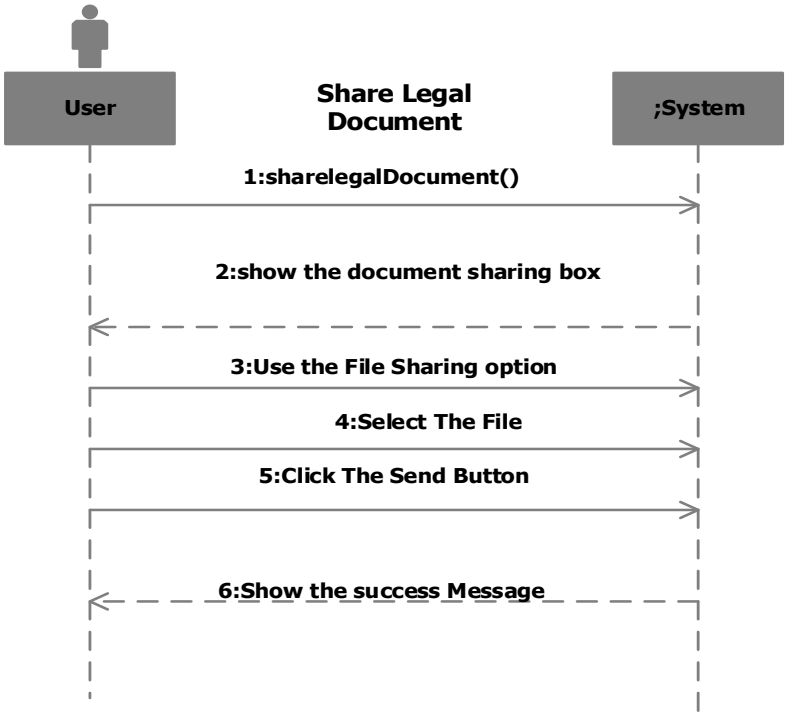


Diagram 3.10: Share Legal Document

System Sequence Diagram Confirm Availability:

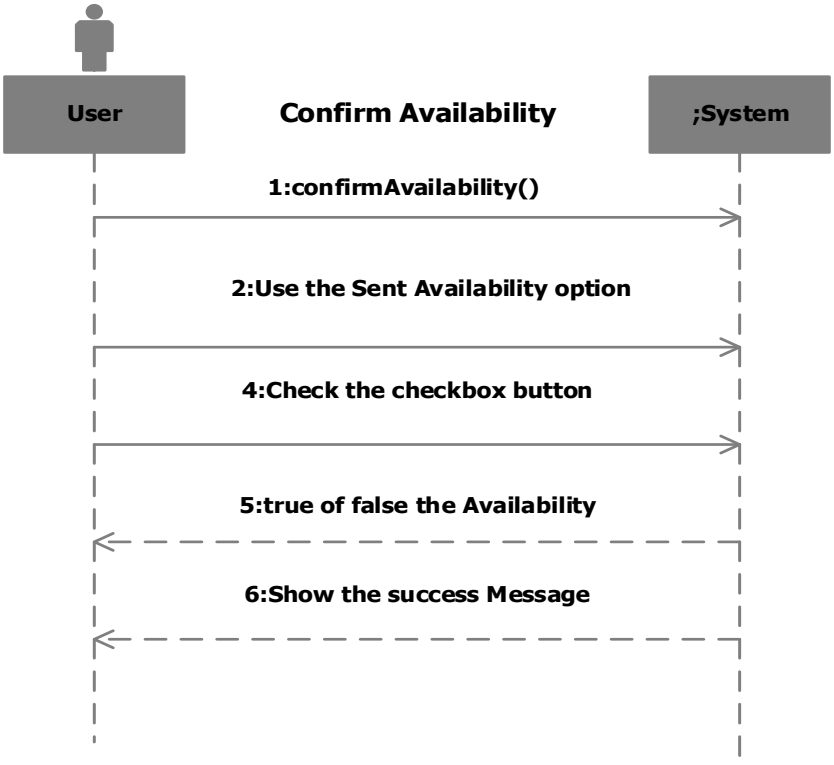


Diagram 3.11: SSD-Confirm Availability

System Sequence Diagram Send Location:

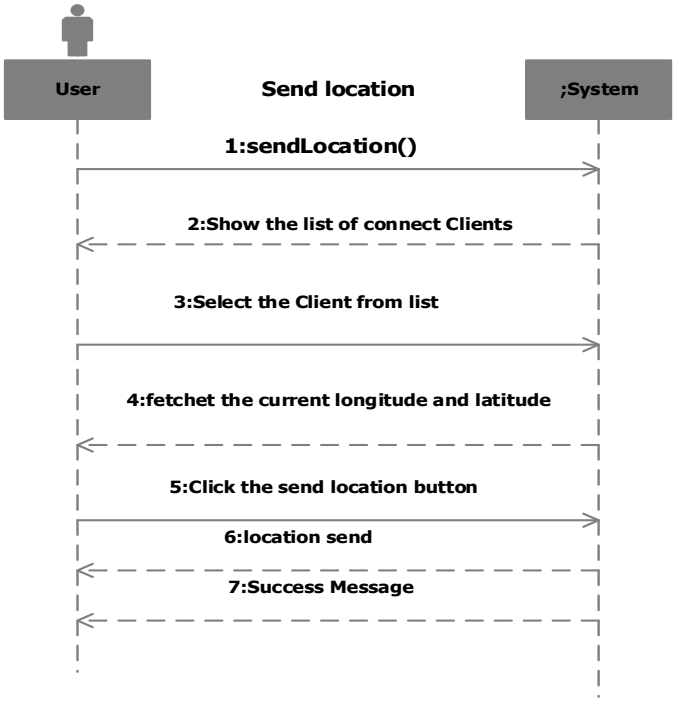


Diagram 3.12: SSD-Send Location

System Sequence Diagram Send Notification:

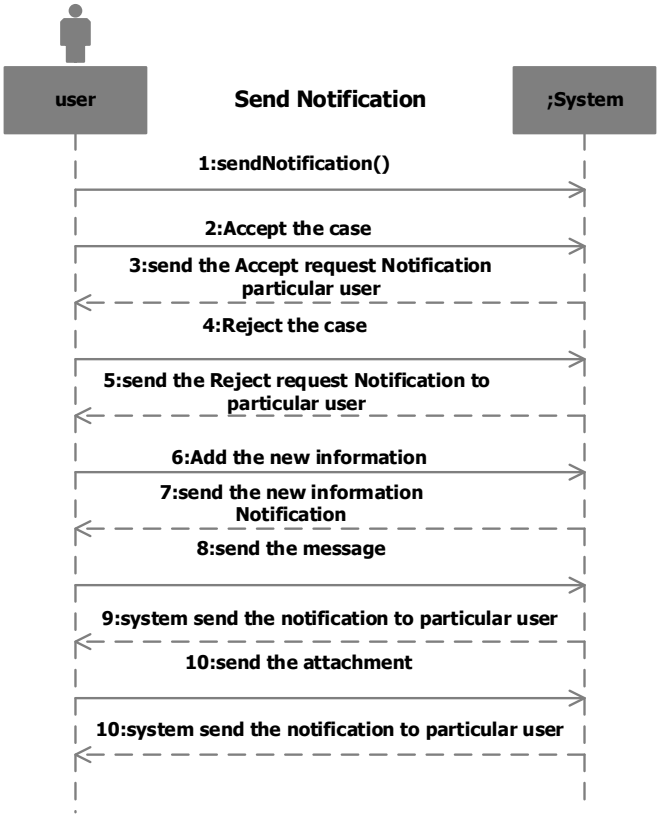


Diagram 3.13: SSD-Send Notification

System Sequence Diagram Verify Case:

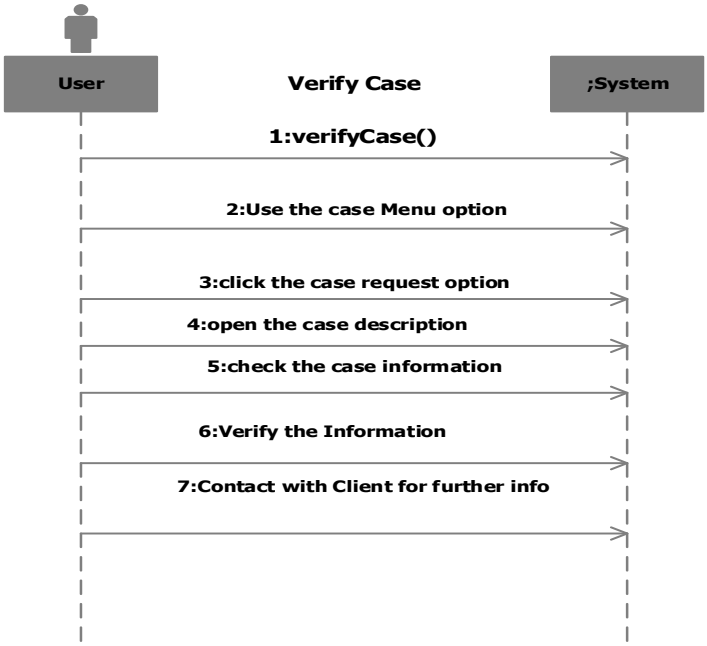


Diagram 3.14: SSD-Verify Case

System Sequence Diagram Save Case History:

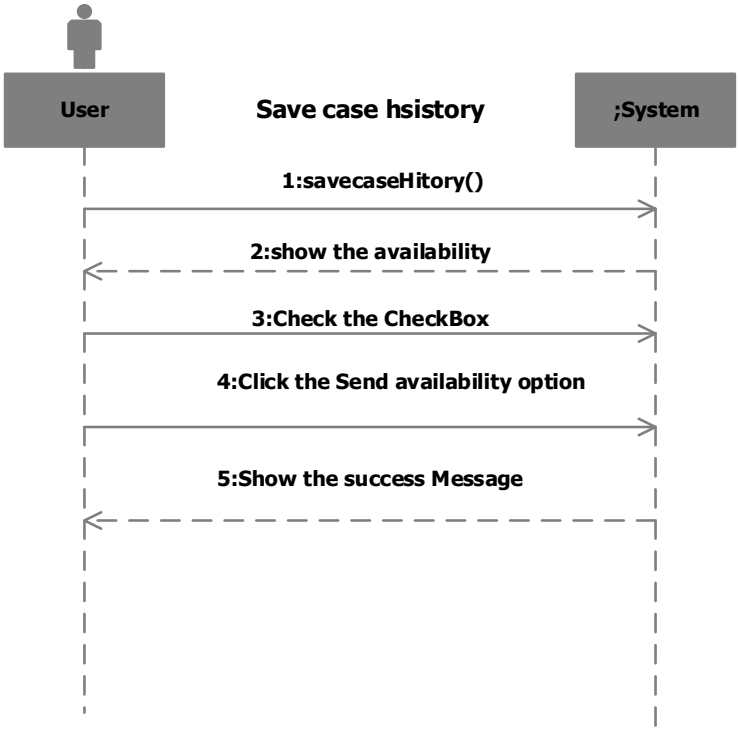


Diagram 3.15: SSD-Save Case History

System Sequence Diagram Communicate with connects:

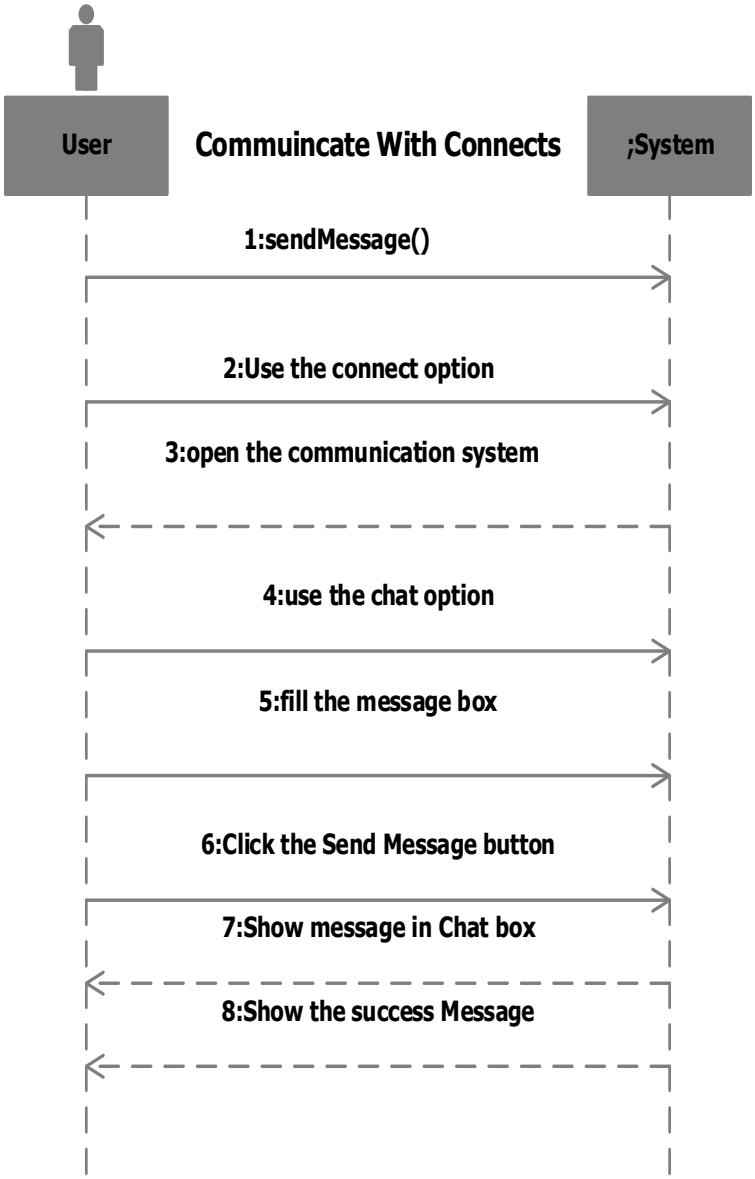


Diagram 3.16: Communicate with Connects

State Dynamic:

The dynamic state is the part of the information state which can change during the use of the application while the static states remain the same. The operation state will be staying remain the same. Otherwise updating the dynamic information state and selecting furthermore. When the dynamic state changes the information as well as operation state will also be changed.

Activity Diagrams:

The system sequence diagram is based on the “Unified Modeling Language” this diagram visualizes the operation of the user with the system against every use case. How the user will operate the system and how the system will respond to the user.

Activity Sign Up:

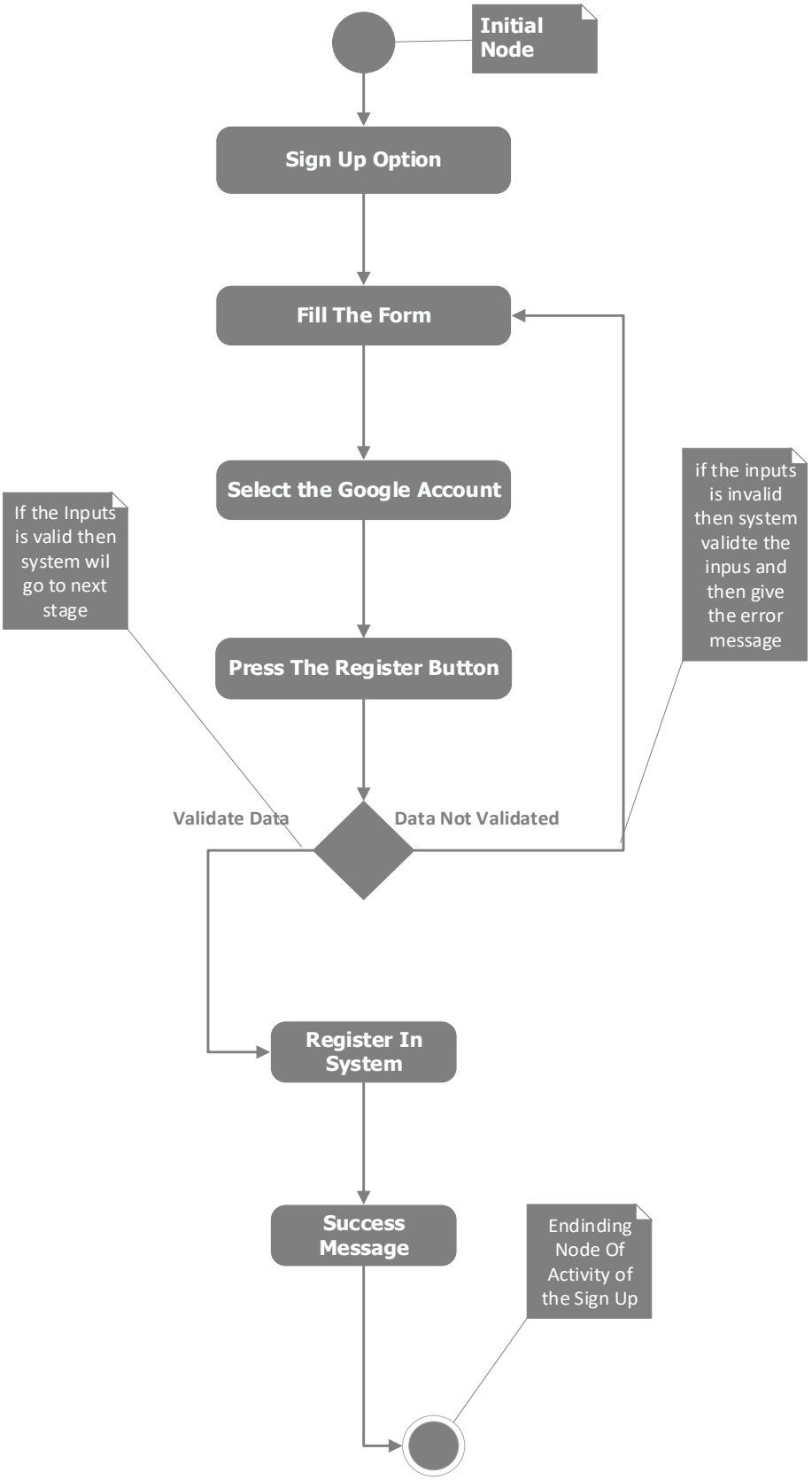


Diagram 3.17: Activity-Sign up

Activity Sign In:

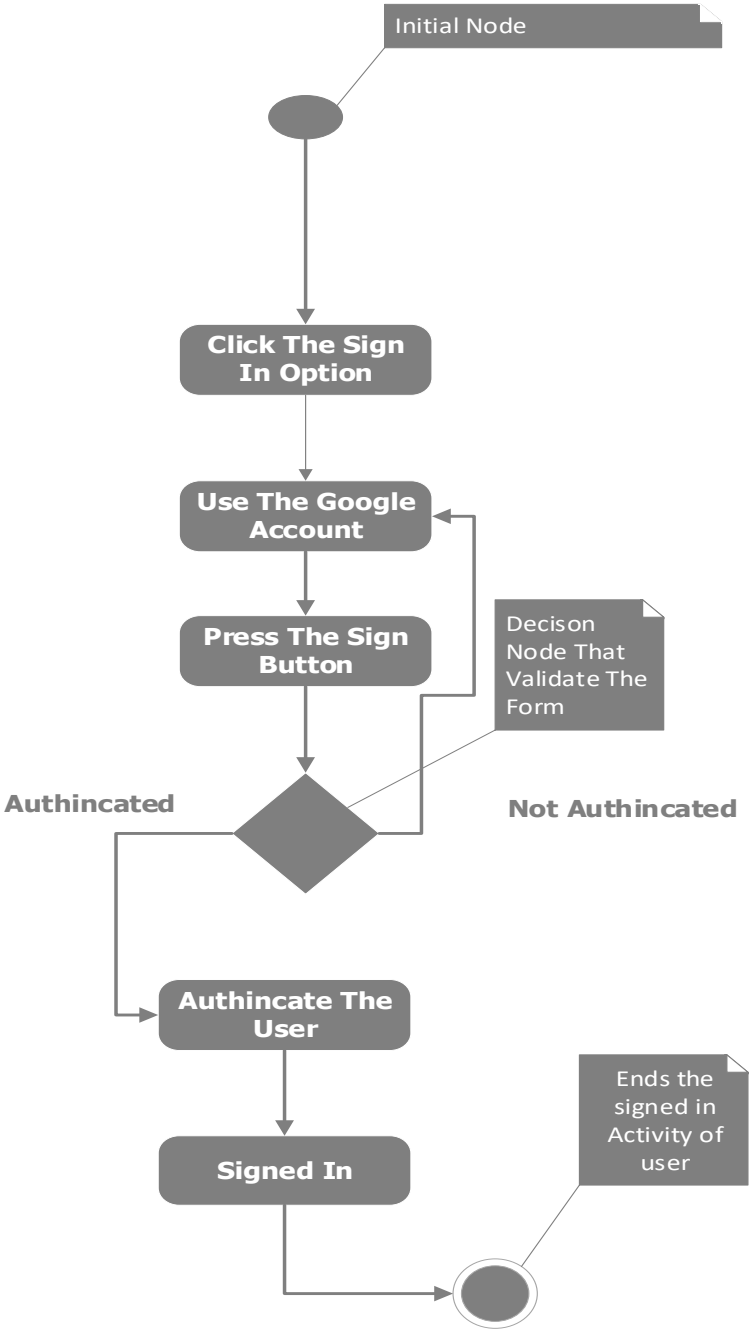


Diagram 3.18: Activity -Sign In

Activity Submit case:

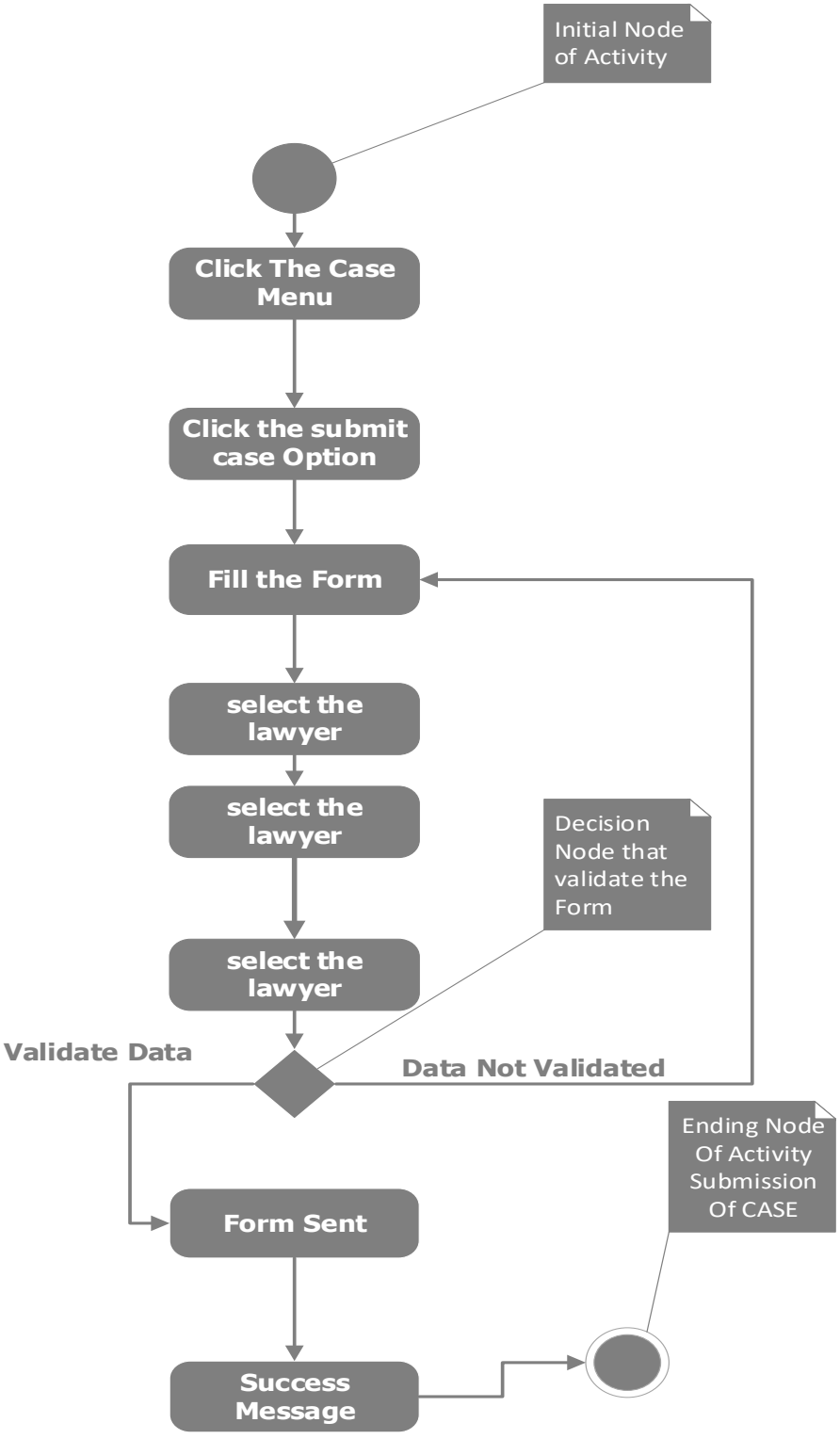


Diagram 3.19: Activity –Submit case

Activity Verify Case:

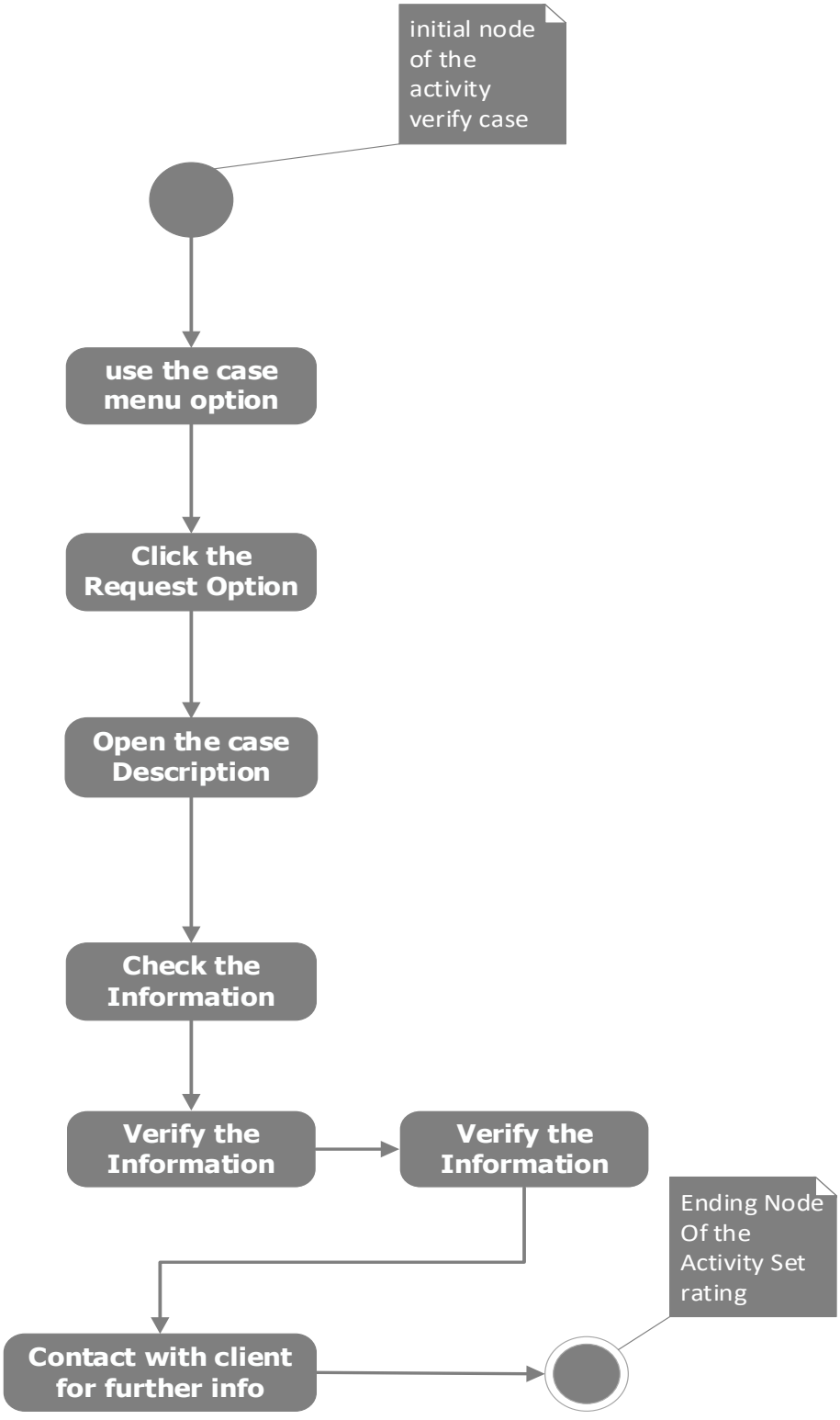


Diagram 3.20: Activity –Verify Case

Activity Manage Account:

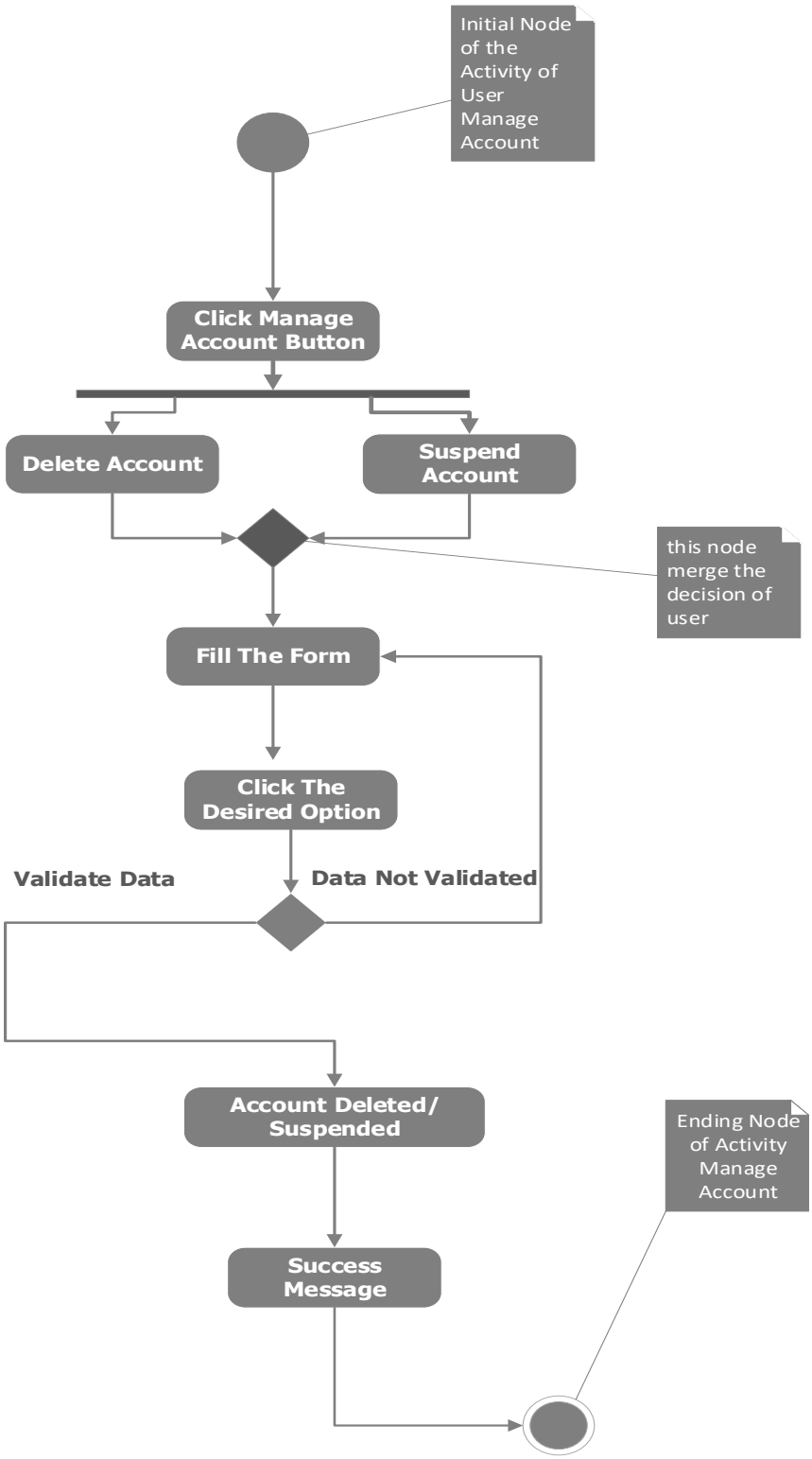


Diagram 3.21: Activity –Manage Account

Activity Manage Contacts:

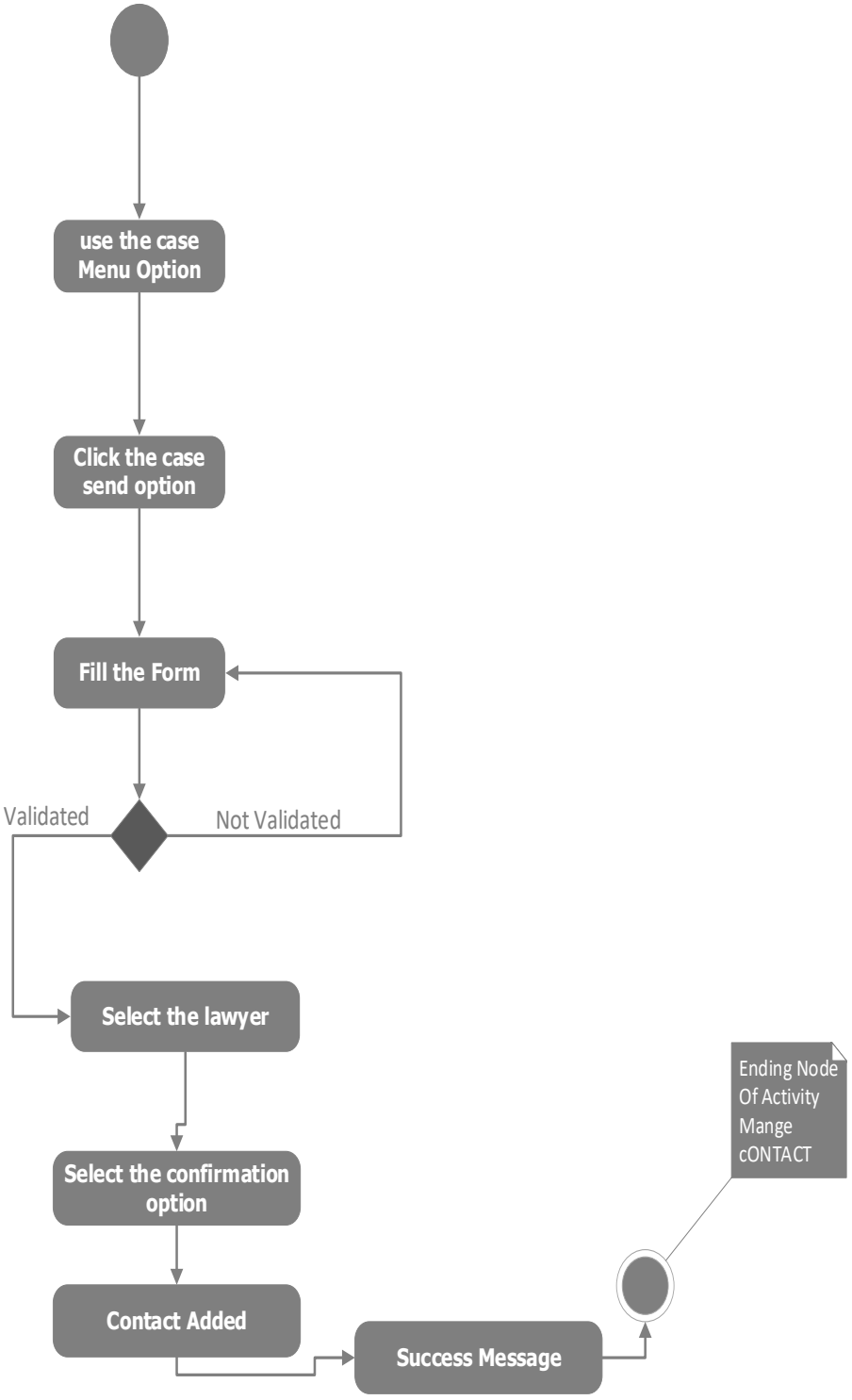


Diagram 3.22: Activity –Manage Contacts

Activity Send Location:

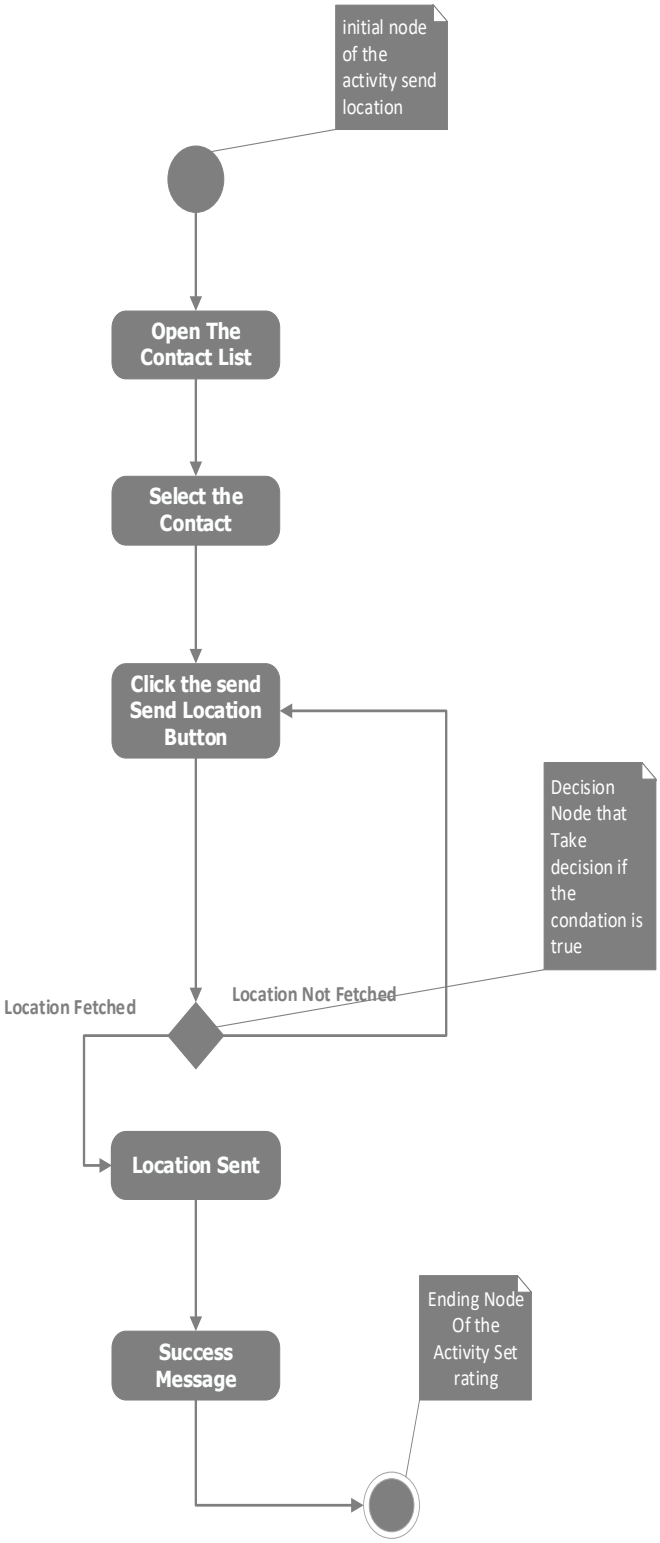


Diagram 3.23: Activity –Send Location

Activity Send Notification:

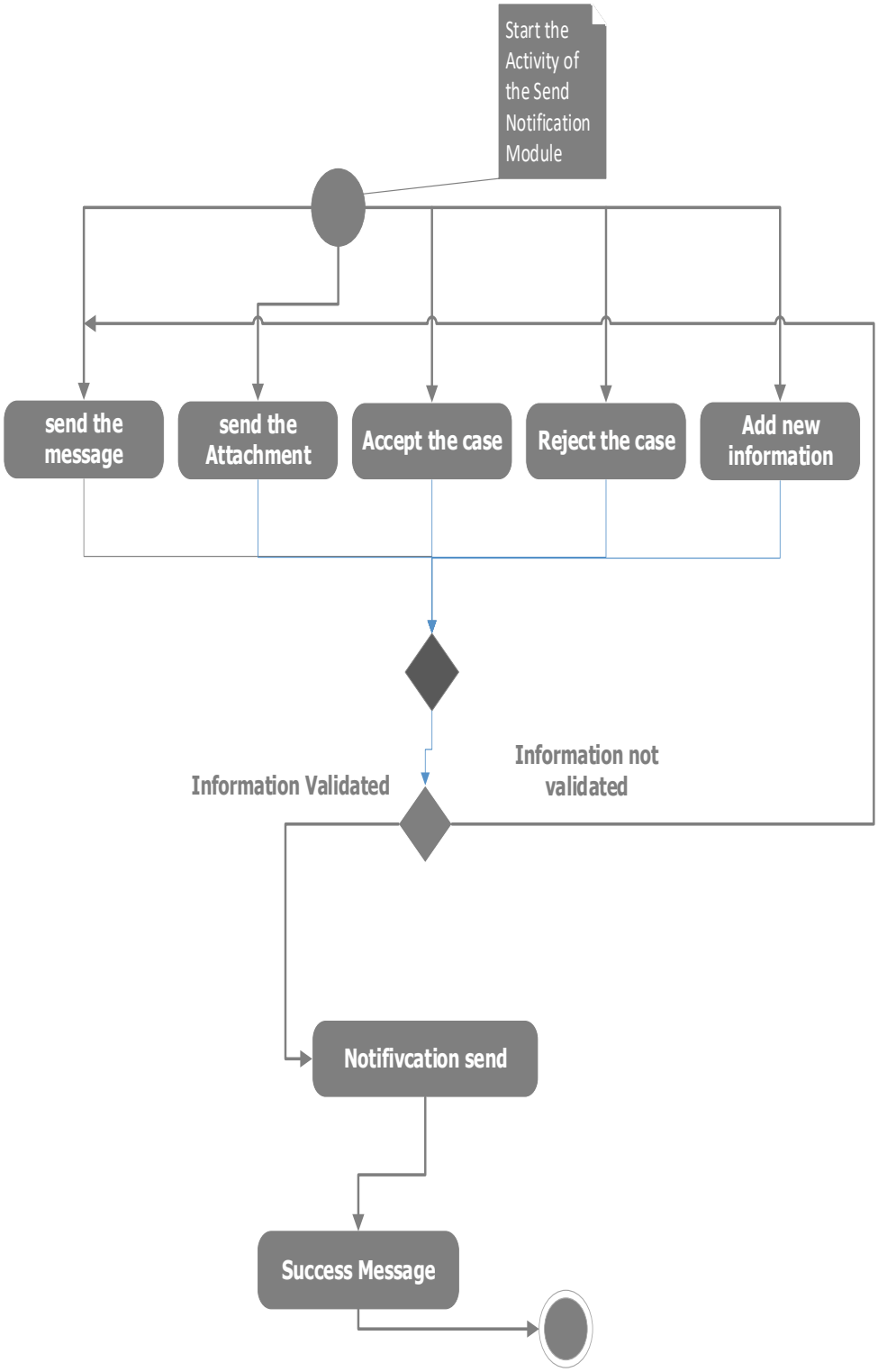


Diagram 3.24: Activity –Send Notification

Activity saves Case History:

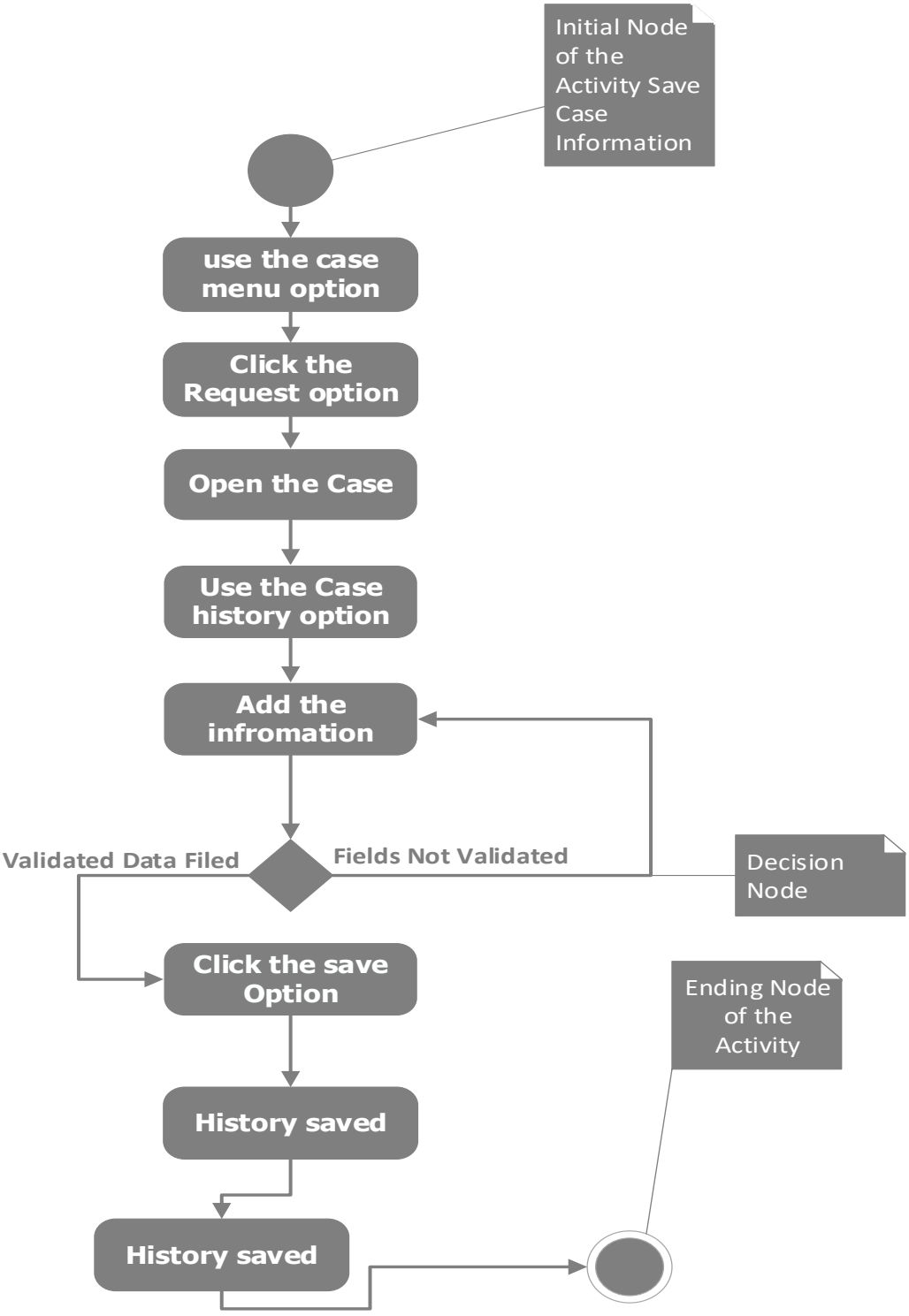


Diagram 3.25: Activity –Save Case History

Activity Send Legal Document:

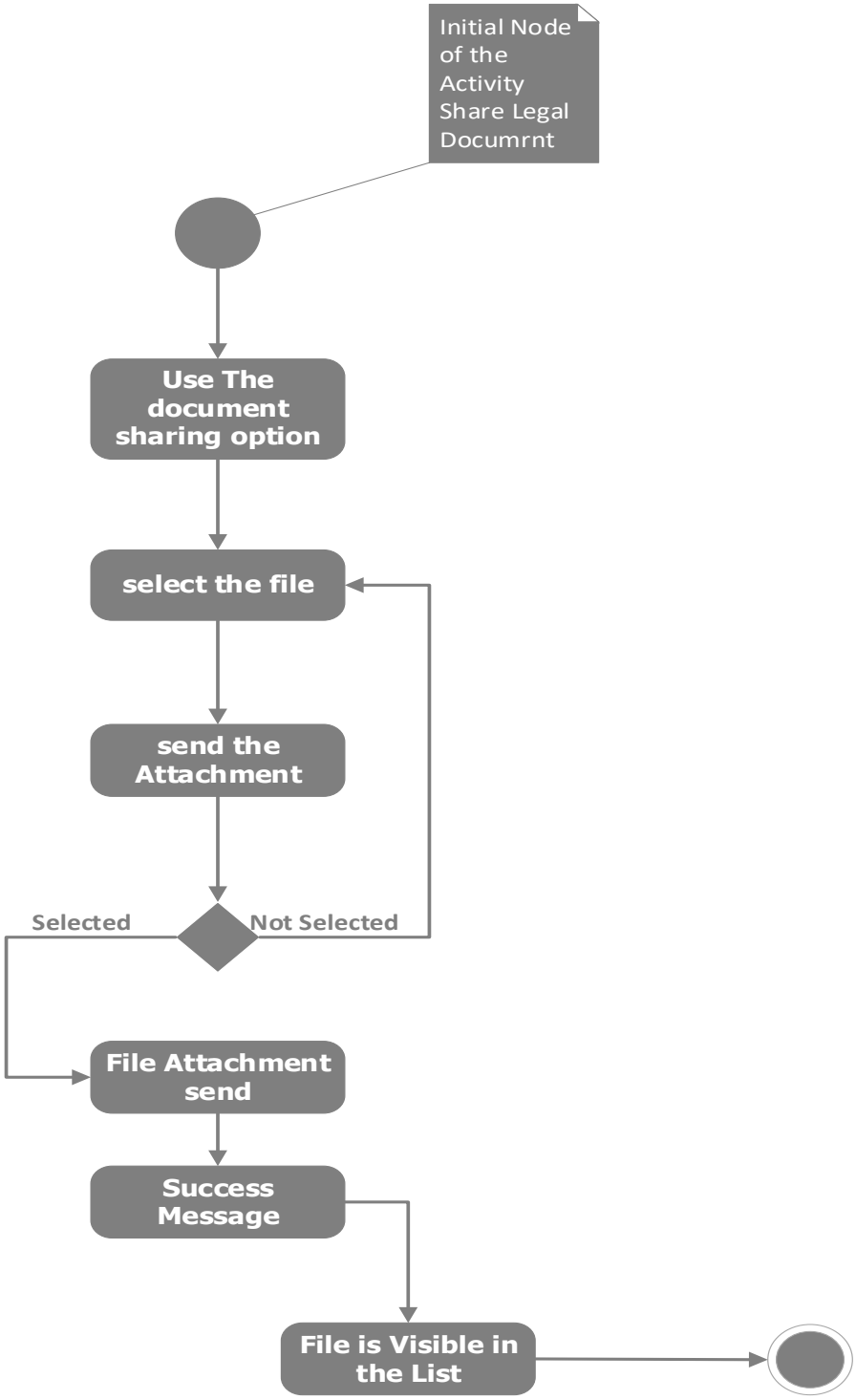


Diagram 3.26: Activity –Send Legal Document

Activity Send Availability:

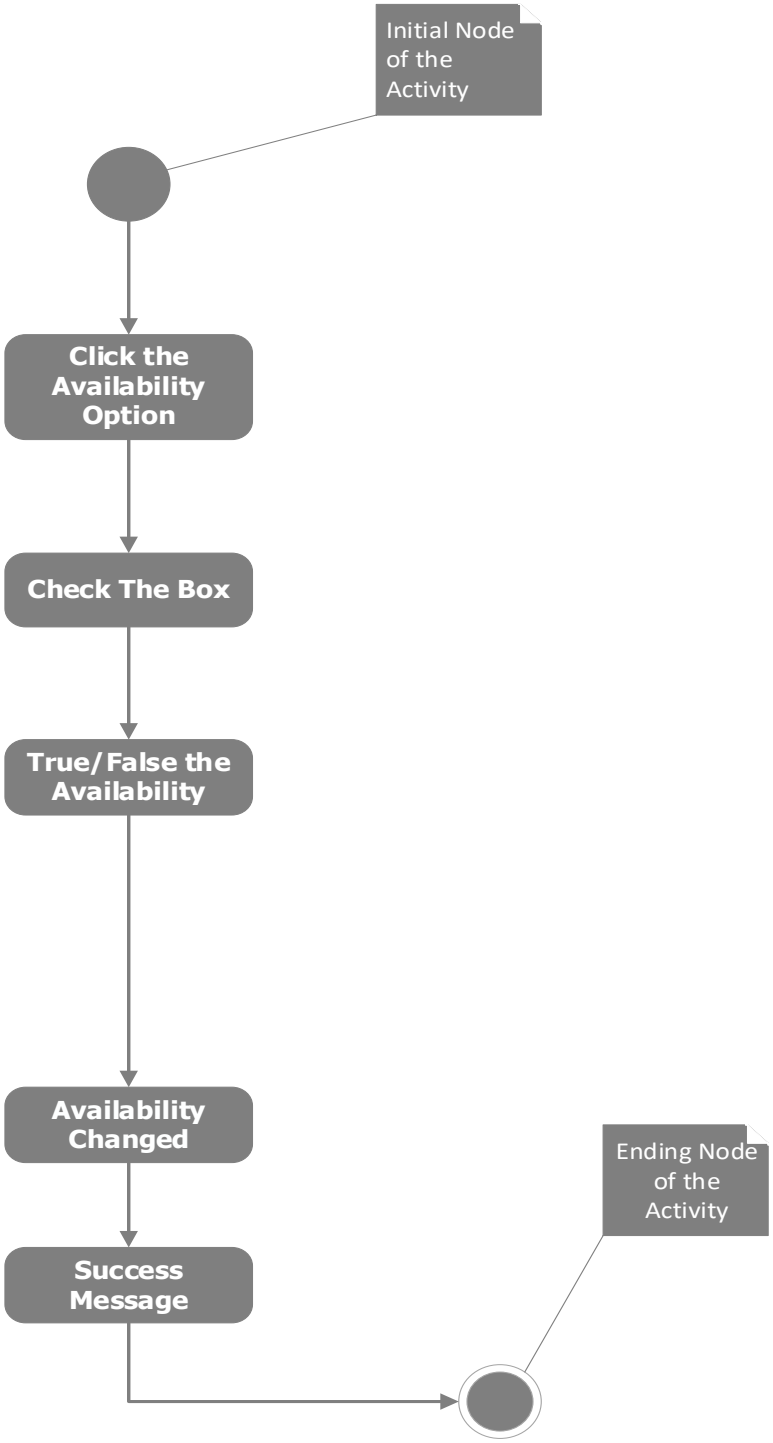


Diagram 3.27: Activity –Send Availability

Activity Set Rating:

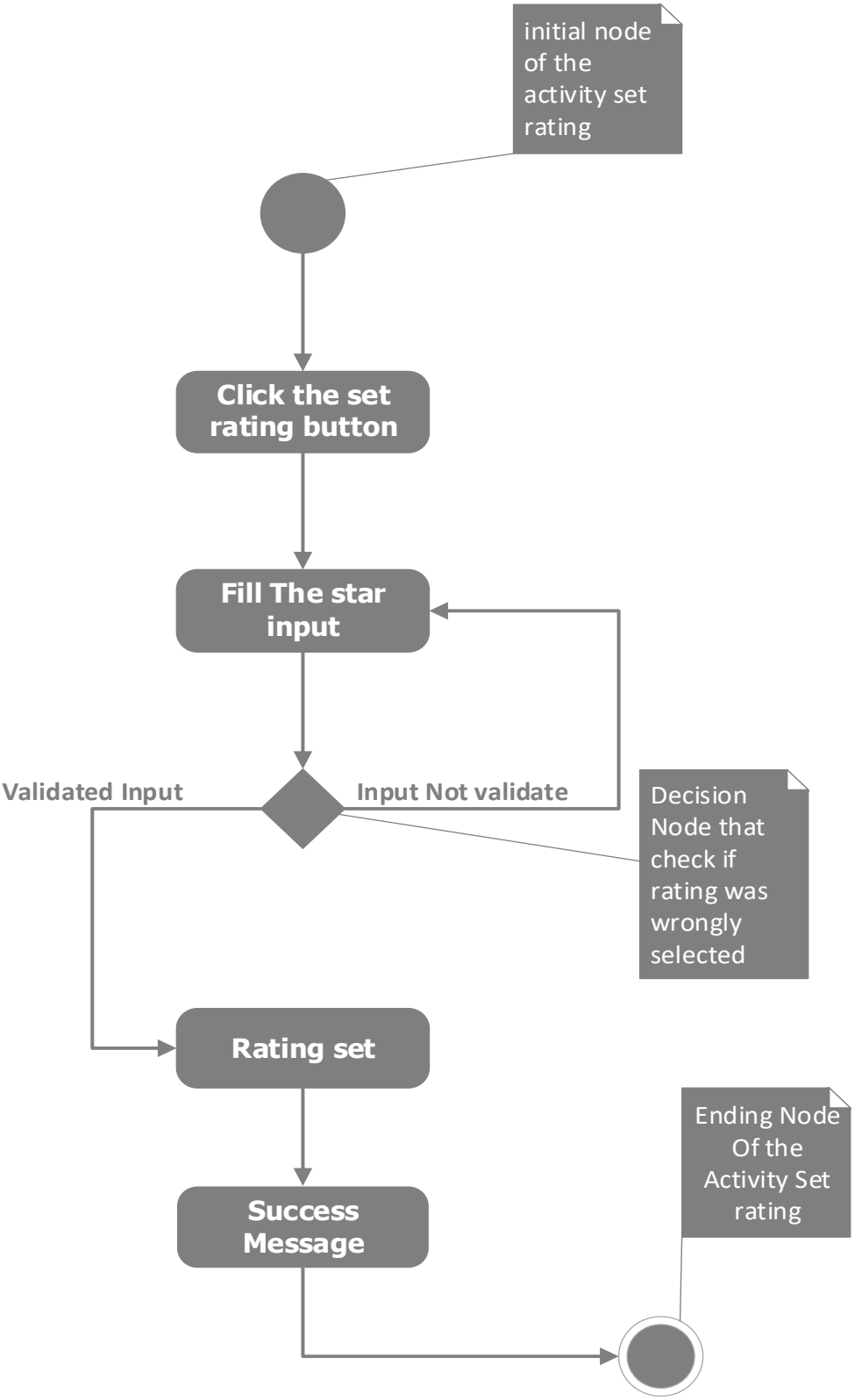


Diagram 3.28: Activity –Set Rating

Sequence Diagrams

The sequence diagram shows the operation of the method how the system calls the methods and how the data flow from one method to others and how the data is loaded from the server using methods as shown below diagrams:

Sequence Diagram Manage Account:

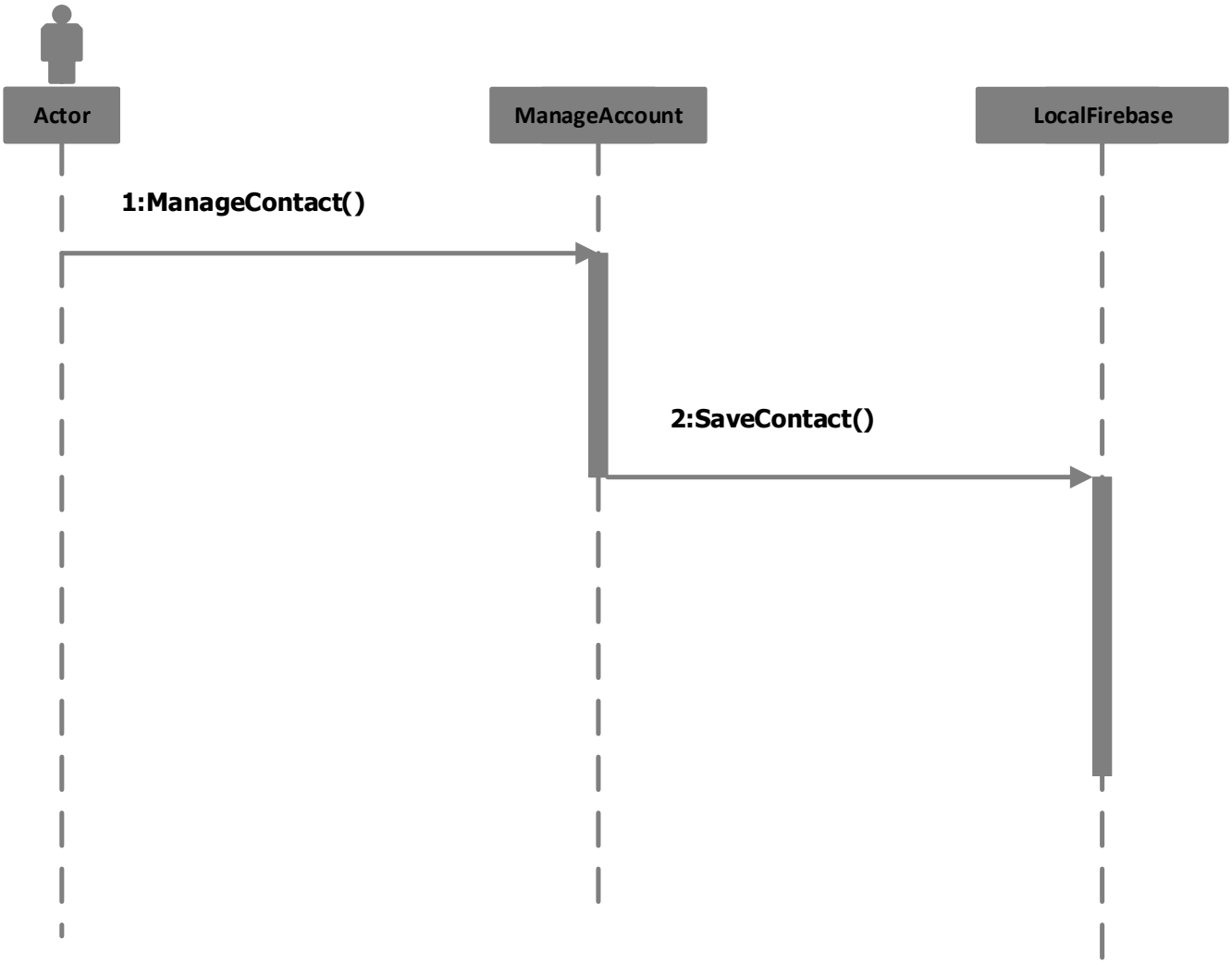


Diagram 3.29: Manage Account

Sequence Diagram Manage Contacts:

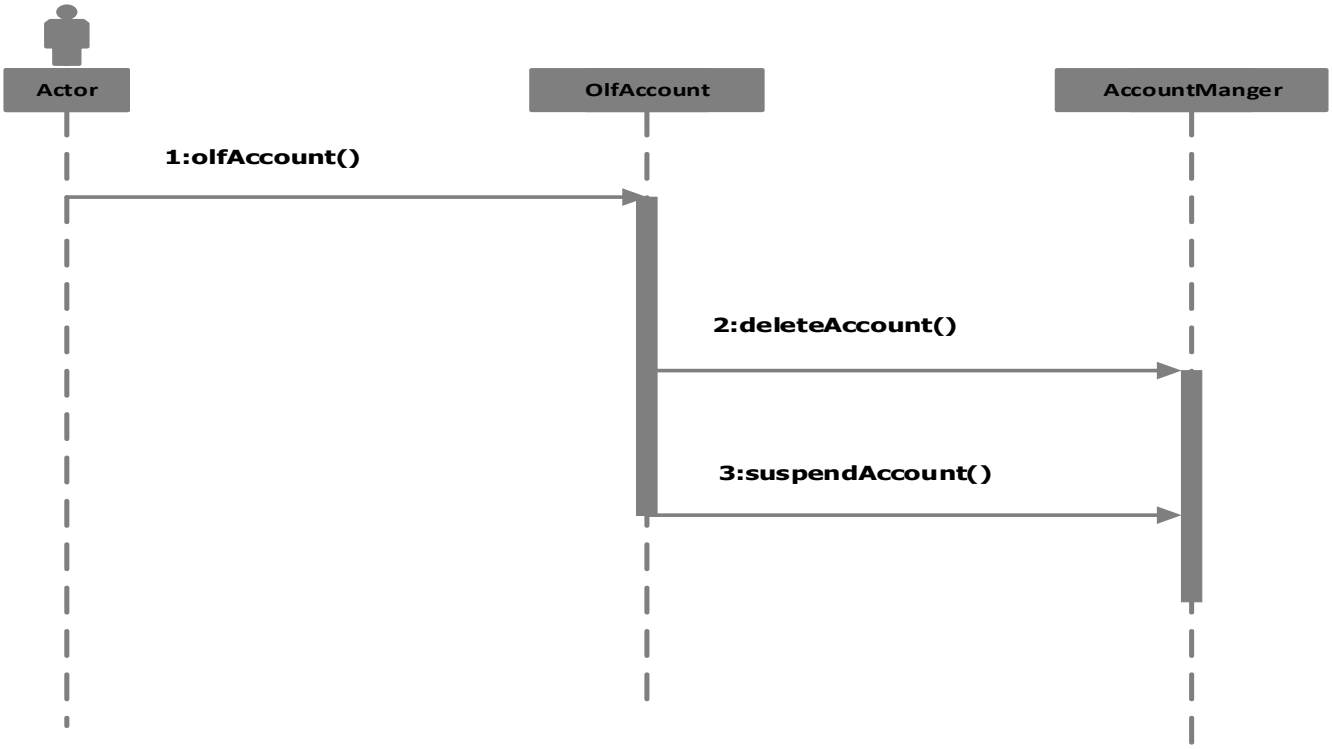


Diagram 3.30: Manage Contacts

Sequence Diagram Save Case History:

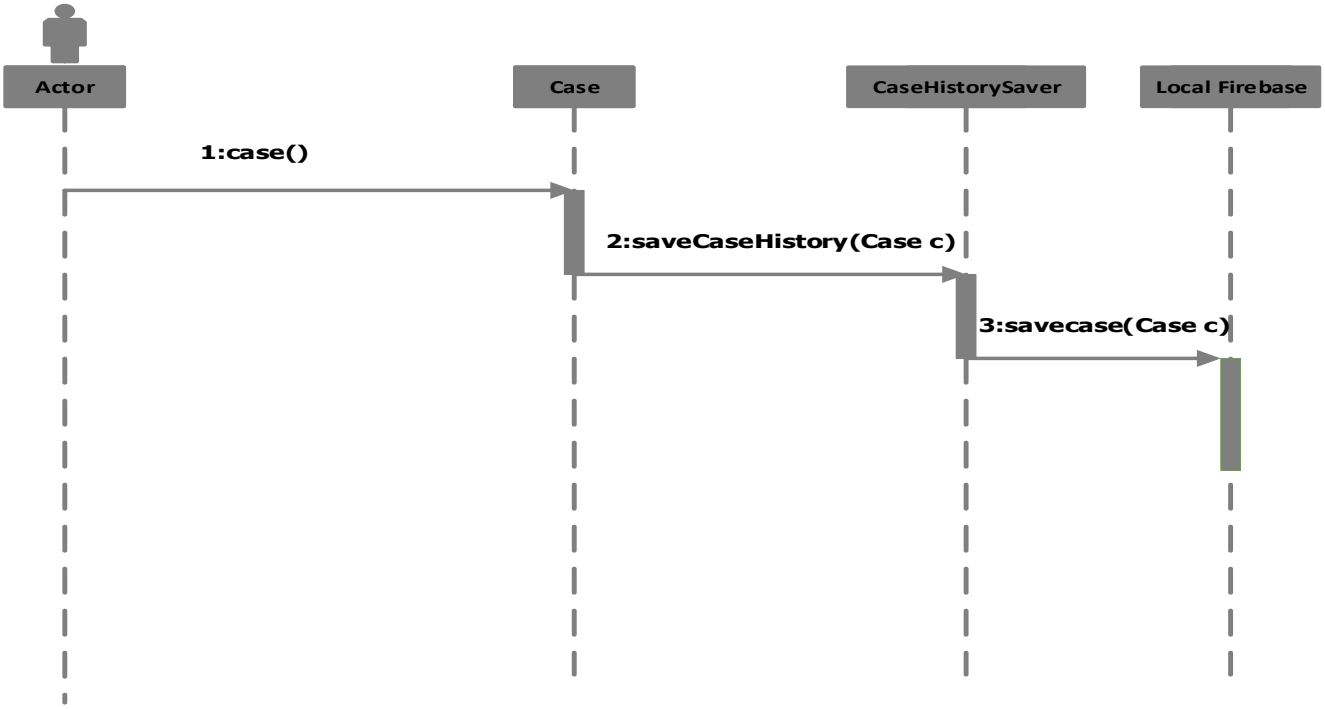


Diagram 3.31: Save Case History

Sequence Diagram Send location:

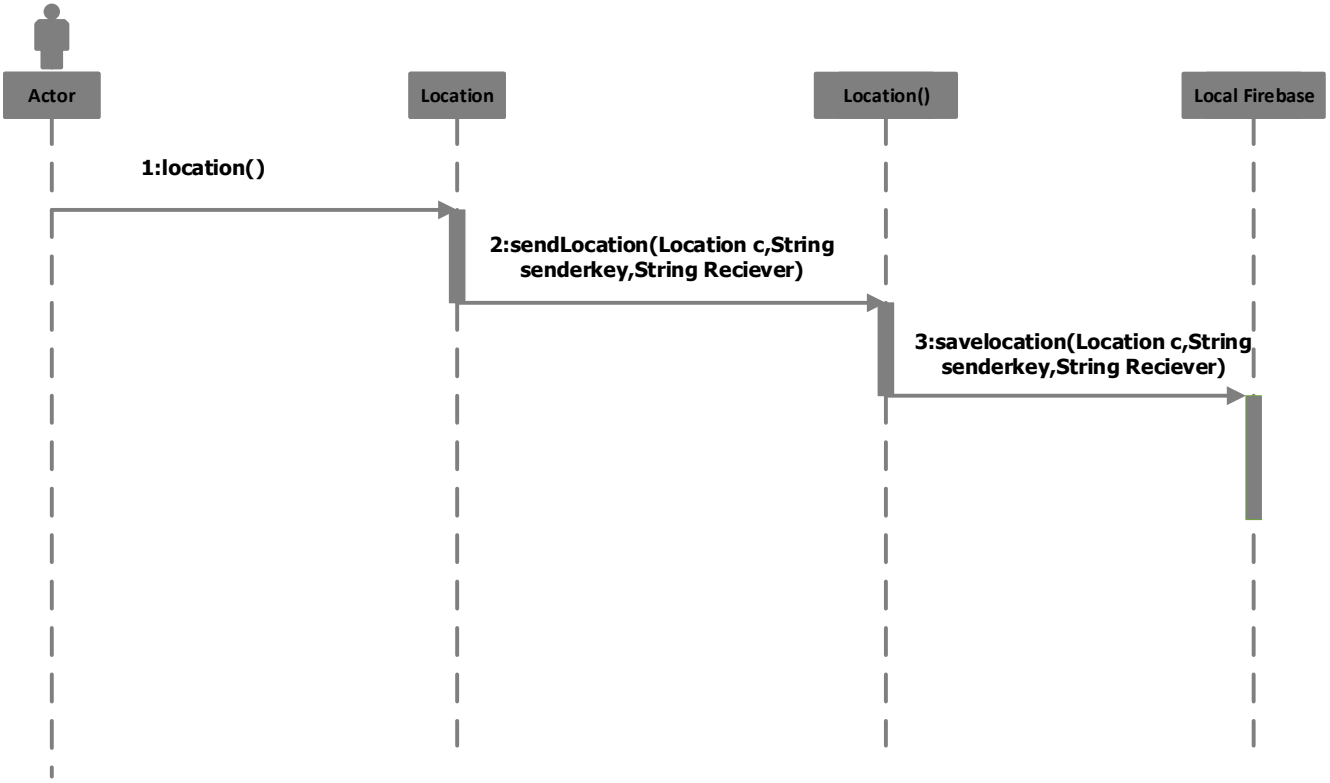


Diagram 3.32: Send Location

Sequence Diagram verifies case:

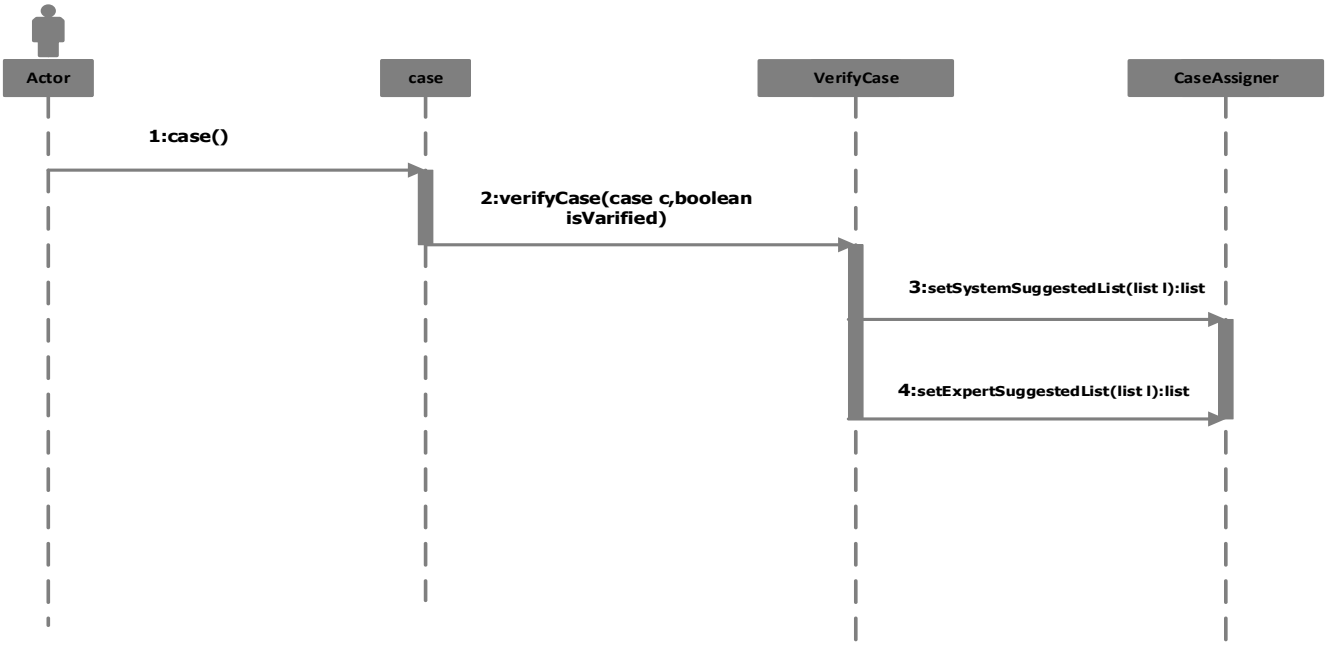


Diagram 3.33: Verify Case

Sequence Diagram Communication Manager:

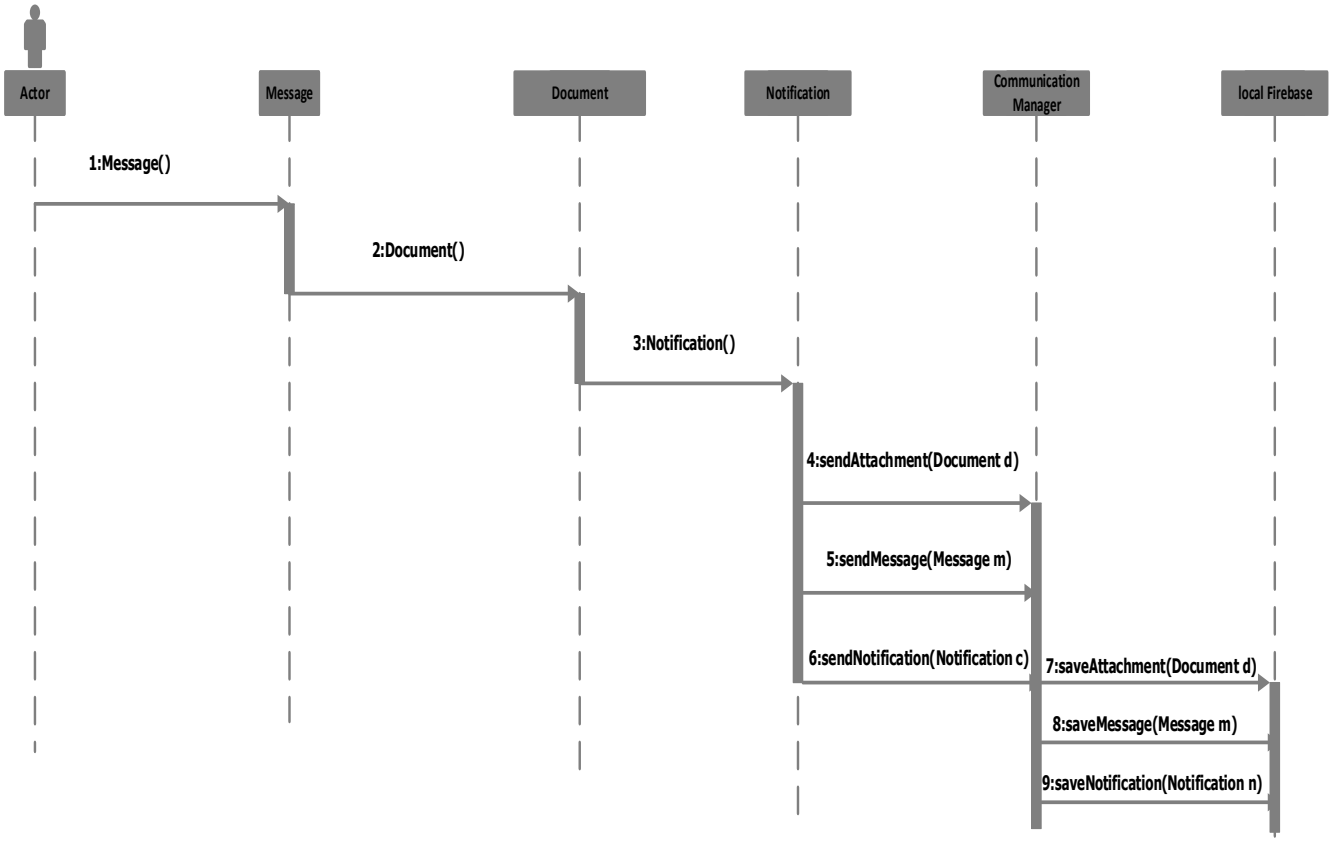


Diagram 3.34: Communication Manager

Sequence Diagram Set Rating:

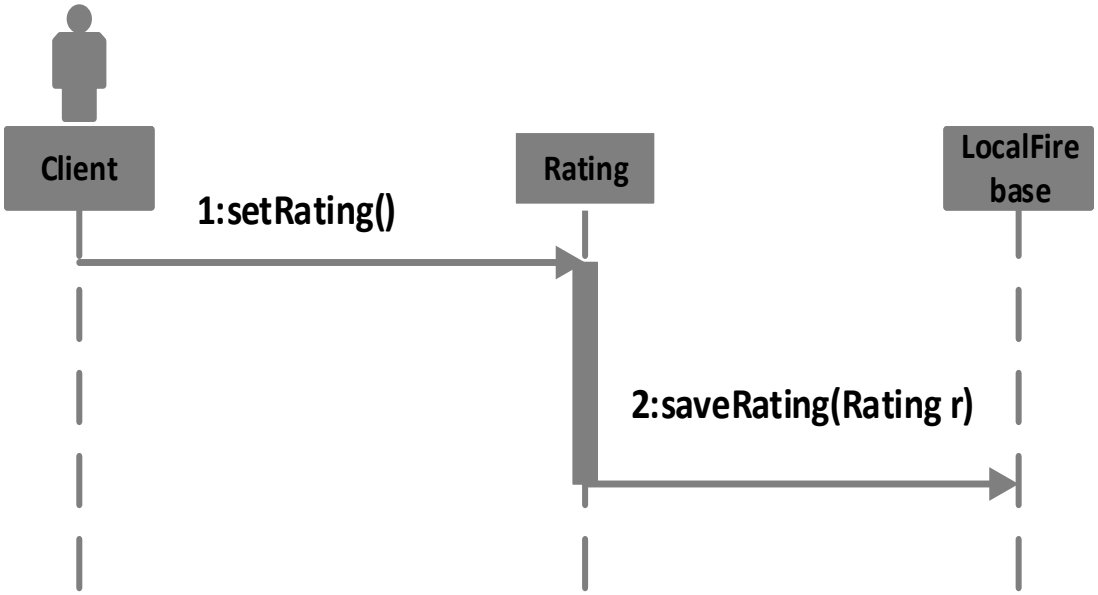


Diagram 3.35: Set Rating

Sequence Diagram Submit Case:

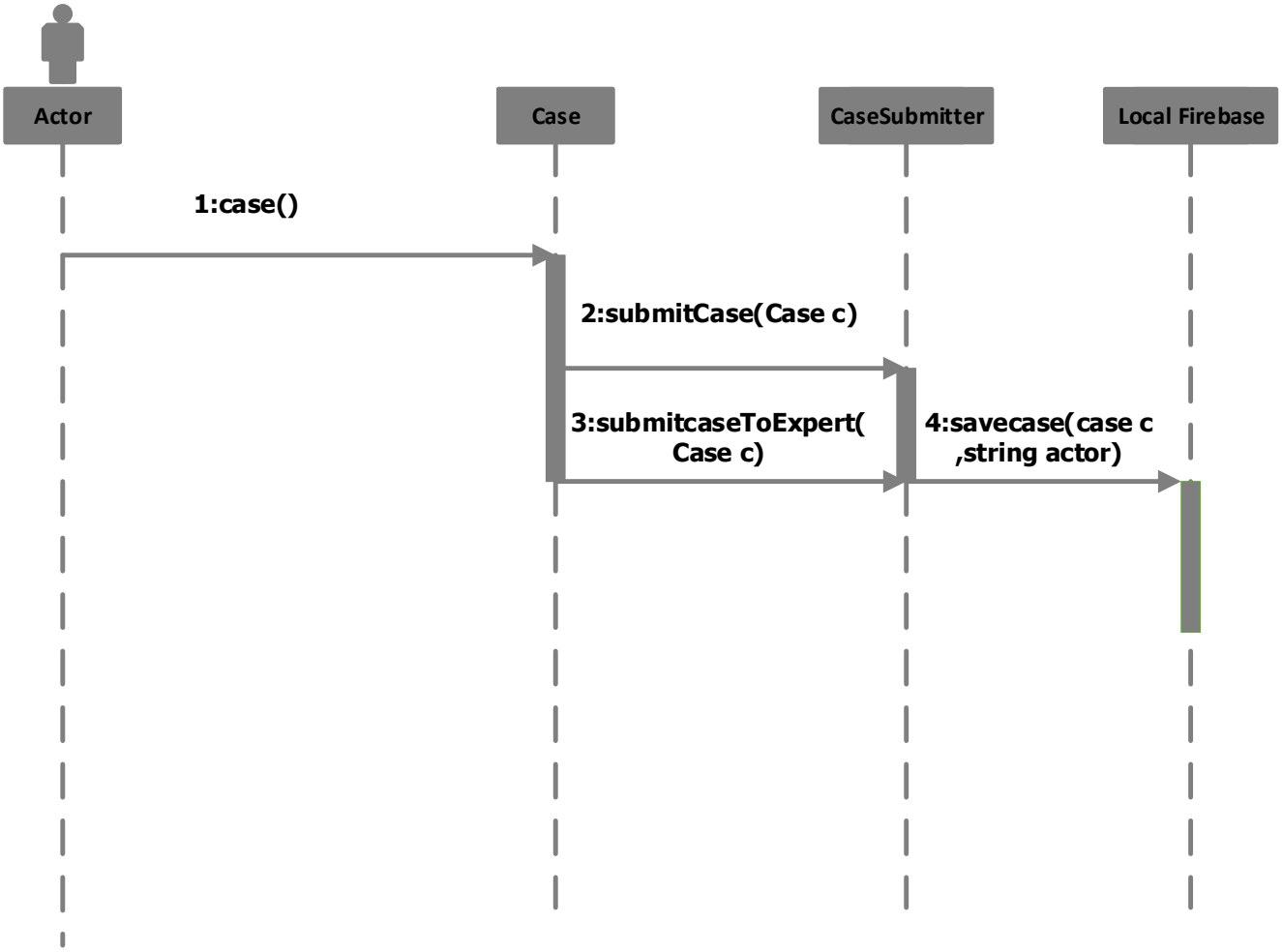


Diagram 3.36: Submit Case

Software System Design Document

Download the document for practicing and writing the Software system Design document of the Project.
https://github.com/SardarMudassarAliKhan/Software-Design-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

19

Development and Configuration Management

Overview

In this chapter, we will explore explores development and configuration management, emphasizing Android Studio's versatility, coding standards, object-oriented principles, and comprehensive development planning for project organization and execution.

Introduction:

In this chapter, we will discuss the development and configuration management and the elements where this production is perfectly useful. And this production is important for comprehension. We discuss on the other hand if any change occurs in the future to improve the efficiency of the system. Some of the following development standards and development used in this product are given below.

- Which tools and framework APIs are used during the development of the system?
- Which libraries are used during the development of the system?
- What are the coding style and standards being followed in the development of the product?

Android studio:

Android Studio is used during the development of the system. Android studio is the most advanced and open-source platform for the development of android systems. Android studio has more the 26 APIs with the most advanced programming features that will help the software engineers to build efferent and best products using the android platform.

Coding standard:

Different types of coding standards are used during the development of software systems. Every programing language provides coding standards and the programmer's engineers must follow that standard during the development of the system. If every programmer wants to use its coding standards, then it is difficult to maintain the code in the future and difficult to manage some are the coding standards used during the development of this system.

Model View Presenter:

Model-View-Presenter is the Software engineering principle used during the development of the system. MVP provides the standard for the development of software products. This principle helps the software engineers to separate the business logic and separate the software product interfaces.

Function Call:

Function Calls are used in the following ways:

- There are no spaces used between the function name, parenthesis is used for parameters, and curly brackets are used for declaring the body of the project.
- Spaces and Commas are used between the parameters of the function.
- There is no space is used between the last parameters and at the end of the parenthesis.

Class definition:

There are opening braces used for the declaration of the class underscore are used for the separation of words for the name of class like (University Employee) for the making of the class meaningful. Capital letters are used for the name class, and we avoid underscore at the start of the class.

Object-Oriented Programming:

An object-oriented programming style is used in the development of this project. Object-oriented programming provides a modular structure that will help the Engineers and users to understand

the structure of the system. The modular structure provides a better understandability of the code and code reusability is high in this style of cladding.

Code Comments:

Java style and C# Razor type comments are used e.g. (@****@ or /* */).

Line length and Control Structure:

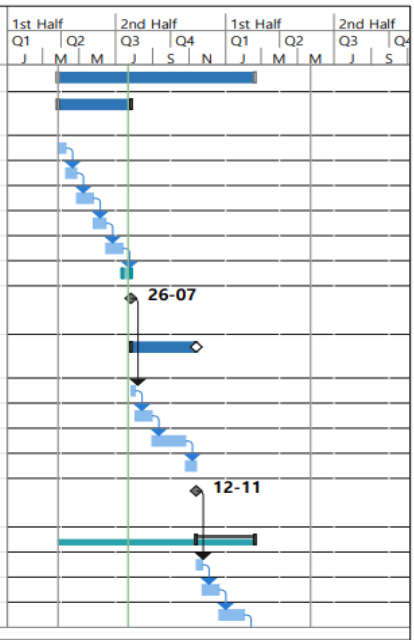
- For the batter understanding and readability of the code range of the character is 70-90 characters per line.
- There is no Tab used but intended of the 4 spaces are used
- In the control statements, one-line space is used between control statements, and parenthesis are used

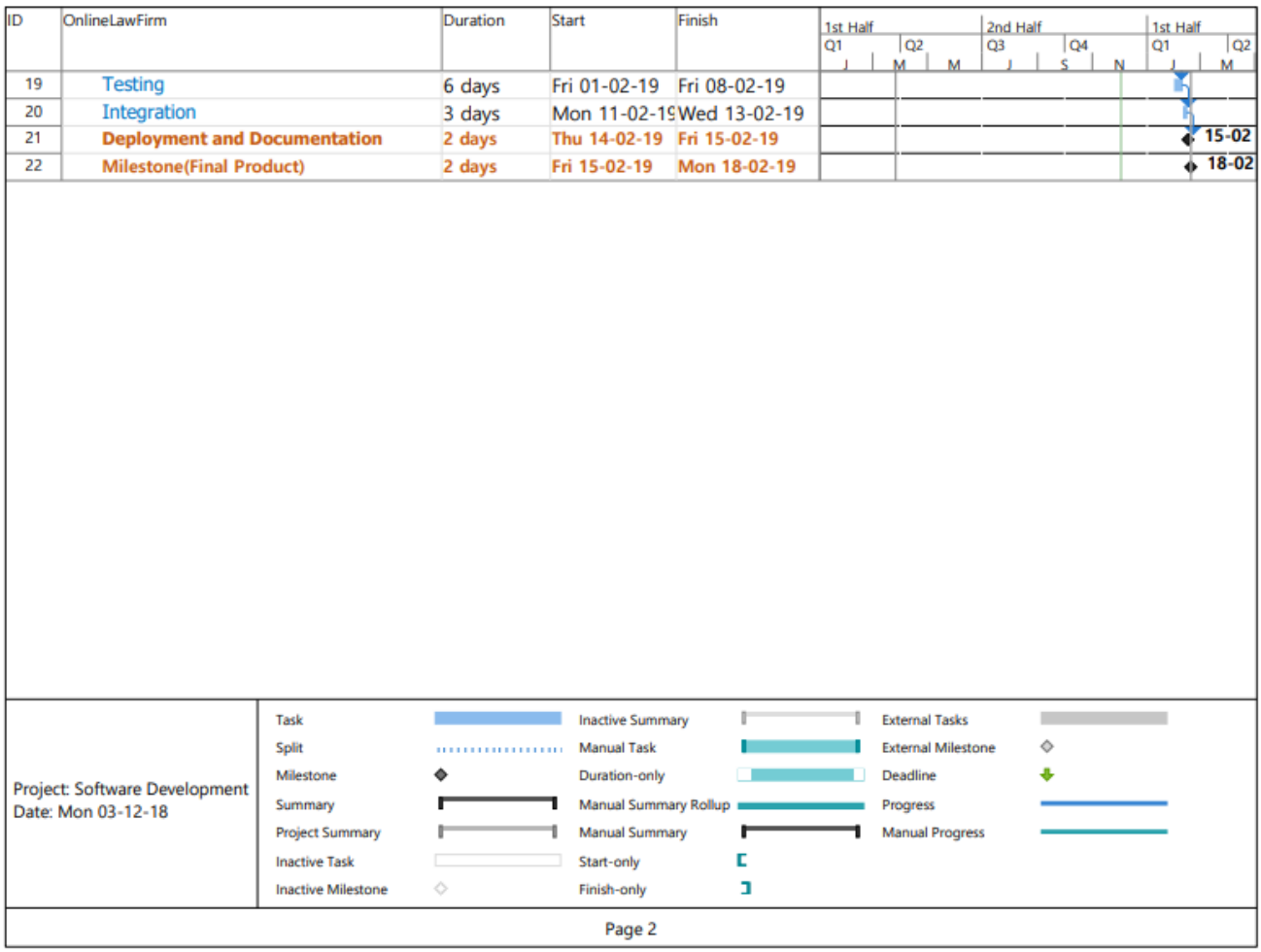
Development Plane:

In the development plane, the overall strategies of the project have been placed full-time spam of the project is placed in this development plane. The giant chart shows all the sequence of activities concerning time and duration. The development plan includes the sequence of iteration and details of alliteration and their sub-tasks and the duration of subtasks. Details of Milestone and detail of Final iteration in which complete system will have given to stakeholder as a shown in Figure 4.1 (Development Plane).

Figure 4.1 (Development Plane)

ID	OnlineLawFirm	Duration	Start	Finish	Milestone
0	**Software Development**	**235 days?**	**Tue 27-03-1**	**Mon 18-02-19**	
1	**Iteration 1 (Interfaces, Authentication, Profile**	**88 days**	**Tue 27-03-18**	**Thu 26-07-18**	
2	Feasiability study	9 days	Tue 27-03-18	Sun 08-04-18	
3	Requirement Phase	14 days	Mon 09-04-18	Thu 26-04-18	
4	SRS Document	20 days	Fri 27-04-18	Thu 24-05-18	
5	Design Document	15 days	Fri 25-05-18	Thu 14-06-18	
6	Implementation Phase	20 days	Fri 15-06-18	Thu 12-07-18	
7	Testing	10 days	Fri 13-07-18	Thu 26-07-18	
8	**Milestone(Interfaces ,Authetication , Profile**	**88 days**	**Tue 27-03-18**	**Thu 26-07-18**	**Yes**
9	**Iteration 2 (Interfaces ,System's Core Modules)**	**77 days**	**Fri 27-07-18**	**Mon 12-11-18**	
10	Requirement Phase	5 days	Fri 27-07-18	Thu 02-08-18	
11	Design Phase	20 days	Fri 03-08-18	Thu 30-08-18	
12	Implementation Phase	40 days	Fri 31-08-18	Thu 25-10-18	
13	Testing	12 days	Fri 26-10-18	Mon 12-11-18	
14	**Milestone(Interfaces,System Core Module)**	**77 days**	**Fri 27-07-18**	**Mon 12-11-18**	**Yes**
15	**Iteration 3 (Remaining Feature of Ap**	**70 days**	**Tue 13-11-18**	**Mon 18-02-19**	
16	Requirement Phase	8 days	Tue 13-11-18	Thu 22-11-18	
17	Design Phase	20 days	Fri 23-11-18	Thu 20-12-18	
18	Implementation Phase	30 days	Fri 21-12-18	Thu 31-01-19	





ID	OnlineLawFirm	Duration	Start	Finish
19	Testing	6 days	Fri 01-02-19	Fri 08-02-19
20	Integration	3 days	Mon 11-02-19	Wed 13-02-19
21	**Deployment and Documentation**	2 days	Thu 14-02-19	Fri 15-02-19
22	**Milestone(Final Product)**	2 days	Fri 15-02-19	Mon 18-02-19

Text Editor (Android Studio):

Android studio is the official Integrated development environment used for the development of the android operating system. Android studio is the most advanced and open-source platform for the development of android systems. Android studio has more the 26 APIs with the most advanced programming features that will help the software engineers to build efferent and best products using the android platform. Android Studio provides an easy way of coding for the development of android operating systems.

20

Quality Assurance

Overview

In this chapter, we will explore the introduction emphasizes the critical role of quality assurance in software development. The test plan outlines testing levels from module to post-execution. Test cases detail scenarios for various functionalities, including managing contacts, submitting cases, confirming availability, sending notifications, setting ratings, communicating with contacts, and saving case history. A link is provided for the Software Quality Assurance Document for feedback submission.

Introduction:

Testing the functionalities is the most impotent part of the testing also for the development of the product because with quality assurance we check whether the functionalities work properly in the way that we want. Quality assurance is very important for finding errors in the application and to deal with bugs in the product. Quality assurance helps the software engineers to find fix and remove errors and bugs very efficiently and make the product more efficient and flexible. In quality assurance, we check which tools are best for the development of the product and how the software engineer will find and fix the bugs from the application using that tool.

Test Plan:

In the test plan, we plan that we define the objective and task we define the scope of the testing, and check which testing strategy is best for the quality assurance in the development of the product. Here in the test plan following the testing level, we will cover quality assurance for “Online LawFirm” in the testing level we will assure the modeling testing to post-execution testing.

- Module testing
- Integration testing
- System testing
- Acceptance testing
- Deployment testing
- Post execution testing

Module testing:

In module testing, we will test each module and ensure that every module and check whether the functionalities work properly in the way that we want. We created a separate and small testing procedure for every single module and tried to test each module of the application separately. Testing is done on the module level by dividing the module into chunks and checking the quality of the module separately. The following module is being tested under the umbrella of Module testing.

- Accounts Module
- Manage Account Module
- Manage Contact Module
- Case Manager Module
- Set Rating Manager Module
- Send Notification Manager Module
- Communication Manger Module
- Send Location Manager Module
- Confirm Availability Module
- Case History Saver Module
- Share Legal Documents Module

Integration testing:

After we tested each module separately then we tried the fundamental principle of integration testing in which we combine individual units are combined and tested as a group to identify the faults and bugs from the result of integration testing. We used test drivers and test stubs to identify the faults and bugs from the group of units and clarify how individual units interact with each other and work as expected.

System Testing:

After Integration testing then we tried the fundamental principle of system testing in which we test a complete software system (OnlineLawFirm) the purpose of this testing is to evaluate the compliance of the software product with the given software requirements.

Acceptance testing:

After Integration testing then we tried the fundamental principle of acceptance testing in which we test the system for acceptability and test the complete software system (OnlineLawFirm) the purpose of this testing is to evaluate the compliance of the software product with the given business requirements and assess that whether this system or product is acceptable for delivery.

Deployment testing:

After acceptance testing then we tried the fundamental principle of deployment testing in which we cover two-phase of deployment testing pre-deployment testing and post-deployment testing in pre-deployment testing we test the individual unit of the software product and overall system testing to find errors and bugs and then fix them after pre-execution testing we will try post-deployment testing we deployed the software product in a real-world environment and test the system working component by component and check that data travel between software and server software system (OnlineLawFirm) the purpose of this testing to evaluate the compliance of the software product with the real work deployment with given software requirements. We will deploy this system on the Google play store and an android system will be used.

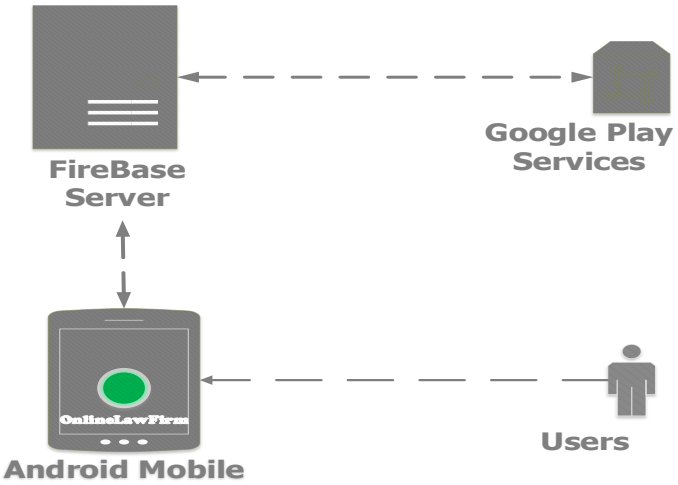


Figure 5.1: Deployment Diagram

Test Cases:

A test case is the set of conditions or variable under which we quality engineer test the software product and determine whether the system works as expected and satisfies the given requirements. Test Cases also help the quality engineer to find the problem in requirements or the design of the application.

Sign Up:

5.1: Sign Up (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF1	Sign Up	**objective:** The objective of this test case is to evaluate the Sign-up Activity	**Pre-condition:**	The user Must have a valid Email Address			
			Step-1	User Enter his First Name	Mudassar	The first Name Entered Successfully	**Pass**
			Step-2	User Enter his Last Name	Ali	The last Name Entered Successfully	**Pass**
			Step-2	User Select the status	Client	Status Entered Successfully	**Pass**
			Step-3	Select the Email Address	Osamajechangir3131 @gmail.com	System show Email and Email Entered Successfully	**Pass**
			Step-4	User Enter the CNIC Number	13101-5968677-1	CNIC Entered Successfully	**Pass**
			Step-5	The user selects his country code	+92	Country Code Entered Successfully	**Pass**
			Step-6	Users enter the postal address	CB-21Street No 7-link road Abbottabad Pakistan	Postal Address Entered Successfully	**Pass**
			Step-7	Users enter education	LLB, BS IN LAW,	Education Entered Successfully	**Pass**
			Step-8	Users enter Specialization	Terrorism	Specialization Entered Successfully	**Pass**
			Step-9	Users enter the Firm Name	Al-Sayed Law Firm	Specialization Entered Successfully	**Pass**
			Step-10	Users enter the Phone Number	03587895698	Phone No Entered Successfully	**Pass**

5.2: Sign Up (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OL F2	Sign Up	**objective:** The objective of this test case is to evaluate the Sign-up Activity	**Pre-condition:**	The user Must have a valid Email Address			
			Step-1	User Enter his First Name	Mudassar	The first Name Entered Successfully	**Pass**
			Step-2	User Enter his Last Name	Ali	The last Name Entered Successfully	**Pass**
			Step-2	User Select Invalid the status	Client	Status Entered Not Successful	**Fail**
			Step-3	Select the Email Invalid Address or Already Registered Email	Osamajechangir3131 @gmail.com	System show Email and Email Entered Not Successful	**Fail**
			Step-4	User Enter the CNIC Invalid Number	13101-5968677-1	CNIC Entered Not Successful	**Fail**
			Step-5	The user selects his country code	+92	Country Code Entered Not Successful	**Pass**
			Step-6	Users enter the postal address	CB-21Street No 7-link road Abbottabad Pakistan	Postal Address Entered Successfully	**Pass**
			Step-7	Users enter education	LLB, BS IN LAW,	Education Entered Successfully	**Pass**
			Step-8	Users enter Specialization	Terrorism	Specialization Entered Successfully	**Pass**
			Step-9	Users enter the Firm Name	Al-Sayed Law Firm	Specialization Entered Successfully	**Pass**
			Step-10	The user enters the Phone Invalid Number	03587895698	Phone No Entered Not Successful	**Fail**

Sign In:

5.3: Sign in (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF3	Sign In	**Objective:** The objective of this test case is to evaluate the Sign-in Activity	**Pre-condition:**	The User must have an account			
			Step-1	User Select the Email Address	Mudassarali2395@gmail.com	System Show the Email Address and Email is selected Successfully	**Pass**

5.4: Sign in (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF4	Sign In	**Objective:** The objective of this test case is to evaluate the Sign-in Activity	**Pre-condition:**	The User must have an account			
			Step-1	User Select Invalid Email Address	Osamajechangir3@gmail.com	System Show the Email Address and Email is selected Not Successful	**Fail**

5.5: Sign in (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF5	Sign In	**Objective:** The objective of this test case is to evaluate	**Pre-condition:**	The User must have an account			

		the Sign-up Activity					
			Step-1	User Select Other Email Address that is not registered in the system	Mudassarali2395 @gmail.com	System Show the Email Address and Email is selected Not Successful	**Fail**

Manage Contacts:

5.6: Manage Contacts (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF6	Manage Contact	**Objective:** To Evaluate the Functionality of Manage Contact Activity	**Pre-Condition:**	• The desired contact must be connected to the user • The user must be signed In			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the case Requests Option	Case Requests Option	Case Menu Requests option will be shown to the system Successfully	**Pass**
			Steps 3	User Select the case Requests to accept Option	Case Description option open to the user	Case Description option open successfully	**Pass**
			Step 4	User Select the case Requests to accept Option	case Requests accept Option selected	Contact Added to Connects To system successfully	**Pass**

5.7: Manage Contacts (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF7	Manage Contact	**Objective:** To Evaluate the Functionality of Manage Contact Activity	**Pre-Condition:**	• The desired contact must be connected to the user • The user must be signed In			
			Steps 1	User Select the Case Menu	Case Menu Option open to the user	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the case Requests Option	Case Requests Option open to the user	Case Menu Requests option will be shown to the system Successfully	**Pass**
			Steps 3	User Select the case Requests Reject Option	Case Description option open to the user	The case Description option does not open successfully	**Fail**
			Step 4	User Select the case Requests to accept Option	case Requests accept Options selected and open to the user	Contact Added to Connects To system successfully	**Pass**
			Step 5	User Select Invalid Option with no Internet	case Requests accept Option selected	case Requests accept Option not selected	**Fail**

Submit Case:

5.8: Submit Case (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF8	Submit Case	**Objective:** To Evaluate the Functionality	**Pre-Condition:**	• The desired contact must be connected			

		of Submit Case Activity		to the user • The user must be signed In			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the case send case Requests Option	Case send Requests Option open to user	Case Menu Requests option will be shown to the system Successfully	**Pass**
			Steps 3	User Fill the Case Registration form	Case Registration option open to the user	Case Registration option open successfully	**Pass**
			Step 4	User Select the case Requests send Option	case Requests send Option selected	case Added in history and send successfully	**Pass**
			Step 5	User Select the case Requests to send option and Profile Option of targeted	case Requests send case Option and Profile selected	case Requests send case Option and Profile selected Successfully	**Pass**

Submit Case Failure:

5.9: Submit Case (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF9	Submit Case	**Objective:** To Evaluate the Functionality of Submit Case Activity	**Pre-Condition:**	• The desired contact must be connected to the user • The user must be signed In			

			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the case send case Requests Option	Case send Requests Option open to user	Case Menu Requests option will be shown to the system Successfully	**Pass**
			Steps 3	User Fill the Case Registration form And fill invalid data	Case Registration option open to the user	Case Registration option open successfully	**Fail**
			Step 4	User Select the case Requests send Option	case Requests send Option selected	Case not Added in history and send not successfully	**Fail**
			Step 5	User Select the case Requests to send option and Profile Option of targeted	case Requests send case Option and Profile selected	case Requests send case Option and Profile selected Successfully	**Pass**

Confirm Availability:

5.10: Confirm Availability (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF10	Confirm Availability	**Objective:** The objective of this test case is to evaluate the Confirm Availability Activity	**Pre-condition**	The User must have an account The desired contact must be connected to the user The user must be signed In			
			Step-1	User Select the Confirm Availability Option Check the option	User Select the Confirm Availability Option	User Select the Confirm Availability Option to send to the	**Pass**

					Visible to the user	user Successfully	

5.11: Confirm Availability (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Expected Results	Status
OLF11	Confirm Availability	**Objective:** The objective of this test case is to evaluate the Confirm Availability Activity	**Pre-condition:**	The User must have an account The desired contact must be connected to the user The user must be signed In			
			Step-1	User Select the Confirm Availability Option to uncheck the option	User Select the Confirm Availability Option Visible to the user	User Select the Confirm Availability Option does not send to the user Successfully	**Fail**

Send Location:

5.12: Send Location (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF12	Send Location	**Objective:** To Evaluate the Functionality of Send Location Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**

			Steps 2	User Select the case history Option	case history Option open to user	case history option will be shown to the user Successfully	**Pass**
			Steps 3	User Select the case history Option and send the location option	Send location option open to the user	Current Longitude and Latitude Selected successfully	**Pass**
			Step 4	User Select the case history Option and send the location option	Send location option open to the user	Current Location send successfully	**Pass**
			Step 5	User Select the case history Option and send the location option	Send location option open to the user	Current Location send successfully and added in Case History	**Pass**

5.13: Send Location (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF13	Send Location	**Objective:** To Evaluate the Functionality of Send Location Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**

			Steps 2	User Select the case history Option	case history Option open to user	case history option will be shown to the user Successfully	**Pass**
			Steps 3	User Select the case history Option and send the location option	Send location option open to the user	Current Longitude and Latitude Not Selected successfully	**Fail**
			Step 4	User Select the case history Option and send the location option	Send location option open to the user	Current Location Not sent successfully	**Fail**
			Step 5	User Select the case history Option and send the location option	Send location option open to the user	Current Location Not sent successfully and not added in Case History	**Fail**

Send Legal Document:

5.14: Send Legal Document (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF14	Send Legal Document	**Objective:** To Evaluate the Functionality of Send Legal Document Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**

			Steps 2	User Select the Send Legal Document Option	Send Legal Document Option open to user	Send Legal Document Option will be shown to the user Successfully	**Pass**
			Steps 3	User Select Send Legal Document Option and send the document	Send Legal Document option open to the user	Send Legal Document Option will be shown to the user and Document Selected Successfully	**Pass**
			Step 4	User Select Send Legal Document Option and click the document sent option	Send Legal Document option and document selected	A current Selected document was sent successfully	**Pass**
			Step 5	User Select Send Legal Document Option and click the document sent option	Send Legal Document option and document selected	A current Selected document was sent successfully And added in OnlineLawFirm Folder	**Pass**

5.15: Send Legal Document (Failure)

ID	**Test Case**	**Test Case Summary**	**Test Case Steps**	**Test case Description**	**Test case Data**	**Actual Results**	**Status**
OLF15	Send Legal Document	**Objective:** To Evaluate the Functionality of Send Legal Document Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system	**Pass**

						Successfully	
			Steps 2	User Select the Send Legal Document Option	Send Legal Document Option open to user	Send Legal Document Option will be shown to the user Successfully	**Pass**
			Steps 3	User Select Send Legal Document Option and send the document	Send Legal Document option open to the user	Send Legal Document Option will be shown to the user and Document Not Selected Successfully	**Fail**
			Step 4	User Select Send Legal Document Option and click the document sent option	Send Legal Document option and document selected	Current Selected Document Not sent successfully	**Fail**
			Step 5	User Select Send Legal Document Option and click the document sent option	Send Legal Document option and document selected	Current Selected Document Not sent successfully And not added in OnlineLawFirm Folder	**Fail**

Verify Case:

5.16: Verify Case (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF16	Verify Case	**Objective:** To Evaluate the Functionality of Verify Case Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be	**Pass**

						shown to the system Successfully	
			Steps 2	User Select the Case Description Option	Case Description Option open to user	Case Description Option will be shown to the user Successfully	**Pass**

5.17: Verify Case (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF17	Verify Case	**Objective:** To Evaluate the Functionality of Verify Case Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the invalid Case Description Option	Case Description Option does not open to the user	Case Description Option will not be shown to the user Successfully	**Fail**

5.18: Send Notification (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF18	Send Notification	**Objective:** To Evaluate the Functionality of Send Notification Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be			

				connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select Send the Location	Send Location Option open to user	Send Location Option will be shown to the user and a Notification send Successfully	**Pass**
			Steps 3	User Select Send the Document option	Send Legal Document option open to the user	Send Legal Document Option will be shown to the user and Document Selected, and Notification send Successfully	**Pass**
			Step 4	User except for the case request option and click the Except case	Case Accepted	Notification sent successfully	**Pass**

5.19: Send Notification (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF19	Send Notification	**Objective:** To Evaluate the Functionality of Send Notification Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user			

				• The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select Send the Location	Send Location Option open to user	Send Location Option will not be shown to the user and a Notification send Successfully	**Fail**
			Steps 3	User Select Send the Document option	Send Legal Document option open to the user	Send Legal Document Option will be shown to the user and Document Not Selected, and Notification send Successfully	**Fail**
			Step 4	User except for the case request option and click the Except case	Case Accepted	Notification Not sent successfully	**Fail**

Set Rating:

5.20: Set Rating (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF20	Set Rating	**Objective:** To Evaluate the Functionality of Set Rating Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the ongoing case Option	ongoing case Option open to user	Case Menu ongoing case will be shown to the system Successfully	**Pass**
			Steps 3	The user opens the case	Case description option open to the user	The case description option opens successfully	**Pass**
			Step 4	Lawyer Mark Case Complete Option	Lawyer Mark Case Complete Option and pop up the menu shown to the user	Lawyer Mark Case Complete Option and pop up the menu shown to the user Successfully	**Pass**
			Step 5	Lawyer Mark Case Complete Option-click the desired option Yes/No	Lawyer Mark Case Complete Option and pop up the menu shown to the user desired	Lawyer Mark Case Complete Option and a pop-up menu will be shown the request sent to the	**Pass**

					option Yes/No	user Successfully	
			Step 6	Client Receive the Request and Mark Case as Complete Mark Case Complete Option-click the desired option Yes/No	Client Mark Case Complete Option and pop up the menu shown to the user desired option Yes/No	Client Mark Case Complete Option and a pop-up menu will be shown the request sent to the user Successfully	**Pass**
			Step 7	Client Receive the Request and Mark Case as Complete Mark Case Complete Option-click the desired option Yes/No	Client Mark Case Complete Option and pop up the menu shown to the user desired option Yes/No	Client Mark Case Complete Option and a pop-up menu will be shown the request sent to the user Successfully	**Pass**
			Step 8	Client Receive the Request and Mark Case as Complete Mark Case Complete Option-click the desired option Yes option	Client Mark Case Complete Option and Rating pop up the menu shown to the user set the rating	Client Mark Case Complete Option and set rating pop-up menu will be shown Rating Sent Successfully	**Pass**

5.21: Set Rating (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF21	Set Rating	**Objective:** To Evaluate the Functionality of Set Rating Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user			

				• The user must be signed in			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the ongoing case Option	ongoing case Option open to user	Case Menu ongoing case will be shown to the system Successfully	**Pass**
			Steps 3	The user opens the case	Case description contains invalid data option open to the user	The case description option does not open successfully	**Fail**
			Step 4	Lawyer Mark Case Complete Option	Lawyer Mark Case Complete Option and pop up the menu shown to the user	Lawyer Mark Case Complete Option and pop up the menu shown to the user Successfully	**Pass**
			Step 5	Lawyer Mark Case Complete Option-click the desired option No	Lawyer Mark Case Not Complete Option and pop up the menu shown to the user desired option Yes/No	Lawyer Mark Case Complete Option and a pop-up menu will be shown no request sent to the user Unsuccessful	**Fail**
			Step 6	The client will not receive the Request	Client Mark Case Not Complete Option and pop up the menu shown to the user desired	Client Mark Case Not Complete Option and a pop-up menu will be shown No request sent to the user Unsuccessful	**Pass**

					option Yes/No		
			Step 7	The client will not receive the Request	Client Mark Case Complete Option and pop up the menu shown to the user desired option Yes/No	Client Mark Case Complete Option and a pop-up menu will be shown the request sent to the user Unsuccessful	**Fail**
			Step 8	Client Receive the no Request and Mark Case as Not Complete Option-click the desired option No option	No Rating Box will be shown	Rating Not Set Successfully	**Fail**

Communicate with connects:

5.22: Communicate with Connects (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF22	Communicate with Connects	**Objective:** To Evaluate the Functionality of Communicate with Connects Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the connects Option	The communication Manager will be shown	Communication Manager will be shown Successfully	**Pass**
			Steps 2	User Open the connects	Connects will be open in chant	Communication Manager will be shown, and chat	**Pass**

						activity will be shown Successfully	
			Steps 3	User Select the Type Message Option and type the message	Hello world	The message will be typed Successfully	**Pass**
			Step 4	User Use the send Message option	Hello word Message content sent	Message content sent Successfully and shown in chat activity	**Pass**

5.23: Communicate with Connects (Failure)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF23	Communicate with Connects	**Objective:** To Evaluate the Functionality of Communicate with Connects Activity	**Pre-Condition:**	• The User must have an account • The desired contact must be connected to the user • The user must be signed in			
			Steps 1	User Select the connects Option	The communication Manager will be shown	Communication Manager will be shown Successfully	**Pass**
			Steps 2	User Open the connects	Connects will be open in chant	Communication Manager will be shown, and chat	**Pass**

						activity will be shown Successfully	
			Steps 3	User Select the Type Empty Data Message Option and type the message	No data	The message will be typed Not Successfully	**Fail**
			Step 4	User Use the send Message option	Empty Data will not be sent	Message content not sent Successfully	**Fail**

Save Case History:

5.24: Save Case History (Success)

ID	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF24	Manage Contact	**Objective:** To Evaluate the Functionality of Manage Contact Activity	**Pre-Condition:**	• The desired contact must be connected to the user • The user must be signed In			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**
			Steps 2	User Select the case Requests Option	Case Requests Option	Case Menu Requests option will be shown to the system Successfully	**Pass**

			Steps 3	User Select the case Requests to accept Option	Case Description option open to the user	Case Description option open successfully	**Pass**
			Step 4	User Select the case Requests to accept Option	case Requests accept Option selected	Contact Added to Connects To system successfully	**Pass**
			Step 5	User Use the Ongoing option	Case Description will be shown to the user	History Of Case will be shown to the user successfully	**Pass**
			Step 6	User Use the Add Detail Option	Add detail Option will be shown to the user	Add detail Option will be shown to the user successfully	**Pass**
			Step 7	User Add Detail and Location	Information added In Case History	Information added In Case History Successfully	**Pass**

5.24: Save Case History (Failure)

D	Test Case	Test Case Summary	Test Case Steps	Test case Description	Test case Data	Actual Results	Status
OLF24	Manage Contact	**Objective:** To Evaluate the Functionality of Manage Contact Activity	**Pre-Condition:**	• The desired contact must be connected to the user • The user must be signed In			
			Steps 1	User Select the Case Menu	Case Menu Option	Case Menu will be shown to the system Successfully	**Pass**

			Steps 2	User Select the case Requests Option	Case Requests Option	Case Menu Requests option will be shown to the system Successfully	**Pass**
			Steps 3	User Select the case Requests to accept Option	Case Description option open to the user	Case Description option open successfully	**Pass**
			Step 4	User Select the case Requests to accept Option	case Requests accept Option selected	Contact Added to Connects To system successfully	**Pass**
			Step 5	User Use the Ongoing option	Case Description will be shown to the user	History Of Case will be shown to the user successfully	**Pass**
			Step 6	User Use the Add Detail Option	Add detail Option will be shown to the user	Add detail Option will be shown to the user Unsuccessful	**Fail**
			Step 7	User Add Detail and Location	Information not added In Case History	Information not added In Case History Unsuccessful	**Fail**

Software Quality Assurance Document

Download the document for practicing and writing the Software Quality Assurance document of the Project.
https://github.com/SardarMudassarAliKhan/Software-Testing-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

21

Results and Discussions of Software product

Overview

In this chapter, we will see the outcomes and discussions of Online LawFirm, highlighting its successful functionality and user-friendly interface. Features like registration, case submission, and communication are examined, highlighting their effectiveness. Moreover, future enhancements including the addition of judiciary and Sharia systems are proposed. This chapter emphasizes the significance of results in guiding future development cycles. OnlineLawFirm aims to provide accessible legal assistance, facilitating seamless interactions between clients, lawyers, and law experts.

Introduction:

This chapter covers the results and discussions of the OnlineLawFirm and covers the possible future enhancement related to the system.

Results and Discussions:

The proposed application OnlineLawFirm is working as expected and the functionality of the system is according to the requirement gathered in the first phase of requirement analysis. In OnlineLawFirm the usability is high the system will give results to the user according to the user requirements. The Interface of the system is user-friendly users can easily use the system. OnlineLawFirm is a Realtime android cloud-based application that changes behavior according to the status of the user. Users can sign up according to /her status and find assistance according to /her needs. OnlineLawFirm gives the results to users within one second. The data of the user is saved in firebase every activity of the user is saved in firebase. OnlineLawFirm provides a communication system through which users of the system can communicate with each other. The system provides a Location-Sender in which specific users who have the authorization to use this feature of the system can send a location to a user of the system. Some of the features of the system are mentioned below and we will discuss the results of each feature.

Features Results:

Registration of Client/Lawyer, Experts, Admin:

Registration of the users is by using the registration feature of the system through which users can register /her account in our system. This feature provides a user-friendly interface to users and guides how to register in the system. This feature of the system is working fine and gives results according to the system requirement.

Manage Account:

The manage accounts feature of the system provides the functionality through which the admin can block and unblock accounts in our system. This feature provides a user-friendly interface to users and guides how to block and unblock the accounts in the system. This feature of the system is working fine and gives results according to the system requirement.

Manage Contacts:

The manage contacts feature of the system provides the functionality through which users can Manage connections in our system. This feature provides a user-friendly interface to users and guides how to manage the contacts in the system. This feature of the system is working fine and gives results according to the system requirement.

Submission of Cases from Clients:

Submission of the case modules provides the functionality through which users can submit the case to lawyers or law experts. This feature provides a user-friendly interface to users and guides how to register in the system. This feature of the system is working fine and gives results according to the system requirement.

Verification of Cases:

Law Experts will receive the case according to their area of the field and then verify each case after verification law expert will proceed and reject the case. This feature provides a user-friendly interface to users and guides how to verify the case in the system. This feature of the system is working fine and gives results according to the system requirement.

Assign the case to the lawyer:

Law Experts can assign the cases to lawyers according to their area of the field. This feature provides the results according to the requirement of the system.

Notification system:

Law experts and lawyers send a notification to the Client law expert will receive the case according to their area of the field and then verifies each case after verification, the law expert will send a notification to the client, and the case is processed and rejected. This feature provides a user-friendly interface to users and guides how to send a notification to clients through the system. This feature of the system is working fine and gives results according to the system requirement.

GPS Location:

The system provides the Location-Sender in which specific users who have the authorization to use this feature of the system can send location to the user of the system. This feature provides a user-friendly interface to users and guides how to send GPS location to clients the in system. This feature of the system is working fine and gives results according to the system requirement.

Communication System:

OnlineLawFirm provides a communication system through which users of the system can communicate with each other. This feature provides a user-friendly interface to users and guides how to send messages to users in the system. This feature of the system is working fine and gives results according to the system requirement.

Lawyers' availability confirmation:

After receiving the notification lawyer can confirm their availability for the Case. This feature provides a user-friendly interface to users and guides how to send messages to users in the system. This feature of the system is working fine and gives results according to the system requirement.

Rating System:

The system provides the Location-Sender in which specific users who have the authorization to use this feature of the system can send location to the user of the system.

Sharing of Legal Documents:

OnlineLawFirm provides a document sharing system through which users of the system can share the document. This feature provides a user-friendly interface to users and guides how to share documents with users in the system. This feature of the system is working fine and gives results according to the system requirement.

Save Case History:

OnlineLawFirm provides functionality to the user through which lawyers and law experts can save the case proceeding each and everything is saved in case of history which is related to the case. This feature provides a user-friendly interface to users and guides how to save the case history of the case in the system. This feature of the system is working fine and gives results according to the system requirement.

Future Enhancements:

At this level, OnlineLawFirm has the following Modules

- Accounts Module
- Manage Account Module
- Manage Contact Module
- Case Manager Module
- Set Rating Manager Module
- Send Notification Module
- Communication Manger Module
- Send Location Manager Module
- Confirm Availability Module
- Case History Saver Module
- Share Legal Documents Module
- Lawyer and Law-expert Promotion Module But in the future, we will increase the functionality of the system some of them are:
    - Increase the functionality by adding Judiciary System
    - Increase the functionality by adding Sharia System
    - Increase the functionality by adding Mobile System

Importance of Results and Discussions:

The result and discussion part are very important in development because the results are based on the very authentic document that is prepared after the complete and various testing process after this whole process development team start the discussion on the results and then plan the next development cycle of the system in which we plan to develop new feature and enhance the existing functionality of the system.

Conclusion:

This system was developed with the will to provide ease to society. We aim to help people facing different problems by providing expert advice through our multi-client cloud-based system. The online law firm will provide ease to access and use of online platforms for people seeking legal help. The system will facilitate the law firm to handle these clients and allocate lawyers to their tasks and keep track of the complete legal process (cases). The client, lawyer, and law experts will be registered in our system. All the Interface which is necessary for the system is completed. A menu-driven approach is used for the designing of the system. This system was implemented successfully. The manpower needed for the working of the system is less and it was seeming to be more secure. Thus, this system was developed successfully

Software Result and Discussion Document

Download the document for practicing and writing the Software result and discussion document of the Project.
https://github.com/SardarMudassarAliKhan/Software-Requirement-Management-Document
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

Complete Software Specification (SRS) of the Project Document

Download the document for practicing and writing the complete SRS document of the Project.
https://github.com/SardarMudassarAliKhan/SRS-Document.git
Submit your document for feedback from our Requirement engineer at the email address mudassarali.official@gmail.com

References:

1. https://www.softwareengineerinsider.com/
2. https://www.google.com
3. https://businessanalyst.techcanvass.com/
4. https://www.microsoft.com/en-ww/microsoft-365/visio/flowchart-software
5. https://www.microsoft.com/en-us/microsoft-365/project/project-management-software

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